

(45 Marks)

2 Hours

SECTION 'A' (Multiple Choice Questions) (09)

This section consists of 18 part questions and all are to be answered. Each question carries ½ mark. Copy the part questions in your answerbook. Write only the full against the proper number of the question and its part.

Write the correct answer for each from the given options:

1. Which organelles contain DNA.

- * Golgi bodies and Lysosome
- * Mitochondria and Chloroplast ✓
- * Cell wall and Cell membrane
- * Smooth endoplasmic reticulum and Glyoxysome

2. Which cell organelles take part in detoxification of alcohol.

- * Golgi bodies
- * Mitochondria
- * Retina Complex
- * Photon

3. Which part of chlorophyll is called.

4. Which part of chlorophyll is called.

5. Which part of chlorophyll is called.

6. Which of the most common C₄ plants are:

- * Sugarcane ✓

7. Which of the following function as carboxylase as well as:

- * Cactus
- * Wheat
- * Oxygenase ✓
- * Ligase

8. Which of the following is a reductase.

- * Hydrolase

9. Which of the following gametophyte of fern is called:

- * Antheridium
- * Prothallus ✓

10. Which of the following is a gametophyte of fern.

11. Which of the following mosquito is the vector of:

- * Ebola Virus
- * Measles Virus

12. Which of the following is a virus.

13. Which of the following is a virus.

14. Which of the following fungus Phytophthora causes this disease:

- * late blight of tomato
- * early blight of tomato

15. Which of the following is a blight of potato.

16. Which of the following is a blight of potato.

17. Which of the following eukaryotes evolved from the ancestors:

- * Eukaryote

- * RNA
- * Protobionts ✓

18. Which of the following is the most important cellulose degraders in ecosystem are:

- * Ascomycota ✓
- * Deuteromycota

19. Which of the following is the most important cellulose degraders in ecosystem are:

20. Which of the following is the most important cellulose degraders in ecosystem are:

- xi) The type of alternation of generation in *Ulva* is:
 * Heterotrophic
 * Isomorphic ✓
 * Heteromorphie
 * Autotrophic
- xii) Spores of land plants tough, flexible and resistant to
 are due to:
 * Chitin * Waxes * Sporopollenin ✓
 * Pe
- xiii) Legumes belongs to this family:
 * Solanaceae
 * Cucurbitaceae
 * Fabaceae ✓
 * Brassicaceae
- xiv) Double fertilization is the characteristics of:
 * Bryophyta
 * Angiosperm ✓
 * Gymnosperm
 * Pteridophyta
- xv) Insectivorous plants live in soil deficient in:
 * Nitrogen ✓ * Calcium * Magnesium * Potassium
- xvi) The water potential of pure water is:
 * - 10 * zero ✓ * 10 * 100
- xvii) The growth of pollen tube through style to the ovary is
 movement:
 * Geotropism * Chemotropism ✓
 * Hydrotropism * Phototropism
- xviii) Guttation occurs through:
 * Lenticels * Hydathodes ✓ * Stomata * Bark

SECTION "B" (Short Answer Question) Mark

2. Answer any NINE part questions. All part question carries marks..

i) Why Lysosome is called suicidal sac?

ANSWER: Lysosomes are cell organelles which clean up the cell by digesting foreign materials as well as worn-out cell organelles. Lysosomes are called suicidal sacs because when the cells damage, they burst and digest their own cell.

ii) Why are chloroplast and mitochondria considered as semi-autonomous organelles?

ANSWER: Chloroplasts and mitochondria are considered semi-autonomous organelles because both chloroplasts and mitochondria have their DNA and ribosomes, which enable them to synthesize some of their proteins and they can replicate themselves, independent of the cell's division cycle.

However, they are still dependent on the cell such as, while they can synthesize some proteins, many proteins are encoded by nuclear DNA imported from the cytosol. They rely on the cell's waste management.

to remove waste products. This mix of independence and
 once earns them the designation of semi-autonomous organelles.
 a note on any one of the following:

* C_4 Plants

Plants

ER:

Plants develop another metabolic cycle to fix CO_2 , in the presence of
 even concentration, these plants are called C_4 plants. They develop
 anatomical and physiological modification, develop special tissues around
 bundle called bundle sheath, shift C_3 cycle in these cells from
 cells, while in mesophyll cells they develop another cycle for
 dioxide fix. The CO_2 fix here will transfer to C_3 cycle in bundle
 in this way two cycles are linked together. In the plant atmospheric
 is fixed by a three carbon compound Phospho Enol Pyruvate (PEP)
 of RuBP. As a result of CO_2 fixation a four carbon compound is
 called oxaloacetate (oxaloacetic acid) therefore this cycle is called C_4
 oxaloacetate transfer carbon through malate to RuBP for C_3
 bundle sheath. Among C_4 plants some important C_4 plant
 are sugar cane and corn.

CRASSULACEAN ACID METABOLISM (CAM)

Adaptation to avoid this condition has evolved in succulent plants like
 cactuses and many others. These plants open their stomata during
 night close them during the day, just reverse of normal behavior.
 stomata during day helps these plants to conserve water. During the
 when their stomata are open, these plants take up CO_2 and
 store into a variety of organic acids. This mode of carbon dioxide
 is called crassulacean acid metabolism or CAM.
 plants are called CAM plants (CAM). The CAM plants store these
 acids with CO_2 moving in their vacuoles. During the day, when the
 reactions supply ATP and $NADPH_2$ for Calvin cycle. These acid
 CO_2 to compete with O_2 . In this way ratio of CO_2 maintain inside
 the CO_2 is fixed through C_3 cycle.

Photorespiration considered as wasteful process?

Photorespiration is considered a wasteful process because
 photorespiration energy is not evolved. ATP and $NADH_2$, do not
 contrary ATP and $NADPH_2$ are produced during the light
 and also consumed. During this process CO_2 is released, instead of
 photosynthesis carbohydrates. The growth of plants is also reduced by up to
 50% in photorespiration.
 the symptoms of AIDS?

SYMPTOMS OF AIDS

At the beginning of HIV infection symptoms appear for a short duration. Fever, flu, and headache appear within the duration of six weeks and that person can infect others. In the next stage, the virus may remain average of ten years without any symptoms but the person may have glands swollen in the neck region. While the immune system weakens, infections like tuberculosis (TB), pneumonia and cancer (like Sarcoma) may occur. Some other viral infections may occur like Influenza (flu). Finally, the immune system weakens due to the loss of T lymphocyte cells weight loss, night sweats, diarrhoea, septicaemia-like infections cause the death of a person.

vi) What are the various methods of control harmful bacteria?

ANSWER:**CONTROL OF HARMFUL BACTERIA**

The chemical and physical methods used to control harmful bacteria.

CHEMICAL METHODS TO CONTROL HARMFUL BACTERIA

Bacterial harmful activities can be controlled by certain chemicals such as disinfectants, antiseptics, antibiotics, and antimicrobial chemicals.

1. STERILIZATION

Sterilization is the process of destroying all living organisms and viruses.

2. DISINFECTION

Disinfection is the elimination of microorganisms, but not necessarily endospores, from inanimate objects or surfaces. An ideal disinfectant or antiseptic (chemical agent) kills microorganisms in the shortest possible time without damaging the material treated.

THE USE OF ANTIBIOTICS

Antibiotics are compounds that are mostly derived from biosynthesis. They have given us antibiotics. Antibiotics act in different ways against microorganisms like of other microorganisms.

PHYSICAL METHODS TO CONTROL HARMFUL BACTERIA**BOILING:**

100 °C denatures proteins and alters membranes and destroy bacteria.

Cooking,

DRY-HEAT OVEN:

170 °C for 2 hours, denature proteins and alters membranes.

INCINERATION:

Exposure to flame or destroy bacteria by burning.

AUTOClave:

Heating at very high temperature like 121 °C for 15-40 minutes at 15 psi denature proteins and alters membranes.

IZATION:
 15 seconds 138 °C for 2 seconds (UHT), Denatures proteins and
 prevents it prevents spoilage of milk, apple juice, honey, and other
 liquids

ATION:
 0 °C to 7 °C, Inhibits metabolism, Preservation of food or
 materials.

stops metabolism, may kill microbes, Long-term storage of
 stony cultures, or medical specimens.
 Distinct between Gram positive and Gram negative bacteria.

POSITIVE BACTERIA	GRAM-NEGATIVE BACTERIA
Gram-positive bacteria have a thick (20-80 nanometers) of peptidoglycan in their cell walls.	Gram-negative bacteria have a thinner layer (2-3 nanometers) of peptidoglycan in their cell walls.
Gram-negative bacteria lack an outer membrane, which makes them more susceptible to antibiotics and other external factors.	Gram-negative bacteria have an outer membrane containing lipopolysaccharides (LPS).
Examples of Gram-positive bacteria include Staphylococcus aureus, Bacillus subtilis, and Clostridium difficile.	Examples of Gram-negative bacteria include Escherichia coli (E. coli), Pseudomonas aeruginosa, and Salmonella enterica.

prokaryotes are different from eukaryotes?

PROKARYOTES ARE DIFFERENT FROM EUKARYOTES

Prokaryotic cells are the primitive kind of cell, whose size varies from 0.5-
 5.0 µm. Eukaryotic cells are the modified cell structure containing
 various components in it, their size varies from 2-100µm,
 prokaryotes are generally found in single cell organisms, while
 eukaryotes are found in multicellular organisms.

In prokaryotes, the cell division takes place through conjugation,
 binary fission, transduction, while in eukaryotes, it is through the process
 of mitosis.

In prokaryotes Cell wall present, comprise of peptidoglycan or
 murein (polysaccharide), while in eukaryotes Usually cell wall absent,
 except in plant cells and fungi.

ANSWER: Because fungi have rigid cell walls containing chitin as the fibrillar material. It provides protection from predators.

* Tulsi * Amaltas * Mulethi

ANSWER:

COMMON NAME	BOTANICAL NAME
Tulsi	<i>Ocimum tenuiflorum</i>
Amaltas	<i>Cassia fistula</i>
Mulhethi	<i>Glycyrrhiza glabra</i>
Mako	<i>Solanum nigrum</i>

Write three steps in evolution of seed.

ANSWER:

xii) Write three steps in evolution of seed.

ANSWER:

STEPS OF EVOLUTION

There are three steps in the evolution of a seed.

(i). Origin of heterospory i.e., two types of spores are produced. Microspores grow into sperm-forming gametophyte (male gametophyte) and megaspores grow into egg-forming gametophyte (female gametophyte).

(ii). Development of integument, an enveloping structure for the protection of mega sporangia to form ovule.

(iii). Retention of the mature megaspores in the sporangia to develop gametophyte

The development of seeds has given vascular plants better adaptation to their environment.

xii) Why cold treatment is necessary for germination of seed in some plants?

ANSWER: Cold treatment plays a very important role which is defined as the promotion of flowering by a cold treatment given to imbibed seeds or young plants. The effect of low-temperature treatment applied in the earlier stages of development has an after-effect on the later stages of the development of a plant, many plants require cold treatment for germination many kinds of cereal like wheat and rice exist in two varieties, the annual or spring and biennial or winter variety.

xiii) Differentiate between Monocotyledon and Dicotyledon.

ANSWER:

MONOCOTYLEDONAE	DICOTYLEDONAE
Monocotyledonae Plants produce only one cotyledon (Scutellum) containing seed.	Dicotyledonae Plants produce cotyledon (seed leaves) containing seed.

In Dicotyledonae dorsal and ventral surfaces are same. (bifacial leaf)	In Dicotyledonae dorsal and ventral surfaces are different. (bifacial leaf)
In Dicotyledonae Stem is mostly branched, vascular bundles arranged in a ring, ring usually present giving rise secondary growth to (exceptions palms)	In Dicotyledonae Stem is mostly branched, vascular bundles arranged in a ring, ring usually present giving rise secondary growth
Cymose Inflorescence.	

INFLORESCENCE

Inflorescence, main axis soon stops growing. The flower develop in succession and opening of flower is centrifugal. It has following

Types (Monochasial) cyme
(dichasial) Cyme

Section C (Detailed - Answer Question) (18)

Attempt any TWO question from this section.
Questions carry equal marks.
Write the life cycle of fern OR Flowering plants.

Q:

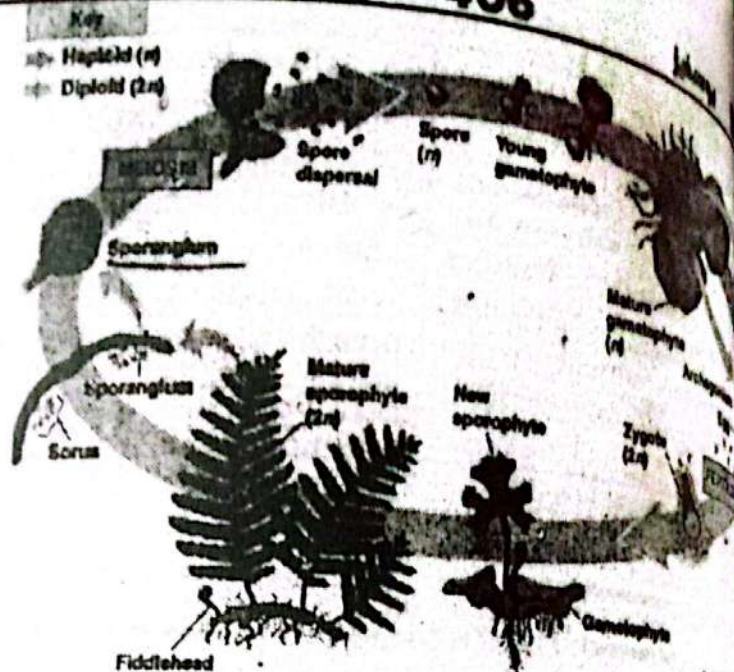
LIFE OF FERN:

Life cycle of fern (*Adiantum*) or *Dryopteris* shows heteromorphic alternation of generation in which the sporophytic phase is dominant. All ferns are homosporous, producing single types of spores.

SPOROPHYTIC PHASE:

Sporophyte ($2n$) which is diploid, consists of adventitious roots, underground stem a rhizome and pinnately compound leaves. For asexual reproduction the leaves give rise to rounded bodies on its surface which are called sori (singular sorus). The sori are green but when they become dark brown. The leaves which bears sori are called soriophylls. Each sorus is the group of sporangia.

Each sporangium consists of a stalk called sporangiophore and a biconvex capsule. The capsule consists of an annulus and stomium. The annular cells thickened whereas the cells are thin-walled. Within sporangium spore mother cells are present. Each spore mother cell divides by meiosis to form four haploid spores. The spores are liberated through stomium. Each spore on germination gives rise to miniature bisexual (monoecious) gametophyte called gametophyte.



GAMETOPHYTIC PHASE:

The gametophyte of fern is called prothallus. The prothallus is an autotrophic, heart shaped, dorsoventrally flattened lying prostrate on a wet substratum. It is fixed to soil with the help of rhizoids which absorb water and nutrients. The prothallus is monoecious having archegonia and antheridia on the same prothallus.

An archegonium consists of a venter with an ovum and a neck and becomes acidic by degenerating neck canal cells at maturity. Each archegonium produces a number of antherozoids (sperms). A number of sperms make chemotactic movement in water and reach the archegonium. One sperm fuses with the ovum to form an oospore (zygote) which is diploid. A new sporophyte develops from the oospore. In the meantime the prothallus degenerates. In this way the life cycle is completed.

LIFE CYCLE OF FLOWERING PLANTS:

Flowering plants show heteromorphic alternation of generations. The haploid gametophyte alternates with the diploid sporophyte. In the life cycle of plants, meiosis produces female and male haploid gametophytes. The gametophyte generation is reduced; the main plant body is the sporophyte. Each of them undergoes cell division by mitosis to produce a male and a female haploid gametophyte. This gametophyte produces male and female haploid gametes by mitotic division. Two gametes fuse to form a diploid zygote which then undergoes mitotic divisions to produce the diploid sporophyte.

SPOROPHYTE GENERATION:

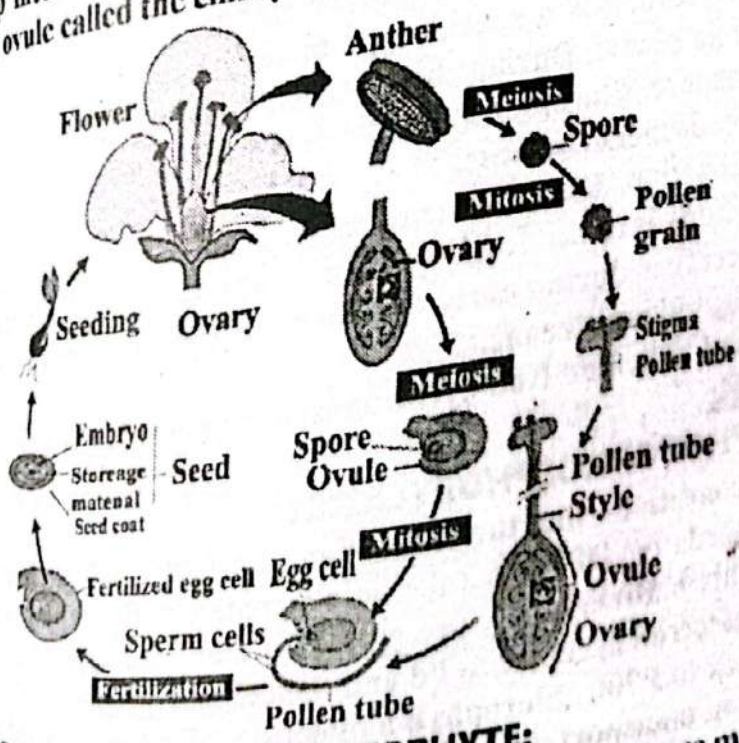
ASEXUAL REPRODUCTION IN ANGIOSPERM:

The main plant body is a sporophyte that consists of a root, stem, and leaves. The root, stem, and leaves are considered vegetative parts while the flowers are the reproductive parts.

is the reproductive part. Flower is a compressed reproductive shoot bearing four whorls of modified (floral) leaves called sepals, petals, stamens (microsporophyll) and megasporophyll (carpel). Microsporophyll (stamen) is the male reproductive structure and megasporophyll (carpel) is the female reproductive structure.

MICROSPOROPHYLLS (STAMENS): Microspores which later develop into pollen grains are produced by microsporophyll or stamens, each stamen consists of a sac-like anther and a filament, usually the anther has four chambers called pollen sacs or microsporangia, containing numerous microspore mother cells. Each microspore mother cell produces four microspores by meiosis so each pollen sac has numerous microspores that develop into pollen grains and rise from here for pollination.

MEGASPOROPHYLLS (CARPELS): A megasporophyll now called carpel produces an ovum in the ovary. Each carpel consists of three parts, stigma, style and ovary, within the ovary one or more ovules (megaspore) develop. The main body of the ovule is called nucellus which is protected by two layers of protective sheaths of cells called integuments. A small pore is present on it, the micropyle, with these it contains one megaspore mother cell which produces four megaspores on meiosis. Out of four megaspores, only one megaspore will be functional and develop into a female gametophyte. The female gametophyte develops in the ovule called the embryo sac.



DEVELOPMENT OF FEMALE GAMETOPHYTE: A megasporangium (Ovule) within the ovary contains megaspore mother cells. Each megaspore mother cell undergoes reduction division to form four haploid megaspores. Three

of them disintegrate and only one survives, which after successive division develops into a seven-cell (eight-nucleated) structure called gametophyte or embryo sac. Out of seven cells three cells are found toward chalazal end and called antipodal cells, two cells are found toward micropylar end called egg apparatus, one central cell with two nuclei termed synergid cells. The comparatively large cell is egg cell and one central cell with two polar nuclei which after fusion called secondary nuclei or definitive nuclei. All the cells except secondary nuclei are haploid. This multicellular structure of the ovule is termed embryo sac which constitutes the female gametophyte of angiosperm, male and female gametophyte dependent on sporophyte.

DOUBLE FERTILIZATION:

After pollination, pollen grains transfer and settle on the stigma of carpel. The pollen tube grows through the tissue of the stigma and style enters the ovary with a tube nucleus at its tip. When the tip of the pollen tube reaches on ovule, it enters the micropyle and then discharges the two sperm into the female gametophyte (embryo sac). One sperm fertilizes the egg to form a zygote ($2n$) which further develops into an embryo sporophyte. By the time fertilization occurs the two polar nuclei of female gametophyte have combined to form a diploid fusion nucleus in which the second sperm unites to form a triploid nucleus structure. This nucleus undergoes a series of divisions and a triploid cell, called endosperm is formed. The endosperms function in the food for the embryo.

Food storage organs provide food to the embryo at the time of germination. Double fertilization is a special type of fertilization which occurs only in angiospermic plants. During this process, a sperm fuses with the ovum to form an oospore. The other sperm fuses with the secondary nucleus to form a triploid endosperm nucleus.

After fertilization, the ovule matures and develops into a seed while the ovary develops into the fruit. The fruit not only helps to protect the seed from desiccation during early development but often also facilitates the dispersal by various means.

4. Describe Cyclic and Non-cyclic phosphorylation during light reaction.

ANSWER:

CYCLIC PHOTOPHOSPHORYLATION

In some conditions like the saturation of NADP, excited electrons do not move towards NADP, they take an alternative pathway i.e. cyclic electron flow. In this pathway, they use only PS-1 and have a shortcut. The electron from primary electron acceptor to Fd and then to cytochrome complex ultimately comes back to P700 chlorophyll a molecule. ATP has generated coupling of the electron transport chain with chemiosmosis, no further NADP is reduced to produce NADPH, and production of O_2 is also stopped during this.

This process of ATP formation is called cyclic phosphorylation, the generation of only ATP. It happens when the chloroplast has low ATP and the demand for it for the Calvin cycle or NADPH, accumulating in the chloroplast. NADPH may stimulate a temporary shift from non-cyclic to cyclic flow until ATP supply meets the demand.

CYCLOPHOTOPHOSPHORYLATION

quantity of H^+ ions in the thylakoid space as compared to the stroma. It develops an electrochemical gradient, these H^+ ions flow out of the space to the stroma through a channel protein present in the membrane called ATPase synthase complex. Hydrogen ions move through the complex by providing energy which is released during the downward flow of electrons in PS-II. This energy is utilized for the synthesis of ATP from ADP and P_i . This movement of H^+ through the ATPase complex creates a concentration gradient called chemiosmosis or chemiosmotic ATP synthesis. Because chemical and osmotic events join to permit ATP synthesis, the transport of three protons (H^+) through the ATPase complex is required for the production of one ATP molecule. The linear flow of electrons from water to $NADP^+$ coupled to ATP synthesis is called non-cyclic phosphorylation because the electrons pass from water to a final acceptor and never back to their initial source.

Two important events take place during light dependent reaction synthesis.

1. Photolysis of water

2. Electron transport chain i.e. PS II and PS I

3. Reduction of NADP into (NADPH)

4. Cyclic phosphorylation

Explain the structure and composition of Cell Wall.

COMPOSITION OF CELL WALL

Cell wall is present in bacterial, fungal, algal and plant cells. The bacterial cell wall is made up of peptidoglycan, and the fungal cell wall is made up of modified polysaccharide chitin, while the plant and the algal cell wall is made up of (cellulose) already discussed in the chapter on carbohydrates.

Plant cell wall is composed of main cellulose, pectin and other carbohydrates. These materials of cell-wall are always synthesized by the protoplasm, secreted out of the cell and deposited around the outer surface of plasma membrane.

STRUCTURE OF CELL WALL

The outer surface of some cells is covered with a non-living, stiff layer called cuticle. A plant cell-wall is mainly differentiated into three layers,

1. Middle lamella
2. Primary wall,

3. Secondary wall

1. MIDDLE LAMELLA

The first formed cell-plate work as cementing layer between two cells and is called the middle lamella. It is a common layer between two cells. These two cells will separate when the middle lamella will be dissolved. It is mainly made up of calcium and magnesium pectates. This layer is dissolved during the cytokinesis of cell-division.

2. PRIMARY CELL WALL

The primary layer of the cell wall is the first product of the cell, the middle lamella. The primary layer is synthesized by protoplast deposits on either side of the middle lamella. In young dividing and enlarging cells this layer is thin and elastic. It becomes thick and rigid when the cell reaches maturity. It contains hemicellulose (a polysaccharide that has beta (1-4) glycosidic linkages) and cellulose arranged in a crisscross manner and pectin. The crisscross arrangement of cellulose increases the strength of the cell wall. At some places in the cell wall, the deposition of wall material does not take place. These places are known as plasmodesmata (singular, plasmodesma) through which the cellular content of neighboring cells remain in communication with each other.

3. SECONDARY CELL WALL

The layer of wall developed in between the plasma membrane and the primary cell wall. It does not deposit in every plant cell, only deposit in some hard tissues i.e. sclerenchyma. The cells become dead at their maturity. Secondary wall deposits after complete maturation of primary cell wall. It is very thick and rigid due to the deposition of lignin, inorganic salts and waxes. It plays important role in support of plants i.e. sclerenchyma, scleroid, xylem vessels, and tracheids, which contain secondary cell-wall. 5. Define Ascent of Sap. Describe the cohesion and tension theory of water movement through xylem.

ANSWER:

ASCENT OF SAP

The upward movement of water from the absorptive surface (roots) to the transpiring surface (leaves) against the downward pull of gravity is known as the ascent of sap.

COHESION TENSION THEORY OF WATER MOVEMENT

The upward movement of water from roots up to leaves takes place through the xylem depending upon two factors. These are transpiration pull and physical properties of water i.e., adhesion and cohesion. Water molecules tend to adhere to cellulose molecules on the walls of xylem vessels. Extensive hydrogen bonding in water gives rise to the property of cohesion. It also gives water a high tensile strength defined as the ability to resist the pulling force. The water molecules in xylem vessels form a continuous column.

Transpiration pulls the results chain of events that starts when leaves begin to absorb solar radiation in the morning. Sunlight raises the temperature of leaves, so the water begins to evaporate from the moist walls of mesophyll cells. The evaporated water is immediately replaced by water inside the cell. As water is replaced with water from neighbouring cells deeper in the leaf, eventually, water is pulled from the xylem to meet the loss of water. Thus, water in the xylem is placed under tension which is transmitted to the root through vessels. This downward transmission of tension is because of the cohesive property of water columns in vessels and tracheids. The water column moves upward by mass flow due to transpiration pull. Adhesion of water molecules to hydrophilic walls of Xylem cells. The small diameter of vessels and tracheids are important factors in overcoming the force of gravity.

What are Bacteria? Describe the structure of bacteria with the help of a labelled diagram.

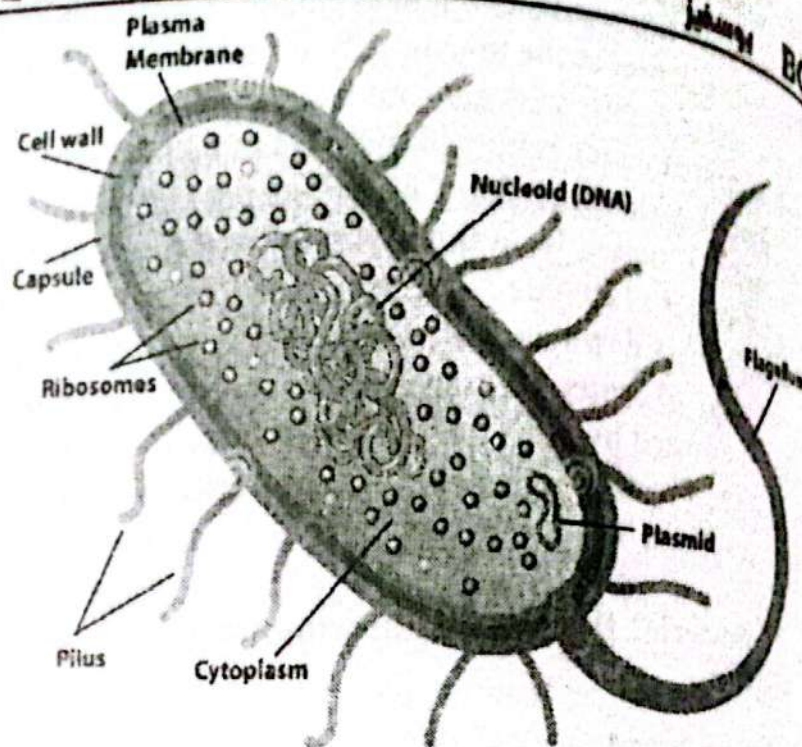
ANSWER:

BACTERIA

Bacteria are the prokaryotic unicellular organisms. Structurally bacterial cell has three distinct regions first appendages that include flagella and pili which help bacteria cells in locomotion and floating respectively, secondly coverings consisting of a capsule the outermost covering in some cells, cell wall and cell membrane and third cytoplasmic region contains the chromosome (DNA), plasmid as the extra circular molecule of DNA used in conjugation and also protect bacteria from antibiotics. Bacteria also contain Ribosomes for protein synthesis, Mesosomes the membranous invaginations which help in DNA replication, cell division respiration and export of enzymes and various inclusions.

STRUCTURE OF BACTERIA

Structurally bacterial cell has three distinct regions first appendages that include flagella and pili which helps bacteria cell in locomotion and floating respectively, secondly, cell coverings consisting of capsule the outermost covering in some cells, the cell wall and cell membrane and third cytoplasmic region that contains the cell chromosome (DNA), plasmid as extra circular molecule of DNA used in conjugation and also protect bacteria from antibiotics. Bacteria also contain Ribosomes for protein synthesis, Mesosomes the membranous invaginations which help in DNA replication, cell division respiration and the export of enzymes and various inclusions. Inclusions are considered to be nonliving components of the cell which do not perform metabolic activities and are not bounded by membranes. The most common inclusions are glycogen, lipid droplets, crystals, and spores.



BOTANY 2023

TIME: 2 Hours

(45 Marks)

SECTION 'A' (Multiple Choice Questions)(09)

1 Choose the correct answer for each from the given options:

1. Cell organelle concerned with detoxification of alcohol:

- * Cell wall
- * Peroxisome✓
- * Mitochondria
- * Ribosome

2. The capacity of a living system to lose water is:

- * Osmosis
- * Water potential✓
- * Neopotential
- * Osmotic pressure

3. The genetic recombination mode of reproduction found in bacteria:

- * Binary fission
- * conidia formation
- * conjugation ✓
- * budding

4. The suicidal activity of lysosome is called:

- * Somolysis
- * Autolysis✓
- * Apolysis
- * Cryolysis

5. RUBISCO is an enzyme which function as carboxylase as well as:

- * oxido-reductase
- * oxygenase✓
- * hydrolase
- * ligase

6. The amoeboid stage of slime mold is called:

- * merozoites
- * slimoid
- * plasmodium✓
- * fruiting bodies

Phaeophytes without roots and leaves are classified in:

- * sphenopsida
- * pteropsida

Resolution of image is known as:

- * Contrast
- * Sedimentation

Infectious structures smaller than virus having single stranded RNA

- * Priors
- * Double stranded DNA virus

In banana plant, flowers covered by large bracts are, called:

- * Capitulum
- * Palea

Induction of flowering in response to the relative length of day and

- * Photoperiodism ✓
- * Photorespiration

The making of DNA from RNA is called:

- * Translation
- * Reverse transcription ✓

Double fertilization is the characteristic feature of these plants:

- * Gymnosperm
- * Feat

Plants growing in salt marshes close to sea are termed as:

- * Xerophytes
- * Halophytes ✓

During this stage bacteria grow and multiply rapidly:

- * LOG phase ✓
- * Decline phase

Fertilization in spermatophytes does not need water because of:

- * Pollen tube ✓
- * Style
- * Stigma

Carnivorous plants grow in habitats with low content of this element:

- * Nitrogen ✓
- * Potassium

Fungi in which sexual reproduction is absent:

- * Basidiomycota
- * Ascomycota

SECTION "B" (Short Answer Question) Marks: 18

Answer any NINE part questions. All questions carry equal marks.

According to Webbed Timmy, how many-veined leaves evolved?

ANSWER:

EVALUATION OF MANY-VEINED LEAVES

According to this theory, many veined leaves (Megaphyllous) evolved much later. These are the evolutionary modifications of the branching system in the primitive plant. The first step in the evolution of leaf was the restriction of forked branches to a single plane. The branching system became flat. The next step in the development was filling the space between the branching and the vascular tissues. The leaf so formed was like the web foot of a duck.

ii) Write the role of any one macro nutrient in plant growth.

ANSWER:

ROLE OF MANGANESE (Mn):

Manganese are absorbed by plants in very minute quantity and become part of enzyme phosphotase (transfer PO₄ groups). Manganese involved in oxidation reduction reaction.

iii) How plants cope with High temperature?

ANSWER: High temperature is more harmful than low temperature. Enzymes are denatured which is disastrous for metabolism. The primary way to cool down the plant is transpiration. At 40°C or above most of the plants cells synthesize heat shock proteins that protect enzymes. Plants use other ways of avoiding overheating such as shiny cuticle, a small leaf surface area, wilting, etc.

iv) Name various divisions of fungi with one example each.

ANSWER:

VARIOUS DIVISIONS OF FUNGI

There are four major divisions of fungi on the basis of sexual reproduction or sexual spores.

DIVISIONS OF FUNGI	EXAMPLE
1. Division Zygomycota	Poread mold mucor
2. Division Basidiomycota	Mushroom
3. Division Ascomycota	Yeast
4. Division Deuteromycota	Penicillium notatum

v) Enlist any four adaptations in xerophytes to cope with water shortage.

ANSWER:

ADAPTIONS TO COPE WITH WATER SHORTAGE

1. Some of the xerophytes complete their life cycle within a very short period when sufficient moisture is available.
2. Xerophytes develop very deep vertical roots for the better absorption of water from the soil.
3. Some plants store water in larger parenchymatous cells present in succulent leaves.
4. Some plants reduce the number of stomata.

ANSWER: Because it's a molecular compound that allows the catabolic reactions (ATP with 3 phosphates to ADP with 2 phosphates and AMP with 1 phosphate). Each reaction transfers energy to a specific function and then is recycled by the anabolic process of having Pi (inorganic Phosphate) re-attached to create more ATP. Therefore, it's a very efficient molecule that can store energy and recycled to be used again.

x) How annual rings are formed?

ANSWER: The plants of temperate regions accumulate secondary growth in the form of concentric layers every year which form an annual ring. Each annual ring consists of two zones, the inner zone of spring wood with larger vessels and an outer zone of summer wood or autumn-wood with smaller vessels.

xi) Name various groups of plants on the basis of osmotic adjustment.

ANSWER: Depending upon the availability of water to flowering plants in their natural habitat they are grouped into four categories;

- | | |
|----------------|---------------|
| 1. Hydrophytes | 2. Halophytes |
| 3. Mesophytes | 4. Xerophytes |

xii) Mention the role of Apical meristem.

ANSWER:

ROLE OF APICAL MERISTEM:

1. The activity of apical meristem results in the primary structure of the plant, i.e., the primary growth.
2. They are responsible for the increase in the length of the plant's stem and both the stem and root branches.
3. They are also responsible for the production of lateral appendages such as leaves and floral parts.
4. In certain cases, increase in length also takes place by the activity of intercalary meristem. These are part of the apical meristem which becomes separated from the apex by permanent tissues and left behind. As the apical meristem moves on during growth.
5. The lateral meristems occur at the cambium and phellogen of the stem. Activity of these tissues results in the increase in diameter of stem and the secondary tissues of plants are formed.

xiii) Explain the movement mechanism of Cilia and Flagella.

ANSWER:

MOVEMENT MECHANISM OF CILIA AND FLAGELLA

Flagellar movement occurs as planar waves, it is a contracting wave that passes either from the base to the tip of flagellum or in the reverse direction to produce forward or backward movement. The tiny protein arms project out from each pair of microtubules in the outer ring. These arms attach to the neighboring pair of microtubules and flex, thereby moving the first pair along relative to the second. However the basal body firmly anchors the flagellum.

all microtubules in the entire cilium and flagellum. Therefore, microtubule can slide past one another only if the whole cilium or flagellum is bent.

Significance of differential staining.

IMPORTANCE OF DIFFERENTIAL STAINING

Differential staining is used to distinguish between different dyes at the same time.

It involves two groups of bacteria.

It is used to give two or more different colours to bacterial cells.

Section C (Detailed - Answer Question) (18)

Attempt any THREE question from this section.

All questions carry equal marks.

Q.1. Describe the life cycle of Moss or Fern.

LIFE CYCLE OF MOSS (BRYOPHYTE)

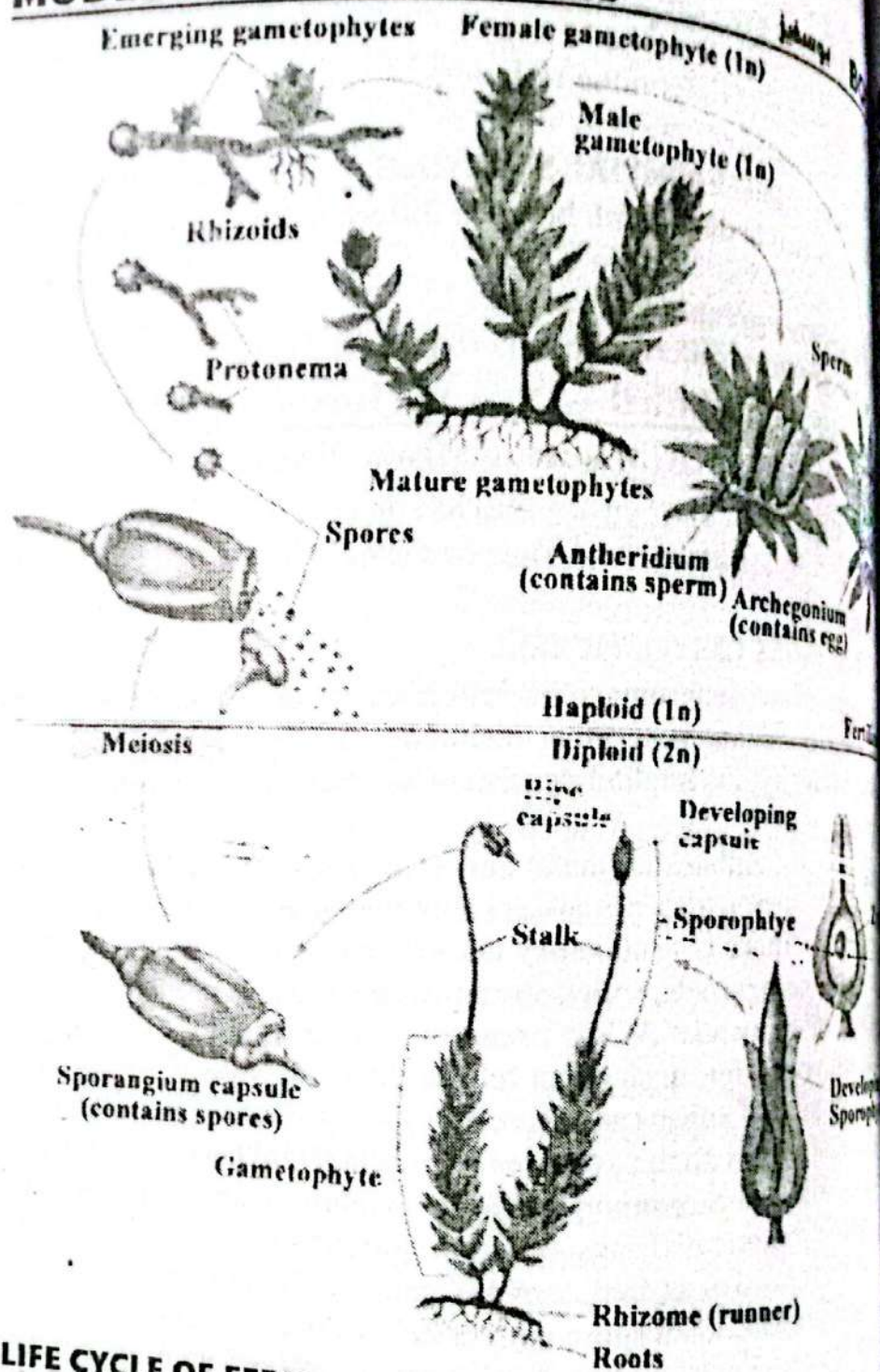
Mosses show heteromorphic alternation of generation with the gametophyte as dominant generation including *Funaria hygrometrica* (a moss).

The gametophyte is haploid consisting of rhizoids, pseudostem and leaves. It is called antheridia (male) and archegonia (female) develop at the base of the pseudo stem which are always dioecious having either male or female organs.

There is protandry because antheridia mature earlier than archegonia. Their antherozoids, which start swimming with the help of their flagella in water or rainwater. When archegonia mature, they have single egg cell and few neck canal cells in the neck. Swimming sperms are attracted by scent of sugar cane secreted by mature archegonium but a sperm fuses with the ovum to form diploid (2n) oospore (zygote).

The zygote develops within archegonium and forms an embryo (2n). This embryo undergoes mitotic divisions to form sporogonium (a sporophyte) which remains attached to the gametophyte. It consists of foot, seta and capsule. Within capsule spores are present. Each spore mother cell divides by meiosis to form haploid spores.

Each spore when dropped at proper place later on germinates into a filamentous body called protonema, which is haploid. It develops from protonema to complete life cycle.



LIFE CYCLE OF FERN:

ANSWER: (See Question No. 3, 2024)

4. Describe oxidative degradation of Pyruvic Acid.

ANSWER:

OXIDATIVE DEGRADATION OF PYRUVIC ACID

The anaerobic respiration as well as aerobic respiration has a common pathway where glucose breaks anaerobically into two molecule of pyruvate (acid) called Glycolysis. It takes place in a series of steps, each catalyze by specific enzyme. All enzymes are found in cytosol with these enzymes, ATP and

Adenine Dinucleotide) are also required. Glycolysis can be

into two phases:

PREPARATORY PHASE:

In this phase Glucose become phosphorylated into glucose-6 fructose phosphate which then isomerized into Fructose-6-phosphate. This 6-phosphate further phosphorylated into biphosphate. The phosphate comes from ATP molecule, 2 molecules of ATP consumed during this phase. Finally, Fructose 1,6-biphosphate breaks into two molecules of phosphorylated triose i.e. Dihydroxyacetone phosphate and 3-phosphoglyceraldehyde (3-Phosphoglycerose).

OXIDATIVE PHASE:

In this phase the molecules of triose phosphate are oxidised as well as electrons and two proton (H^+) are removed from phosphorylated triose and transfer to NAD to reduce it in $NADH_2$. Therefore, it is called oxidative reaction. During this phase 4 ATP molecules are also produced by substrate level phosphorylation at two different steps of glycolysis. Finally, 2 molecules of pyruvate, 2 molecules of $NADH_2$ and 4ATP are formed, as the net gain of 2 ATP takes place.

Pyruvate produces at glycolysis have three metabolic pathways depending to availability of enzyme in organisms. It may be anaerobic or aerobic.

Q Explain the light dependent reaction with reference to Electron transport chain and formation of ATP.

ANSWER:

LIGHT DEPENDENT REACTION

The first phase of photosynthesis where energy of photon is captured and converted into chemical energy. The energy of photon is stored in special molecules i.e. ATP and $NADPH_2$. The energy of ATP and $NADPH_2$ will be used to produce carbohydrate during light independent phase. In chloroplast the light capturing chlorophyll molecules, membrane proteins and electron carriers all together constitute the electron transport chain. Four major groups of complexes are present in the chain. These are photosystem-I (PSI), Photosystem-II (PS-II), the cytochrome b/f and an ATPase complex. Some mobile electron carriers are also present which carry excited electrons between complexes. These mobile carriers are plastoquinone (PQ), plastocyanin and ferredoxin (Fd). Photosystem I and II both contain special chlorophyll-a molecules at their reaction centres. These chlorophyll molecules are identical to all other chlorophyll-a molecules. The change in absorbing spectra are due to their association with different protein bound proteins. The chlorophyll - a molecule at the reaction centre of PS - I has maximum adsorption at 700 nm while that of PS - II

absorb at 680 nm. These reaction center are called P_{700} and P_{680} simply stands for pigment.

ELECTRON TRANSPORT CHAINS:

The light reactions of photosynthesis start from reaction center of P_{680} which consist of chlorophyll-a dimer. When a photon of light is absorbed by these chlorophyll a molecule, the energy of these photons is absorbed and results in the excitation of an electron from ground state to excited state. The excited electron produced within P_{680} is rapidly transferred to primary acceptor phaeophytin and then to plastoquinone (PQ) molecules.

The P_{680}^+ produced by this primary charge separation and e^- transfer is compensated (re-reduced) by e^- from H_2O . The water splitting complex is present on the luminal side of thylakoid membrane and consists of a manganese cluster, Z complex (the immediate electron donor to P_{680}) and an associated protein. The water splitting complex produces $4e^-$ from water molecules and released $4H^+$ and one molecule of O_2 into the lumen. The excited electrons are transferred from primary e^- acceptor plastoquinone (PQ), the PQ molecules which accept two electrons and pick up two protons from the stroma. It carries electrons from the PS-II complex to the Cytb/f complex. This is thought to be the rate limiting step of electron transport. The PQ releases protons into lumen. Finally the e^- transfer to plastocyanin (PC), PC is reduced which is situated in the lumen.

Plastocyanin acts as an electron donor to PS-I, the primary electron acceptor of photosystem-1, passes the photo excited electrons to a secondary electron transport chain, which transmits them to ferredoxin (Fd) a non-haem protein. An enzyme called NADP reductase then transfers the electrons from Fd to NADP, this NADP with $2e^-$ received proton from stroma and reduced to $NADPH_2$. It stores high energy which will provide reducing power for the synthesis of sugar during light independent reaction.

FORMATION OF ATP :

The energy released during movement of excited e^- down the cytochrome system is coupled to build up ATP in an indirect manner. Some of the carrier of the cytochrome system pump hydrogen ion (H^+) from stroma into thylakoid space (lumen). This thylakoid space acts as a reservoir for hydrogen ions because H^+ ion produced by splitting water during the process of photolysis accumulate here.

The high quantity of H^+ ion in thylakoid space as compared to stroma develop electrochemical gradient. These H^+ ions flow out of the thylakoid space to stroma through a channel protein present in membrane called ATPase synthase complex. Hydrogen ions move through this channel providing energy which is released during down movement of electrons from PS-II. This energy is utilized for the synthesis of ATP from ADP and P_i . The movement of H^+ through ATPase complex due to concentration gradient.

chemiosmosis or chemiosmotic ATP synthesis because chemical and event join to permit ATP synthesis. The transport of three proton through ATPase complex are normally required for the production of P molecule. The linear flow of electrons from water to $NADP^+$ to ATP synthesis is called non-cyclic photophosphorylation because electrons pass from water to a terminal acceptor and never back to its source.

Two important events takes place during light dependent reaction of photosynthesis.
 1. Photolysis of water
 2. Electron transport chain i.e. PS II and PS I
 3. Reduction of NADP into ($NADPH_2$)
 4. Photophosphorylation
 5. Growth regulators

are growth regulators? Give their names and explain the role of

ER:

PLANT REGULATORS

Certain chemicals produced by plants have a profound effect on their growth and development. Such chemicals are called plant hormones or phytohormones. They are synthesized by plants in minute quantities and exert their effect either by altering gene expression, activating or inhibiting enzymes or changing properties of the membrane. They are produced in young embryonic tissues as there is no specific organ for production in plants. There are five kind of plant hormones. These

ii. Gibberellins

iii. Cytokinins

v. Ethene

ic acid

AUXINS:

Division and cell-enlargement:

Auxin promotes cell division, cell enlargement and brings about increase in plant height. It stimulates wall loosening factor, for example, elastin to break down cell-wall. If Gibberellins are also present, the effect is stronger. It promotes cell-division if cytokinins are present. Xylem tissues can be broken down when the auxin concentration is equal to the cytokinins.

FUNCTION OF ROOTS:

Auxin initiates development of adventitious roots when applied at the base of stem.

ABSCISION:

In leaves and fruits when auxin production diminishes, a layer of cells is formed at the base of petiole and stalk of fruit. This layer is called abscission layer and causes fall of leaves and fruit with slight jerk

4. GROWTH OF FRUIT:

Auxin is responsible for cell division, the tissues of ovary initiate to auxin and the ovary enlarge to develop this process is called fruit.

5. PARTHENO-CARPY:

Use of auxin also helps in producing parthenocarpic or seedless fruit.

6. APICAL DOMINANCE:

Besides growth promoting function, auxin also has inhibitory effect on growth. Growth of apical bud inhibits growth of lateral buds because of apical dominance. This phenomenon is termed as apical dominance. Removal of apical bud initiates growth of lateral buds with more leaves and axillary buds.

7. WEEDICIDE:

Auxins are selective weed killer. 2-4-dichlorophenoxy acetic acid is used to kill weeds in lawns and cereal crops.

8. FLOWERING:

Auxin plays a minor role in the initiation of flowering. It can delay senescence of flowers in low concentrations.

6. How does E. Munch hypothesis explain the translocation of food in plants?

ANSWER:

EXPLANATION TRANSLOCATION OF FOOD IN PLANTS

Phloem translocation is mainly explained by a theory called the pressure flow hypothesis proposed by Ernst Münch in 1930, which states that the flow of solution in the sieve elements is driven by a pressure gradient produced due to difference in osmotic pressure between sources and sinks. A pressure gradient is produced due to Phloem loading and unloading at source and sink, respectively. A high osmotic pressure generated by active Phloem loading in the sieve elements of a source tissue, causes a decrease in the water potential. Whereas the xylem has high water potential so water moves out of xylem and enter into phloem and produces a high turgor pressure. In the sink tissue, present at other terminal of the translocation pathway, phloem unloading occurs, which produces a low osmotic pressure in the sieve elements of the tissues. As the low osmotic pressure, the water potential of the phloem rises above that of the xylem and water tends to move into the Phloem in response to the water potential gradient, which causes a decrease in the turgor pressure of the Phloem sieve elements in the sink. An equilibrium between the two ends (source and sink) would be reached very soon if the entire translocation pathway were a single membrane bound compartment. The sieve plates that are present in the sieve elements overcome the resistance along the pathway and maintain the pressure gradient in sieve elements between the sources and sinks. The photoassimilates produced in sieve elements are physically pushed along the translocation pathway in bulk flow, much like water flowing through a garden hose.

PAPERS A BOTANY
movement in the translocation pathway is therefore driven by the gradient rather than the water potential gradient. The passive, long-distance translocation in the sieve tube ultimately depends on the short-distance transport mechanism involved in the pressure loading and unloading. These active mechanisms are responsible for the pressure gradient in the first place.
the chemical composition of Plasma membrane. Also describe its role in regulating cell's interaction with environment.

COMPOSITION OF CELL MEMBRANE

Cell membranes have the same basic molecular organization. They consist of a double layer (Bilayer) of phospholipids interspersed with

phospholipids molecules in the plasma membrane are arranged in two layers. Their non-polar hydrophobic ends face each other whereas their hydrophilic ends are associated with carbohydrates protein etc. The plasma membrane also contains several types of lipids like cholesterol sterol etc. In some animal cells cholesterol may contain 50% of lipid molecules in the membrane. Most of the plasma membrane consists of approx. 50% protein by weight, while the carbohydrates portion of the membrane consists of 5 to 10% of the membrane mass.

FUNCTION OF PLASMA MEMBRANE IN REGULATING CELLS ACTION WITH ITS ENVIRONMENT

Plasma membrane performs several functions like the platform for channels, enzymes, antigen fluidity etc, but the main and the most important function are protection of cell cytoplasm, to regulate the flow of material in and out of the cell with certain limitation. For transport, there are two main process of transport.

(ii). Active transport.

PASSIVE TRANSPORT:

Transport of molecules by diffusion and osmosis without consumption of energy.

ACTIVE TRANSPORT:

Transport of molecules against concentration gradient by using energy of ATP.

Two other phenomena i.e. endocytosis and exocytosis. Endocytosis is the process of intake of material in bulk by infolding cell membrane. It includes intake of solid material i.e. phagocytosis or intake of fluid or liquid material i.e. pinocytosis, whereas exocytosis is the process of membrane fusion and release of material from the cell. Cholesterol helps to regulate the fluidity over the range of temperature. It also prevents the leakage of proton and sodium ions the plasma membrane.

NEW MODEL QUESTION PAPER
Annual Examination 2023

BOTANY (THEORY)

CLASS XI (SCIENCE PRE-MEDICAL GROUP)

Total Time: 2 Hours

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Q.)

Time: 15 Minutes

NOTE: This Section consists of 18 MCQs and all are to be answered. (9 Marks)
Each MCQ carry 0.5 marks.

1. Choose the correct answer from the given options

1. Organelles other than the nucleus that contain DNA

- I. Ribosomes II. Chloroplast III. Mitochondria
a. I only b. II only c. II and III ✓ d. I and II

2. Select the correct statement

a. PS I and ATP synthase complexes are located in the appressed thylakoid.

b. PS I and NADP reductase are located in the appressed thylakoid membrane.

c. Appressed part contain NADP reductase and ATP synthase.

d. Non appressed (non-stack) having PS I. ✓

3. The Oxygen consumed during cellular respiration is involve in which process or events

a. Glycolysis

b. Accepting electron at the end of electron transport chain ✓

c. The citric acid cycle

d. The oxydation of pyruvate to acetyle CoA

4. Bacteriophages escape from host cell by the activity of

a. Lysozyme ✓

b. Ribozyme

c. Peroxidase

d. Reductase

5. When habitat conditions become harsh and nutrients are exhausted the development in bacteria is initiated called

a. Capsule

b. Cell wall

c. Endospore ✓

d. Mesosome

6. The most important cellulose degraders in eco system are

a. Ascomycota ✓

b. Zygomycota

c. Basidiomycota

d. Deutromucota

Division of Tracheophyta which does not contain true roots and

b. Psilopsida ✓

d. Sphenopsida

which process involved in the promotion of flowering by cold

b. Vernalization ✓

d. Transpiration

periodism

secondary growth

occurs through

tracheids b. Hydathodes ✓

clarity of image is generally known as

b. Contrast

d. Sedimentation

acidification

oxidative decarboxylation of isocitrate form

b. Succinate

d. Fumarate

isotoglutarate ✓

Aconitate

Plant oxidizes sugar in chloroplast during day time without

b. Photorespiration ✓

d. Photophosphorylation

cycle

cycle

some structure are smaller than virus having single stranded RNA

b. Prions

d. Double stranded DNS virus

triple ✓

single strand virus

aerobic bacteria produce all chemicals during respiration except

b. CO₂

c. Water ✓

d. Lactic acid

ethanol

typical structure of obligate parasite, specialized for fixation and

proton

gella

b. Pili

c. Haustoria ✓

d. Root hairs

a banana tree, flowers are covered over by one or many large

bracts called

b. Spadix

c. Capitulum

d. Palea

the hydrostatic pressure in excess of atmospheric pressure is known

water potential

osmotic potential

induction of flowering in response to the relative length of day and

photoperiodism ✓

photoperiodism

photoperiodism

b. Pressure potential ✓

d. Solute potential

b. Photophosphorylation

d. Phototropism

NOTE: Answer any Eleven (11) questions from this section. All questions carry equal marks.

2. (i) Why insectivorous plants use insects as food?

ANSWER: Insectivorous or carnivorous plants typically grow in nutrient deficient habitats. They use insects as food to fulfil their nutrient requirement.

2. (ii) Why Chloroplast is said to be an energy converting organelle?

ANSWER: Chloroplast converts light energy into chemical or food energy by the process of photosynthesis, during photosynthesis chlorophyll captures the energy of sunlight and transfers it to another molecule in the thylakoid membrane. These molecules in turn transfer the energy to ATP and energy carrier molecules like NADPH₂. Due to this flow of energy from one form to another, the chloroplast is an energy-converting organelle.

2. (iii) Describe the role and deficiency symptoms of Nitrogen and Potassium in plant.

ANSWER:

ROLE OF NITROGEN

Nitrogen is an essential constituent of proteins, nucleotides, nucleic acids and chlorophyll etc.

DEFICIENCY SYMPTOMS OF NITROGEN

Due to Deficiency, Nitrogen Leaves turn pale yellow due to loss of chlorophyll content and cell enlargement is inhibited. Plants' growth remains stunted.

ROLE OF POTASSIUM

Potassium plays the important role in the stomatal opening and closing, acts as an enzyme activator in the synthesis of certain peptide bonds and in carbohydrate metabolism.

DEFICIENCY SYMPTOMS OF POTASSIUM

Due to the Deficiency of Potassium Leaf turns dull or bluish green, chlorophyll content is reduced, growth is stunted, and the shortening of internodes.

2. (iv) Differentiate between Prokaryotes and Eukaryotes.

ANSWER:

DIFFERENCES

PROKARYOTES	EUKARYOTES
Prokaryotic cells are the primitive kind of cell, whose size varies from 0.5-3µm, they are	Eukaryotic cells are the modified cell structure containing different components in it, their size varies

EL PAP	multicellular organisms.
usually found in single	
prokaryotes, the cell division	In eukaryotes, it is through the process of cell division.
place through conjugation,	
prokaryotes Cell wall present,	In eukaryotes Usually cell wall
presence of peptidoglycan or	absent, if present (plant cells and
peptide (polysaccharide).	fungus), comprises of cellulose
	(polysaccharide).

Why Photorespiration is considered as wasteful process?

Q.2 (iv), 2024

Q.2 (iv), 2024

*** Heterospory**

FERTILIZATION

Fertilization is a special type of fertilization which occurs only in sperms plants. During this process, a sperm fuses with the ovum to form a zygote. The other sperm fuses with the secondary nucleus to form endosperm nucleus.

HETEROSPORY

Heterospory is a situation in which a plant bears two different types of spores. The smaller spore is the microspore which forms the male gametophyte. The larger spore is the megaspore which forms the female gametophyte.

What do you mean by bacterial growth? Describe its phases.

BACTERIAL GROWTH

Bacterial growth is defined as an increase in the number of bacteria in a population rather than in the size of individual bacteria. The growth of a bacterial population occurs in a geometric or exponential manner. Exponential describes a very rapid increase.

LAG PHASE

Lag phase describes a kind of slowness or delay. Immediately after inoculation into a fresh nutritive medium, the bacterial population initially remains unchanged. Although there is no apparent cell division occurring, the cells are growing in volume or mass.

EXPONENTIAL (LOG) PHASE

Exponential phase of growth is a pattern of balanced growth where all cells are dividing regularly and rapidly by binary fission. The cells divide at a constant rate depending on the composition of the growth medium and

the conditions of incubation. The growth of bacteria is very high during the phase.

3. STATIONARY PHASE

The log phase of bacterial growth is followed by the stationary phase in which the size of a population of bacteria remains constant, because the bacterial population has simply stopped. In this phase rate of reproduction is equal to the rate of death.

4. DEATH PHASE

If incubation continues after the population reaches the stationary phase, a death phase follows, in which the viable cell population declines. During this phase the rate of death is higher than the rate of reproduction. This phase ends with the complete exhaustion of nutrients.

2. (viii) Why Protoctista considered as polyphyletic kingdom?

ANSWER: Because they have many characteristics of their ancestors. Different groups of protists do not share a common ancestor with each other.

2. (ix) Give botanical name of any four of the following :

*Wheat *Mako *Barley *rice *Amaltas *Mulhethi

ANSWER:

COMMON NAME	BOTANICAL NAME
Wheat	Triticum
Mako	Solanum nigrum
Barley	Hordium
Amaltas	Cassia fistula
Mulhethi	Glycyrrhiza glabra

2. (x) Describe the classification of bacteria on the basis of their shapes

ANSWER:

CLASSIFICATION OF BACTERIA

Bacteria are classified into the following five groups according to their shapes

1. COCCI:

These are spherical and non-flagellated. Cocci exist in different arrangements as a single cell, in pairs of two called diplococci, in groups of four called tetrads, in chains as streptococci, in clusters as staphylococci, in cubes consisting of eight cells as sarcina.

2. BACILLI

Bacilli are rod-shaped maybe flagellated bacteria they exist in a single form, paired form streptobacillus, as diplobacillus; arranged in chains called streptobacillus. A few other are called Palisades long and slender and their arrangement resembles the picket fence.

HEL PAPERS XI **STANY**

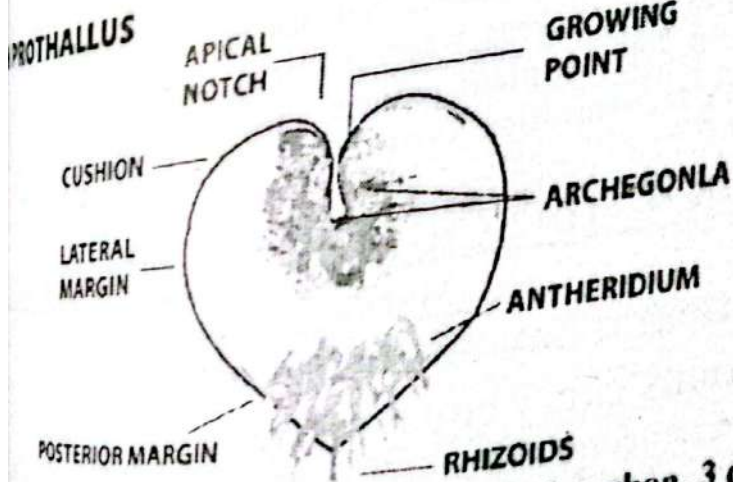
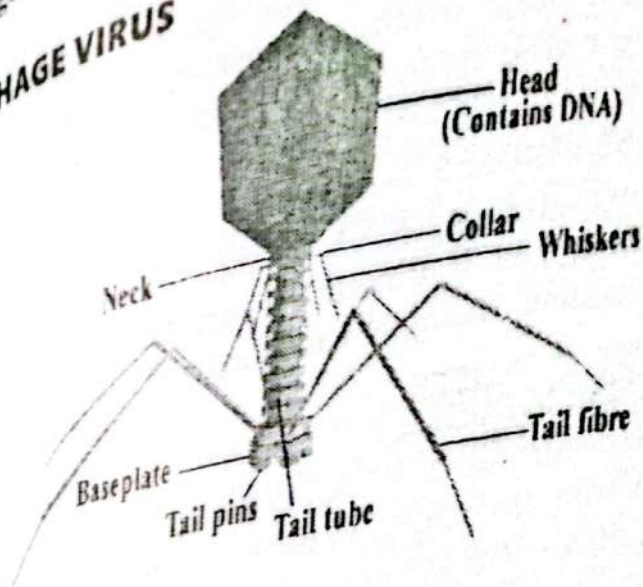
SPIRILLA
 (one spirillum for a single cell) are flagellated and deeply curved from one end of the body.

SPHINCTER
 are slightly curved bacteria like comma shape with flagella.

ARCHETES
 are long, slender, and flexible cork screw-like bacteria.

Draw a well labelled diagram of the followings:
 * Fern prothallus

PHAGE VIRUS



How many ATP and NADPH requires to fix 1 carbon, 3 carbon, 6 carbon and 12 carbon.

Carbon	Required ATP	Required NADPH
1 carbon	3 ATP	2 NADPH ₂
3 carbon	6 ATP	6 NADPH ₂
6 carbon	12 ATP	12 NADPH ₂

MODEL PAPERS XI (S) 430

12 Carbon

24 ATP

36 NADPH

2. (xiii) What features allow fungi to survive in all environments possible?

ANSWER: Because fungi have rigid stronger and thicker cell walls containing chitin as the fibrillar material. It provides extra support and protection from predators.

2. (xiv) Why osmotic adjustment is beneficial for plants?

ANSWER: Osmotic adjustment is beneficial for plants' useful tolerance in response to drought conditions or saline stress, many plants accumulate organic solutes to lower the osmotic potential.

2. (xv) What do you mean by positive and negative sense virus?

ANSWER:

POSITIVE SENSE VIRUSES

They have a single-stranded RNA. Positive sense means that their function as mRNA and is directly translated by the host cell without involving transcription. Example Coronavirus, Dengue virus, and Hepatitis C virus.

NEGATIVE SENSE VIRUSES

These have single-stranded RNA. When they enter the host cell, they prepare mRNA from their RNA in the host cell for any translation. Rabies virus, and Paramyxovirus are examples of this type.

2. (xvi) Why desert plants reduce their leaf size?

ANSWER: In desert areas, water is scarce. So, in order to adapt to water scarce areas, the desert plant reduced their leaves to spines. Spines reduce water loss by transpiration significantly by reducing the surface area of leaves.

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: 7x2=14)

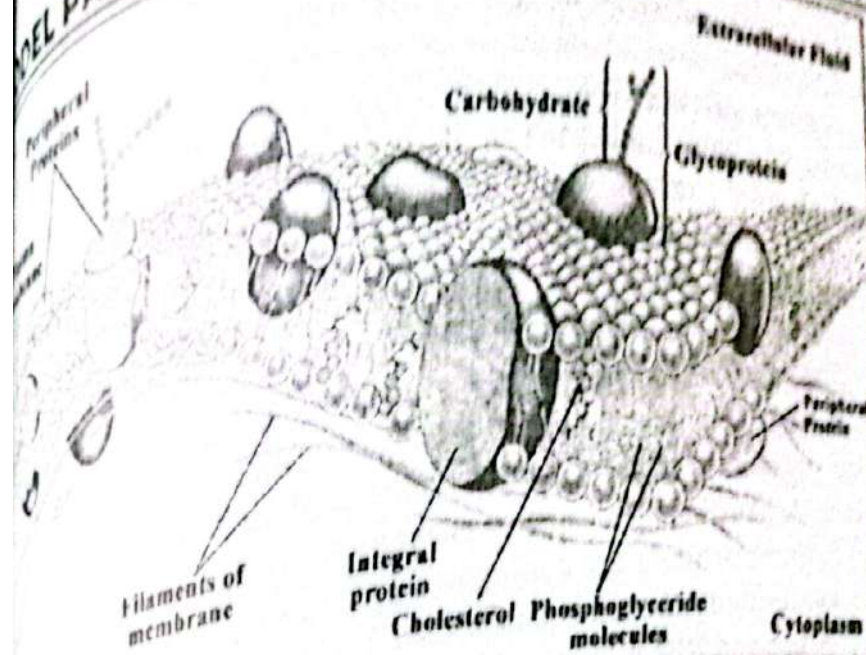
Note: Attempt any three (2) questions from this section. All questions carry equal marks:

Q 3. Explain structure and properties of Plasma membrane with diagram.

ANSWER:

(i) STRUCTURE OF PLASMA MEMBRANE

Number of biologists presented different models of cell-membrane. One of the models is sandwich model. According to this model the cell membrane is composed of lipids bilayer sandwiched between inner and outer layers of protein. This basic structure is called the unit membrane and is present in all the cellular organelles. The modern technology has revealed that lipid bilayered is not sandwiched between two protein layers.



FLUID MOSAIC MODEL

(1972) Singer and Nicolson proposed a working model of plasma membrane known as fluid mosaic model. In the fluid mosaic model the lipid bilayer is retained as the core of the membrane. These lipid molecules are present in a fluid state able to rotate, translate and vibrate these molecules moving laterally within their layers of membrane. Proteins are also present in a layer of phospholipids.

The structure and arrangement of membrane protein in the fluid mosaic model are like ice bergs in the sea. The protein occurs as a "mosaic" of continuous particles that penetrate deeply into and even completely through the lipid sheet. The components of plasma membrane are mobile and capable of coming together to engage in various type of transient or permanent interaction.

Proteins associated with the lipid bilayer can be divided into groups.

b) Peripheral proteins

Integral proteins

INTEGRAL PROTEINS (INTRINSIC PROTEINS):

These are proteins that are directly incorporated within the lipid bilayer. Some of these proteins are believed to provide a channel through which the soluble substances, such as ions, can pass back and forth between the extracellular and intracellular compartment.

PERIPHERAL PROTEINS (EXTRINSIC PROTEINS):

These are proteins located entirely outside the lipid bilayer on either the extracellular or cytoplasmic surface, exhibit a loose association with the membrane surface.

Some proteins which may possess lipid (lipoprotein) or carbohydrates (glycoprotein) side chains are arranged as mosaic within the cell-membrane.

FUNCTION OF PLASMA MEMBRANE

The plasma membrane performs several functions like platform receptor, channels, enzymes, antigen fluidity etc, but the main and the most important function are protection of cell cytoplasm, to regulate the flow of solutions and material in and out of the cell with certain limitation. The limitations are checked by in flow of materials across the membrane. It is necessary to maintain suitable pH, ionic concentration for enzyme action and excrete toxic substances etc. The lipid bilayer controls the fluidity at variable temperature conditions by increasing or decreasing unsaturated fatty acids in them. The fluidity also increases the flexibility of the membrane. It also controls the movement of polar molecules and ions. The differentially or selective permeability is due to presence of specific channel proteins which permit only specific molecule to pass through them. The protein is carrier protein embedded in phospholipid layer. Some extrinsic proteins also work as enzymes e.g. ATPase complex to synthesize ATP. Some proteins are conjugated proteins work as receptor for different hormones and other molecules. While other proteins work as antigen like RH protein of RBCs.

Q 3. (OR) Light independent reaction (C₃ cycle) of photosynthesis in detail.

ANSWER:

LIGHT INDEPENDENT REACTION OR CALVIN BENSON CYCLE

The second phase of photosynthesis, where carbohydrate molecules are formed by fixing atmospheric carbon dioxide. This part of photosynthesis does not require light energy directly, it requires chemical energy of ATP and NADPH₂ and therefore it is light independent reaction previously called dark reaction. The details of this phase were discovered by Melvin Calvin and his colleague Benson therefore it is also called Calvin Cycle. During this cycle CO₂ is reduced to triose phosphate i.e. phosphoglycerate or dihydroxyacetone phosphate and subsequently via other metabolic pathways to hexoses, sucrose and starch. The first stable product formed during this cycle is a three carbon containing acid i.e. 3-phosphoglyceric acid (3-phosphoglycerate abbreviated by 3-PGA) therefore, this cycle is also called C₃ cycle and the plants which carry this cycle only called C₃ plants. The Calvin cycle starts with a phosphorylated five carbon sugar i.e. Ribulose 1,5-bisphosphate and ends on this sugar, therefore it is called a cyclic process. Calvin cycle is divided into three distinct phases for the convenience

i) CARBOXYLATION:

It is the fixation of atmospheric carbon dioxide with five carbon sugar.

ii) REDUCTION:

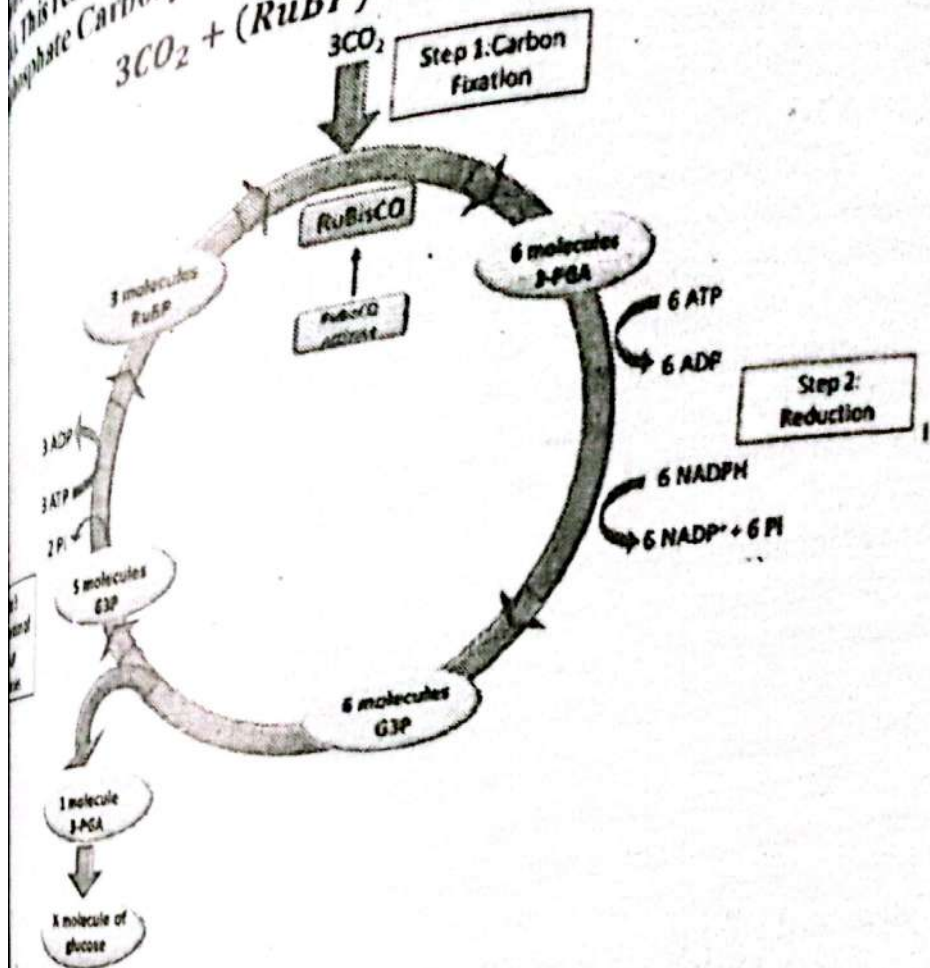
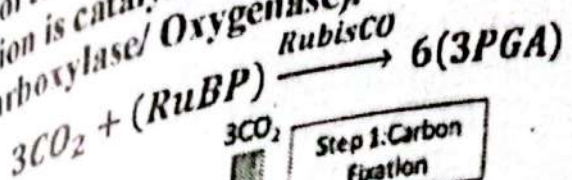
Reduction of three carbon containing acid takes place to form triose.

iii) REGENERATION:

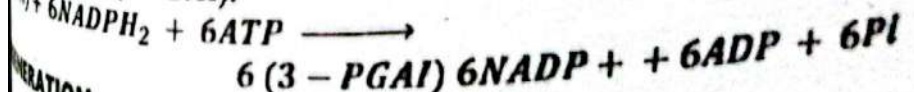
Where reduced carbon utilize to regenerate 5 carbon sugar.

CARBON FIXATION:

The first and key step of Calvin cycle where carbon dioxide is fixed with Ribulose 1,5-biphosphate (RuBP), as a result of this fixation a 6 carbon intermediate compound is formed, which immediately breaks, into two molecules of three carbon containing acid called 3 phosphoglycerate. This reaction is catalyzed by the enzyme called RuBisCO (Ribulose biphosphate Carboxylase/ Oxygenase).

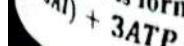


REDUCTION: This phase comprises of a series of reactions. Simply, during this phase 3-phosphoglycerate (3-PGA) reduce to glyceralate 1,3-biphosphate by using ATP from light reaction and then to Triose phosphate by oxidation of NADPH from light reaction, these triose phosphates are phosphate and 3-carbon aldehyde (3-PGAI).



REGENERATION:

Many carbon rearrangements take place during this phase, 5-carbon Triose sugar (3-PGAI) in three interlinked C₃ cycles rearrange to form 3 molecules of 5-carbon sugar (Rub 1-5, P₂) while one molecule of 3-carbon sugar is formed as a gain of these three cycles.



During this cycle three molecules of CO_2 fix with three molecules of RuBP which produce six molecules of Triose sugars. These six molecules of Triose rearrange to three molecules of five carbon sugar (RuBP) and one molecule of Triose as net gain. Therefore, only one molecule of three carbon Triose phosphate is produced which can (a) re enter the cycle to produce hexose or (b) be used for starch synthesis within chloroplast or exported via phosphate translocator to cytosol for sucrose synthesis.

For the net synthesis of Triose molecule, the Calvin cycles consume a total of nine (09) molecules of ATP and six (06) molecules of $NADPH$.

Q 4. What are Growth regulators? Name and discuss five in detail.

ANSWER:

GROWTH REGULATORS

Certain chemicals produced by plants have a profound effect on subsequent growth and development. Such chemicals are called hormones or phytohormones. They are synthesized by plants in minute concentration and exert their effect either by altering gene expression, activating or inhibiting enzymes or changing properties of the membrane. They are produced in young embryonic tissues as there is no specific site for their production in plants. There are five kind of plant hormones. They are:

- | | | |
|--------------------|------------------|-----------------|
| i. Auxins | ii. Gibberellins | iii. Cytokinins |
| iv. Abscissic acid | v. Ethene | |

I. AUXINS :

Auxins is a class of plant growth substances both natural and synthetic first revealed by Fritz-Went (1926).

They were the first of the major plant hormones to be discovered. They are a major coordinating signals in plant development. Indole-acetic acid (IAA) is the principal type of auxin of higher plants, synthesized at the apical stem and root (apical meristem).

In addition to (IAA) other naturally occurring auxins are: 4-chloro-indole acetic acid, phenyl acetic acid (PAA) and indole-3 butyric acid (IBA). where the synthetic auxins include naphthalene acetic acid (NAA), 2,4-dichloro- phenoxy acetic acid (2, 4-D) and others. Auxins co-ordinate development at all levels in plants, from the cellular level to organs and ultimately the whole plant.

II. GIBBERELLINS:

It is a group of chemicals that promote cell division and cell elongation. It was first noticed in *Gibberella fujikuroi* fungus which infected rice seedlings and produced a disease called bakanae (foolish seedling). The infected seedlings were elongated and ultimately fell over without producing grains. Even extracts from the fungus when applied to rice seedlings produced the same disease.

that a definite chemical compound is responsible for disease. and T. Hayashi succeeded in isolating the active substance from and was named Gibberellin, after the name of genus. Its ability to grow attracted the scientists who have isolated 70 different types of Gibberellin, many of them occur naturally in higher plants. Major sites of production are roots, stem and leaves.

CYTOKININS:

A group of substances both natural and synthetic, which react with to induce cell division. Originally obtained from coconut milk, the sources are Herring sperm DNA and yeast extract. One of the occurring cytokinin is zeatin, which was obtained in pure form from immature corn grains. Kinetin, a synthetic cytokinin has same effect as that of zeatin.

ABSCISIC ACID:

It is a growth promoting hormones like auxin, gibberellins and abscisic acid (ABA) is growth inhibitor produced by plant during environmental conditions such as drought condition and at the onset of dormancy in buds and seeds, causes stomata to close, and primordia into scale which protects the buds and promotes dormancy.

ETHYLENE:

An important role of ethane (a gas), is that it triggers ripening of fruit. It increases permeability of cell membrane, which allows enzymes responsible for breaking down of chloroplast with the result that red and yellow colours are released and fruit assumes ripened colour. It contributes to leaf abscission and breaks dormancy of buds and seeds in some species. It also initiates dormancy in some plants e.g. pineapple.

Q. Explain the life cycle of Moss with the help of diagrams.

ANS. (See Question No. 3, 2023)

Q. Describe structure of Bacterium with diagram.

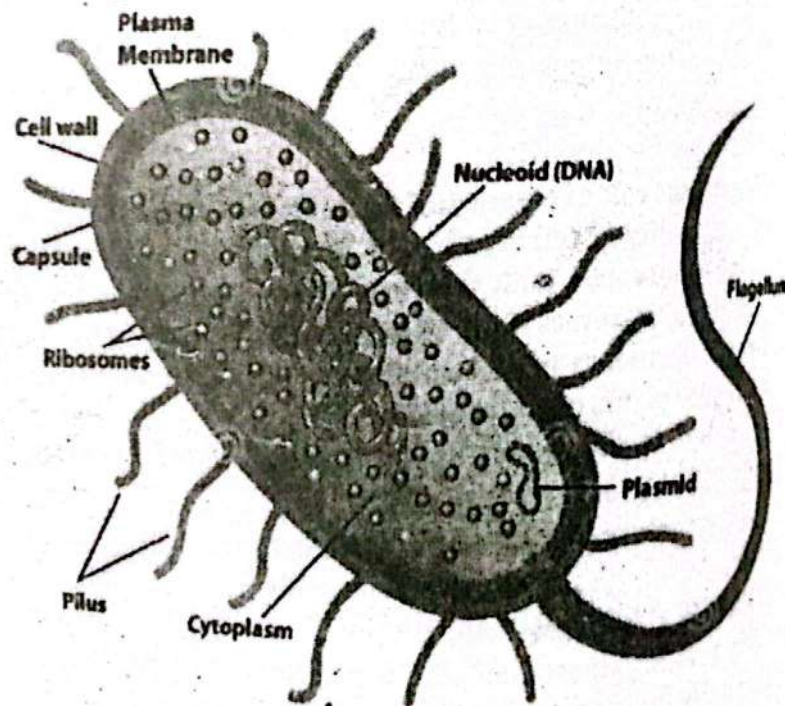
ANS.

STRUCTURE OF BACTERIUM

Bacteria are prokaryotic unicellular organisms. The structurally bacterial cell has three distinct regions first appendages that include flagella and pili which help bacteria in locomotion and floating respectively, secondly, capsule consisting of capsule the outermost covering in some cells, the third cell membrane and third cytoplasmic region that contains the chromosome (DNA), plasmid as the extra circular molecule of DNA used for reproduction and also protect bacteria from antibiotics. Bacteria also have Mesosomes for protein synthesis, Mesosomes the membranous structures which help in DNA replication, cell division respiration and other metabolic enzymes and various inclusions. Inclusions are considered to

MODEL PAPERS XI (S) 436

be nonliving components of the cell that do not perform metabolic functions and are not bounded by membranes. The most common inclusions are glycogen, lipid droplets, crystals, and pigments. Generally, bacterial cells have two protective coverings the outer capsule and the inner cell membrane. Some bacteria have a third protective covering called glycocalyx which separates the bacterial cell from its surroundings. Glycocalyx may be condensed to form capsules which is soft and sticky according to the surrounding environment. The capsule form of glycocalyx is relatively tightly associated with the cell wall, it provides sticky or gummy nature to the cell wall, whereas slime is loosely attached to the cell wall and gives slimy and slippery nature to the bacterial cells, which is formed by polysaccharides, polypeptides and hyaluronic acid in different species. Most capsules are hydrophilic and protect bacteria from desiccation. The capsule protects bacteria from ingestion of WBCs and increases bacterial pathogenicity in host organisms. Another function of glycocalyx is to promote adhesion with each other and form a biofilm. Biofilm bonding becomes more resistant and harder to kill.



YOUSUIFS MODEL QUESTION PAPER 1
Annual Examination 2023

BOTANY (THEORY)

CLASS XI (SCIENCE PRE-MEDICAL GROUP)

Total Time: 2 Hours

Total Marks: 45

PAPERS XI (5)
SECTION 'A' (COMPULSORY)
MULTIPLE CHOICE QUESTIONS (M.C.Qs)
(9 Marks)

This Section consists of 18 MCQs and all are to be answered.
Each MCQ carry 0.5 marks.

Choose the correct answer from the given options:

Primary objective of cell fractionation is to:
Study the structure of cell membranes.
Study the enzymes outside the organelles.
Measure the size of various organelles.
Measure the major organelles so that their particular functions can be determined.

Chlorophyll-a is almost identical to chlorophyll-b but slight difference between them is enough to give:
More energy during the light reaction.
Different absorptive spectrum.
Different product during the Calvin cycle.

Parasitrophic flagellates escape from host cell by the activity of:
(a) Cilia
(b) Ribozyme
(c) Glyoxisomes
(d) Peroxisomes

Bacteria show several types of movements such as flexing, spinning, tumbling and creeping are called:
(a) Spirochetes
(b) Spirochaetes
(c) Cocci
(d) Coccus

Members of green algae have the pigment combination:
(a) Chlorophyll a and b
(b) Chlorophyll a and c
(c) Chlorophyll a and phycobilins
(d) Chlorophyll a b and c

Dominant generation of living vascular plant existing as free plants is:
(a) Sporophyte
(b) Gametophyte
(c) Pteridophyte
(d) Pteridophyte

Where plants primarily obtain essential nutrients:
(a) Air
(b) Soil
(c) Light
(d) Light

Which is one of the main energy transformers of cells?
(a) Endoplasmic reticulum
(b) Golgi complex
(c) Mitochondria
(d) Mitochondria

Ability of a microscope to increase the size of an image is:
(a) Magnification
(b) Contrast
(c) Contrast
(d) Objective

Link between living and non-living

MODEL PAPERS XI (S) 438

- (a) Bacteria (b) Fungi (c) Protozoans (d) Virus
- (xi). Vascular plants are classified under:
(a) Bacteria (b) Algae (c) Bryophytes (d) Tracheophytes
- (xii). It is a Heterosporous plant:
(a) Lycopodium (b) Fern
(c) Selaginella ✓ (d) Marchantia
- (xiii). Number of ATPs produced by the complete oxidation of a molecule is:
(a) 36 (b) 38 ✓ (c) 37 (d) 40
- (xiv). This compound connects glycolysis and Kreb's cycle:
(a) Pyruvic acid ✓ (b) Acetyl coenzyme A
(c) Citric acid (d) Oxaloacetate
- (xv). Photosynthetic Bacteria use this instead of water:
(a) H_2SO_4 (b) HCl (c) H_2S ✓ (d) NH_3
- (xvi). It is a wasteful process in C_3 plants which decreases crop yield:
(a) Photosynthesis (b) Respiration
(c) Photo-Respiration ✓ (d) Transpiration
- (xvii). Excess water is released at night in the form of droplets through:
(a) Stomata (b) Cuticle (c) Hydathodes ✓ (d) Lenticels
- (xviii). During this stage the bacteria grow and multiply rapidly:
(a) LAG phase (c) LOG phase ✓
(c) Stationary phase (d) Decline phase

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 11x2=22)

NOTE: Answer any Eleven (11) questions from this section. All questions carry equal marks.

2. (i) Why N_2 is included in mineral nutrient although it is not mineral?
ANSWER: Because Nitrogen is the main constituent of plants. In plants it is converted into amino acids and then amino acids are used for different metabolic activities.

2. (ii) Differentiate between peroxisome and glyoxysome.

ANSWER:

PEROXISOME	GLYOXYISOME
Peroxisomes are single membrane bound organelles found in most of the eukaryotic cells.	Glyoxysomes are specialized peroxysomes that are present in plants and filamentous fungi.
Breakdown of very long chain fatty acids through beta-oxidation	Catalysis of the formation of acetyl CoA from fatty acids stored within germinating seeds

PAPERS XI (3)	BOTANY
Present in most eukaryotic cells fungi, plants, animals, etc.	Present in plant cells and in filamentous fungi
to convert fats into	Notably found in germinating seeds
	Able to convert stored fats into carbohydrates

Why phytochrome pigments are important to photoperiodism?
Phytochrome pigments are important to photoperiodism
is useful for determining flowering by the balance between the two
Phytochrome pigment, which in some way switches on the genes
for flowering.

Why Chloroplast cell is called energy converting cell organelle?
Chloroplast converts light energy into chemical or food energy
of photosynthesis, during photosynthesis chlorophyll captures
of sunlight and transfers it to another molecule in the thylakoid
These molecules in turn transfer the energy to ATP and other
carrier molecules like NADPH₂. Due to this flow of energy from one
another, the chloroplast is an energy-converting organelle.

Why CAM plant close stomata in day time?
Closing the stomata of CAM plants during the day helps these
to conserve water. During the night, when their stomata are open,
plants take up CO₂ and incorporate it into a variety of organic acids.

Why heterogamy is important for plants?
Heterogamy is important for plants because it produced two
types of gametes i.e. One is male (motile) and the other is female
(non-motile) full of stored food. The male and female parts of the flower
at the same time, so that the individual flower can pollinate itself.
Explain the bacterial diseases in man with their sign, symptoms
and prevention.

WER:

Q2A

AND SYMPTOMS

Nausea and vomiting, sleepiness or lethargy, Dehydration, Muscle
Electrolyte imbalance, thirst, Dry skin, dry mucous membrane

TMENT AND PREVENTION

hydration solution (ORS) and take antibiotics

Q2B

AND SYMPTOMS

fever and skin rash, weakness, abdominal pain, constipation,

TREATMENT AND PREVENTION

Proper wash hands drink clean water, eat properly cooked food visit a doctor if symptoms persist.

TUBERCULOSIS

SIGNS AND SYMPTOMS

Cough that produces phlegm, fatigue, a fever, chills and a loss of appetite, nausea or vomiting and headaches

TREATMENT AND PREVENTION

Staying away from crowdly and congested places wearing a mask and sanitizing yourself.

PNEUMONIA

SIGNS AND SYMPTOMS

Coughing with blood and mucus, fever, sweating and chills, shortness of breath, chest pain, fatigue, loss of appetite, nausea or vomiting and headaches

TREATMENT AND PREVENTION

Properly wash hands, eat clean and properly cooked food, do exercise, quit smoking.

2. (viii) Why asexual and sexual ulva plants are called as sporophyte and gametophyte?

ANSWER: Because thallus in Ulva is of two types, the one with 2n chromosomes is called a sporophyte and the other with 13 chromosomes is called a gametophyte. Morphologically both gametophyte and sporophyte are exactly alike so ulva perform both sexual and asexual reproduction

2. (ix) Why bryophytes are called non vascular plants?

ANSWER: Bryophytes lack true roots, stems, or leaves. They are called non-vascular plants because of the absence of vascular tissues (xylem and phloem). these tissues function for the conduction of food, water, and minerals.

2. (x) Explain the bacterial disease in plant with their sign symptoms treatment and prevention.

ANSWER:

BACTERIA LEAF SPOTS

SIGNS AND SYMPTOMS

Necrosis, sometimes leaf spots will grow together creating large black blotches on leaves completely black. Shoots, buds and flowers can also become black.

TREATMENT AND PREVENTION

Eradication of pathogen by using pathogen – tested seeds and propagated materials and proper spray.

SYMPTOMS
withering, browning, spotting, or drying of leaves, flowers,
parenchyma cells damage.

PREVENTION
control less effective than biological control methods

SOFT ROTS
SYMPTOMS
becomes macerated (soft and watery)

PREVENTION
of plant seeds, control weeds, avoid harvesting under wet

WILT DISEASES IN PLANTS
SYMPTOMS
caused by blocking of the vessels of host plant by masses of
result plant wilt and dies

PREVENTION
bacterial wilt depends on control of insect beetle vectors, Use of
Prions and Viroids.

VIROIDS	PRION
are the tiniest infectious consisting only of a circular, single- replicating RNA protein coating.	Prion is an aberrant or misfolded protein that causes fatal sickness in animals and humans by infecting normal variations of the same protein with its misfolded structure.
made up entirely of single- circular RNA.	Prion is made up only composed of Protein.
are smaller than virus	Prion are smaller than viroids.
infect only higher plants (hepatitis D virus in humans is similar to viroid)	Prions infect animals causing neurological degenerative diseases
Cherry Spindle Tuber (Chrysanthemum are two common)	Examples: Mad cow disease in cows and Scrapie disease in sheep and goat, Creutzfeldt-Jakob Disease (CJD), Kuru.

Cyclic photophosphorylation helpful in photosynthesis?
Cyclic Photophosphorylation is the process in which
accomplish the conversion of ADP to ATP for immediate

energy for the cells. Cyclic photophosphorylation is required to maintain pools of phosphorylated intermediates of the Calvin cycle. Cyclic photophosphorylation is required all the time as it can generate ATP at a low cost.

2. (xiii) Name various classes of Protozoa and how each of them move?

ANSWER:

Members of this group of protozoa are divided into five classes which differ in their means of locomotion.

CLASSES OF PROTOZOA	LOCOMOTION
Class Sarcodina (Rhizopoda)	Have pseudopodia as locomotory structure
Class Flagellata (Mastigophora)	Members of this class have one or more flagella as locomotory structure
Class Ciliata (Ciliophora)	Have cilia as locomotory structure
Class Suctoria	Have cilia at early stage of life but become sedentary at adult stage
Class Sporozoa	No locomotory structure at any stage of life

2. (xiv) Why osmotic adjustment is beneficial for plants?

ANSWER: (See Question No. 2(xiv) New Model Paper)

2. (xv) Differentiate lytic and lysogenic life cycle of bacteriophage.

ANSWER:

LYTIC CYCLE	LYSOGENIC CYCLE
Lytic cycle is a type of a viral reproduction mechanism, which results in the lysis of the infected cell.	Lysogenic cycle is a viral reproduction mechanism where the viral DNA is integrated into the host genome.
In the lytic cycle, viral DNA does not integrate into the host DNA.	In the lysogenic cycle, viral DNA integrates into the host DNA.
Host DNA is hydrolyzed during the lytic cycle.	Host DNA is not hydrolyzed during the lysogenic cycle.
Viral DNA replication occurs independently from the host's DNA replication in the lytic cycle.	Viral DNA replication occurs along with the host's DNA replication in the lysogenic cycle.
The productivity of viral DNA in the lytic cycle is high due to the independent replication of the viral DNA.	The productivity of the viral DNA in lysogenic cycle is lower than that of the lytic cycle since the viral replication occurs along with the host's DNA replication.

does not allow genetic recombination in the host

Lysogenic cycle allows genetic recombination of the host bacterium.

Plants require minerals to grow leaves, stems, and roots. Minerals are essential for seed, flower, and fruit production. Plants use minerals to produce chlorophyll and carry out photosynthesis, the process by which they convert sunlight into usable energy.

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: $7 \times 2 = 14$)

Attempt any three (2) questions from this section. Questions carry equal marks:

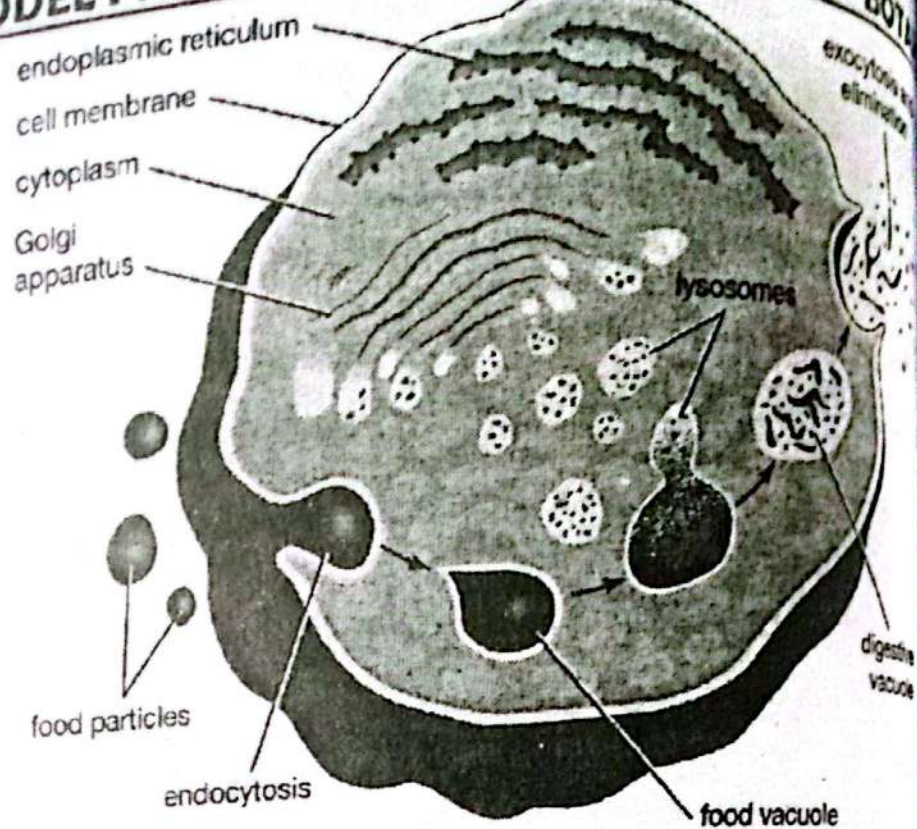
Explain the structure and functions of lysosomes.

STRUCTURE OF LYSOSOME

Lysosomes are spherical, single, membrane-bounded bodies, a few micrometers in diameter, originated by Golgi-bodies, containing hydrolytic enzymes. They occur only in the cytoplasm of animal cells and function in the digestion of material taken into cells by phagocytosis. Normally, they act as destroyers of foreign particles and worn-out cellular organelles.

Each formed lysosome before starting its functions called the primary lysosome. They contain about 40 different types of hydrolytic enzymes the enzymes during performing their function attach to the membrane of the material like endocytosis, phagocytosis or autophagocytosis. These lysosomes are generally called secondary lysosomes but specifically, they are called phagosomes and autophagosomes respectively. They also perform autophagy, the process by which unwanted structures within the cell are engulfed and digested by lysosomes. This is the self-eating process of the cell.

It sometimes eliminates old cells or unwanted cells at the embryonic stage. According to their genetic information, this planning process is called programmed cell death. During this self-destruction process, the membrane of the cell is ruptured at a particular time. As a result, the hydrolytic enzymes are released free in the cell and the cell undergoes chemical breakdown or lysis. This process of a cell to destroy itself by digesting its own macromolecules, so that the cell is referred to as a "suicidal sac" this process is called autolysis.



FUNCTIONS OF LYSOSOMES

1. They take part in the digestion of food obtained through phagocytosis known as intracellular digestion.
2. Lysosomes release their enzymes to the outside through exocytosis to perform extracellular digestion.
3. Lysosomes destroy the obstructing structures.
4. In sperms, lysosomes provide enzymes for breaking the limiting membrane of eggs.
5. Leucocyte granules are derived from lysosomes.
6. Lysosomes cause the breakdown of aging and dead cells.

Q 3. (OR) What is cellular respiration? Explain types of respiration in detail.

ANSWER:

CELLULAR RESPIRATION

Every living cell requires energy to carry out their functions. This energy comes from fuel molecules such as glucose. In cellular respiration glucose molecules are oxidized either in the absence of oxygen or in the presence of O_2 . Carbon-carbon bonds of glucose molecules break to release energy. Some of this energy is stored in the form of ATP while other is lost. Thus a cell transfer energy from glucose to ATP through reduction and oxidation process. This break down of glucose molecules by redox reaction to synthesize ATP is called respiration. This ATP provides energy for metabolism wherever required by removing its terminal phosphate liberating energy ADP and P (Phosphate) are formed.

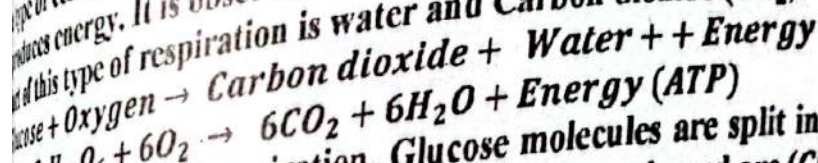
Types of respiration processes have been observed

ANAEROBIC RESPIRATION OR FERMENTATION

anaerobic respiration was originally defined by W. Pauster as respiration in the absence of air (O_2). It is an alternative term used for anaerobic respiration. Products of anaerobic respiration are either ethyl-alcohol or lactic acid. Type of fermentation where alcohol is formed is termed alcoholic fermentation and the fermentation where lactic acid is formed is called lactic acid fermentation. A small but significant minority of organisms obtain energy by anaerobic respiration. Many micro-organisms including yeasts and some bacteria can respire anaerobically. Anaerobic respiration as well as aerobic respiration has a common phase where glucose breaks down chemically into two molecules of pyruvate (pyruvic acid) called pyruvate.

AEROBIC RESPIRATION

A type of cellular respiration that takes place in the presence of oxygen and produces energy. It is observed in both plants and animals and the end product of this type of respiration is water and Carbon dioxide (CO_2).



During, in this type of respiration, Glucose molecules are split in the presence of oxygen and the end or by-products which are released are (CO_2), and energy in the form of ATP. The total energy released in this process is 2900 KJ which is used to produce ATP molecules. It is observed in all types of multicellular organisms.

Describe the cohesion tension theory of water movement through plants. What supplies the cohesion and what is the source of tension? How do these two interact to move water through a plant?

ANSWER:

COHESION TENSION THEORY OF WATER MOVEMENT

Upward movement of water from the absorptive surface (roots) up to the transpiring surface (leaves) against the downward pull of gravity is known as the ascent of sap.

Upward movement of water from roots up to leaves takes place through xylem depending upon two factors. These are transpiration pull and adhesive properties of water i.e., adhesion and cohesion. Water molecules are attracted to cellulose molecules on the walls of xylem vessels. Extensive hydrogen bonding in water gives rise to the property of cohesion. It also gives water a high tensile strength defined as the ability to resist the pulling force. Water molecules in xylem vessels form a continuous column. Transpiration pulls the results chain of events that starts when leaves begin to transpire. Solar radiation in the morning. Sunlight raises the temperature of

leaves, so the water begins to evaporate from the moist walls of the cells. The evaporated water is immediately replaced by water from the xylem which is replaced with water from neighbouring cells deeper in the plant. Ultimately, water is pulled from the xylem to meet the loss of water. Water in the xylem is placed under tension which is transmitted to the vessels. This downward transmission of tension is because of the cohesive property of water columns in vessels and tracheids. The water column moves upward by mass flow due to transpiration pull. Adhesion of water molecules to hydrophilic walls of Xylem cells. The small diameter of vessels and tracheids are important factors in overcoming the force of gravity.

Q 4. (OR) Write down general characteristics of plants.

ANSWER:

GENERAL CHARACTERISTICS OF PLANTS

1. Plants are multicellular eukaryotes.
2. They are Photosynthetic autotrophs and contain photosynthetic pigment specially chlorophyll, which enable the plants to convert light energy into food (chemical energy).
3. Rigid cell wall of plant cell primarily made up of cellulose
4. Life cycle of show heteromorphic alternation of generation. The zygote is retained and develops into embryo
5. Plant cuticle, composed of insoluble lipid polymers and waxes help to prevent diffusion of water to the drier atmosphere and also filter out the substantial component of UV radiation from sunlight.

Some others characteristics included are:

Plants produce some secondary products like sporopollenin, lignin. Sporopollenin tougher than lignin composed chiefly of carotenoids. It protects them from attack by grazing organism and microorganism. It makes spores of land plants tough and flexible and resistant to biodegradation. Lignin is important to provide strength to support terrestrial plants.

In terrestrial environment gases enter through small pores in the surface. Stomata are present in modern hornworts, mosses, stomata in lenticel and vascular plants as well as simple, ancient fossil plants like Rhynia.

Q 5. List down the physical and chemical methods to control bacteria.

ANSWER:

CHEMICAL METHODS TO CONTROL BACTERIA

There are different chemicals that are used to control bacteria:

1. Phenol
2. Halogens
3. Heavy Metals
4. Soaps and Detergents
5. Aldehydes
6. Ethylene Oxide

8. Food Preservatives
2. Dry-Heat Oven
4. Autoclave
6. Refrigeration
8. High-Pressure Processing
10. Simple Desiccation
12. Ionizing Radiation
14. Nonionizing Radiation

OSUIFS MODEL QUESTION PAPER 2
Annual Examination 2023
BOTANY (THEORY)
CLASS XI (SCIENCE PRE-MEDICAL GROUP)

Total Marks: 45

Time: 2 Hours

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)
(9 Marks)

15 Minutes

This Section consists of 18 MCQs and all are to be answered.
 MCQ carry 0.5 marks.

Choose the correct answer from the given options:
 animals cells, hydrolytic enzymes are packed to prevent general
 of cellular components. Which of the following organelles
 in a the compartmentalization?

- (b) Lysosome✓
 (d) Glyoxysome

linear flow of electrons from water to NADP^+ coupled to ATP

- photophosphorylation
 cyclic photophosphorylation✓
 chemiosmotic photophosphorylation
 oxidative photophosphorylation

that severely damage motor neurons and causes paralysis

- (b) Dengue
 (c) Polio✓
 (d) Herpes
 is caused by the bacteria:
 (b) Clostridium

- (c) *Vibrio* ✓
 (v). Protoctista having isomorphic alternation of generation:
 (a) *Chlorella*
 (c) *Euglena*
 (vi). Hepaticae, anthocerotae and moss are the classes of:
 (a) Engiosperms
 (c) Tracheophyte
 (vii). The cause of colour of leaf turns into dull or bluish green in deficiency of:
 (a) Nitrogen
 (c) Potassium
 (viii). The volume enclosed by the plasma membrane of plant cell is often much larger than the corresponding volume in animal cells. The most reasonable explanation for this observation is that:
 (a) Plant cells are capable of having a much highly surface to volume ratio than animal cells.
 (b) Plant cells are capable of having convoluted plasma membrane than animal cells.
 (c) Plant cells contain a large vacuole that reduces the volume of the cytoplasm. ✓
 (d) Animal cells are more spherical, while plant cells are elongated.
 (ix). This enzyme can fix both CO_2 and O_2 :
 (a) Deoxyribose
 (c) PEPG
 (b) RUBISCO ✓
 (d) PEPIS
 (x). Vexillary aestivation is a characteristic of this family:
 (a) Fabaceae ✓
 (c) Mimosaceae
 (b) Caesalpinaceae
 (d) Solanaceae
 (xi). Insectivorous plants trap insects to obtain:
 (a) Lipid
 (c) Cellulose
 (b) Enzymes
 (d) Nitrogenous Compounds
 (xii). Process which decomposes private into three molecules of CO_2 :
 (a) C_3 cycle
 (d) C_4 cycle
 (b) Benson and Calvin cycle
 (c) C_3 cycle
 (d) Benson and Calvin cycle
 (xiii). The beak-like out growth of ascogonia is called:
 (a) Antheridium
 (c) Hyphae
 (b) Trichogyne ✓
 (d) Mycelium
 (xiv). The miniature bisexual gametophyte of fern is called:
 (a) Prothallus ✓
 (c) Sporophyte
 (b) Gametophyte
 (d) Seed
 (xv). A classification unit of inbreeding organism is:
 (a) Species ✓
 (b) Genus
 (c) Family
 (d) Phylum

ametophytic generation is dominant in:

- (b) Bryophytes ✓
- (d) Gymnosperms

gymnosperms

metophytes

Oxygen becomes attached to RUBP during:

- (b) C4 Pathway
- (d) Photorespiration ✓

C4 pathway

Benson

This disease is caused by arbovirus:

- (b) Yellow fever ✓
- (c) Measles
- (d) Rabies

Measles

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 11x2=22)

Answer any Eleven (11) questions from this

All questions carry equal marks.

Osmotic adjustment is beneficial for plants?

Ans: Osmotic adjustment is beneficial for plants' useful drought response to drought conditions or saline stress, many plants use organic solutes to lower the osmotic potential.

Mitochondria is called power house of cell?

Ans: The folds formed by the inner membrane of mitochondria are in a fluid-like organic matrix, with a number of compounds and in it, due to the production of a high amount of energy for all cell and their activities, this energy is stored and utilized wherever it is required by an organism.

Desert plants reduce their leaf size?

Ans: (See Question No. 2(xvi) New Model Paper)

Plant cell wall is rigid?

Ans: The secondary cell walls of the plant are very thick and rigid due to deposition of lignin, inorganic salts and some waxes. It plays a role in support of plant.

What is fermentation? Also name its types.

Ans:

It was originally defined by W. Pauster as respiration in the absence of O_2 . It is an alternative term used for anaerobic respiration. Types of anaerobic respiration are either ethyl-alcohol or lactic acid.

FERMENTATION

In fermentation where alcohol is formed is termed as alcoholic fermentation and the fermentation where lactic acid is formed called lactic acid fermentation.

Life cycle of plants shows alternation of generation?

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ANSWER: Alternation of generation is the type of life cycle in which an asexually reproducing generation alternates with a sexually reproducing generation. It is an adaptation in plants' life cycle. The gametophyte generation which is haploid produces male and female gametes by mitosis. Male and female gametes fuse to form a zygote which grows into the diploid sporophyte generation. The sporophyte produces spores by meiosis which are dispersed and germinate into new haploid gametophytes.

2. (vii) Explain the economic importance of bacteria.

ANSWER: Bacteria have widespread economic importance as they are used in various processes. Milk is converted into curd by bacterial action. The milk's lactose is converted into lactic acid, which gives the characteristic sour taste of the curd. Cheese production starts with milk coagulated by lactic acid bacteria and the curd formed is filtered to separate the whey. Some bacteria are used to exploit to produce antibiotics.

2. (viii) What features allow fungi to survive in all environment where possible?

ANSWER: (See Question No. 2(xiii) New Model Paper)

2. (ix) Write the botanical names of any four of the following:

* Brinjal * Tomato * Pear * Sweet pea * Rice * Mulhethi

ANSWER:

COMMON NAME	BOTANICAL NAME
Brinjal	<i>Solanum melongena</i>
Tomato	<i>Lycopersicon Esculentum</i>
Pear	<i>Pyrus pyrifolia</i>
Sweet pea	<i>Lathyrus Odoratus</i>
Mulhethi	<i>Glycyrrhiza glabra</i>

2. (x) Explain the photosynthesis mechanism in bacteria.

ANSWER:

PHOTOSYNTHESIS MECHANISM IN BACTERIA

Bacterial photosynthesis is based on a cyclic photophosphorylation mechanism and only one pigment system (PS-I) is involved. During the process, bacteriochlorophyll absorbs light and this light energy rises the chlorophyll molecule to an excited state. Excited bacteriochlorophyll releases an electron and becomes positively charged. It serves as a strong oxidizing agent and electron acceptor.

2. (xi) Define Viroids.

ANSWER:

VIROIDS

Viroids are simple nonenveloped single-stranded RNA infectious molecules with some double-stranded regions. They can be transmitted by pollen.

Smallest viroid discovered so far is 220 nucleotides long while the largest virus causing the infection is around 2000 bases in size. Viroids are RNA polymerase. The Potato spindle tuber viroid was identified as a viroid and now the known species of viroids are about 33 in total. Viroids don't interrupt protein synthesis but they only replicate their specific RNA molecules. Viroids mainly cause plant diseases and plants show distorted growth. There is no evidence of human ATP is called common energy currency of living system?

(See Question No. 2(ix), 2023)

Various divisions of Fungi.

OF FUNGI

Four major divisions of fungi, which are divided on the basis of reproduction.

2- Ascomycota

4- Deuteromycota

Explain the deficiency symptoms of Nitrogen in Plants.

(See Question No. 2(ii) New Model Paper)

What are the symptoms of AIDS?

(See Question No. 2(v), 2024)

Explain Ascent of Sap.

OF SAP

Explain movement of water from the absorptive surface (roots) up to the transpiring surface (leaves) against the downward pull of gravity is the ascent of sap.

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: 7x2=14)

Attempt any three (3) questions from this section. Questions carry equal marks:

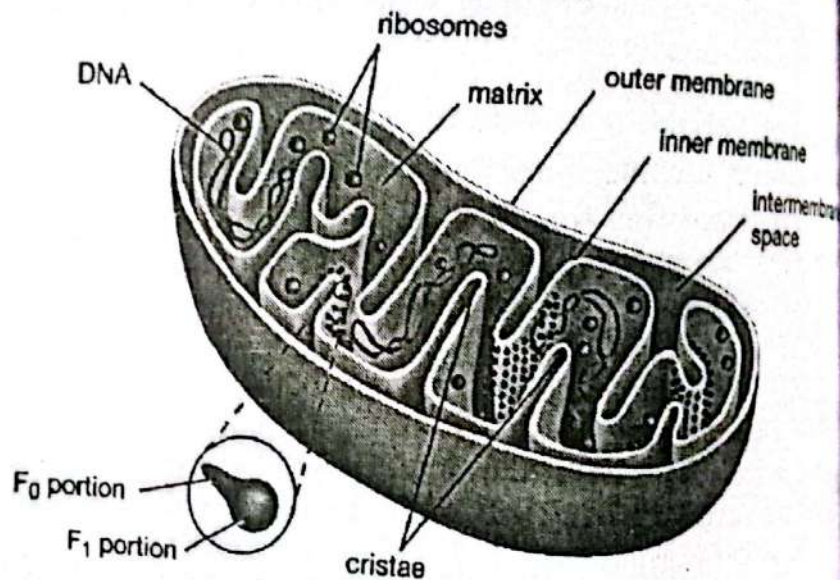
1. Draw the structure of mitochondria with suitable diagram

STRUCTURE OF MITOCHONDRIA

Mitochondria or chondriosome are universally present in the cytoplasm of eukaryotic cells. They appear as minute granular, vesicle rodlets, thread or ring-like structures depending on the nature of the cells, but usually, it is considered that they are bean-shaped. They are seen to be in constant motion in living cells. They are the centres of aerobic respiration. Each mitochondrion is

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approximately about 0.5 to 1.0 nm in diameter and about 10 nm long. They are double membrane-bounded organelles. Both membranes are made up of lipids and proteins. The outer membrane is smooth and has pores made up of a protein called Porins. These pores are responsible for the transport of molecules across the membrane; therefore this outer membrane is permeable for all. The inner membrane forms irregular folds called cristae, which increases the surface area to attach the number of proteins like ATPase, cytochrome, NAD, FAD etc. These complexes and molecules are electron carriers, which metabolize carbohydrates (starch), lipids and amino acids (protein) into CO_2 and H_2O with energy in the form of ATP which is stored in mitochondria. The folds formed by the inner membrane are filled with a fluid-like organic matrix, with a number of compounds and enzymes in it, due to the production of a high amount of energy for all cell organs and their activities.



Q 3. (OR) Describes tricarboxylic acid cycle in detail.

ANSWER:

TRICARBOXYLIC ACID CYCLE

The tricarboxylic acid (TCA) cycle, also known as the Krebs or citric acid cycle, is the main source of energy for cells and an important part of aerobic respiration. The cycle harnesses the available chemical energy of acetyl coenzyme A (acetyl CoA) into the reducing power of nicotinamide adenine dinucleotide (NADH). The TCA cycle is a central pathway that provides a unifying point for many metabolites, which feed into it at various points. It takes place over eight different steps:

TRICARBOXYLIC ACID CYCLE

Acetyl CoA (two carbon molecule) joins with oxaloacetate (4 carbon molecule) to form citrate (6 carbon molecule).

Citrate is converted to isocitrate (an isomer of citrate)

Isocitrate is oxidised to alpha-ketoglutarate (a five carbon molecule) which results in the release of carbon dioxide. One NADH is formed.

The enzyme responsible for catalysing this step is isocitrate dehydrogenase.

This is the rate limiting step, as isocitrate dehydrogenase is an allosterically regulated enzyme.

Alpha-ketoglutarate is oxidised to form a 4 carbon molecule. This step is catalysed by enzyme A, forming succinyl CoA. A second molecule of NADH is formed alongside a second molecule of carbon dioxide.

Succinyl CoA is then converted to succinate (4 carbon molecule) and a GTP molecule is produced.

Succinate is converted into fumarate (4 carbon molecule) and a second molecule of FADH₂ is produced.

Fumarate is converted to malate (another 4 carbon molecule).

Malate is then converted into oxaloacetate. The third molecule of NADH is also produced.

The primary role of the TCA cycle is production of NADH and FADH₂ which are used in the electron transport chain.

The cycle also produces molecules that supply various biosynthetic pathways. These enter or exit the cycle at various points depending on the organism.

For example, alpha-ketoglutarate can leave the cycle to be converted into amino acids, and succinate can be converted to haem.

What are growth regulators? Name and discuss five in detail?

Q: (See Question No. 4 New Model Paper)

Q: How leaves are evolved?

Q: What is the function of the leaf?

Q: What is the structure of a leaf?

Q: What is the importance of the leaf?

Q: What is the evolution of the leaf?

Q: What is the structure of a leaf?

Q: What is the importance of the leaf?

Q: What is the evolution of the leaf?

Q: What is the structure of a leaf?

Q: What is the importance of the leaf?

Q: What is the evolution of the leaf?

Q: What is the structure of a leaf?

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Q: What is the structure of a leaf?

Q: What is the importance of the leaf?

Q: What is the evolution of the leaf?

Q: What is the structure of a leaf?

Q: What is the importance of the leaf?

Q: What is the evolution of the leaf?

REDUCTION THEORY:

Another possibility is that a single veined leaf originated by a reduction in the size of the part of the leafless branching system of the primitive plant. This is how the leaf of lycopodium (club mosses) and equisetum (horsetail) came into existence.

Evolution of Many Veined Leaf:

WEBED THEORY:

Many veined leaf (megaphyllous) originated much later. There were several evolutionary modifications of the forked branching system in the evolution of this leaf. The first step in the evolution of this leaf was the restriction of the branches to a single plane. The branching system became flat. The next step in the evolution was filling the space between the branching and the stem with tissues. The leaf so formed looked like the web foot of a duck.

Q 5. What do you mean by normal flora of bacteria?

ANSWER: The human body is not sterile; we become colonized by bacteria from the moment we are born. We are covered with, and contain within our intestines, approximately one hundred trillion bacteria that form the normal flora of our bodies. This normal flora helps to prevent us from being colonized by more dangerous bacteria, which might lead to infection.

THE NORMAL FLORA

Normal flora is the microorganisms that live on another living organism (human or animal) or inanimate object without causing disease. In a human animal, the internal tissues, e.g. blood, brain, muscle, etc., are normally free of microorganisms. However, the surface tissues, i.e., skin and mucous membranes, are constantly in contact with environmental organisms and become readily colonized by various microbial species. The microorganisms regularly found at any anatomical site is referred to as the normal flora. The normal flora of humans consists of a few eucaryotic fungi and protists, but bacteria are the most numerous and obvious microbial components of the normal flora.

SIGNIFICANCE OF THE NORMAL FLORA

These normal floras provide us with many benefits. They prevent colonization by pathogens by competing for attachment & nutrients. They stimulate the production of antibodies. Since the normal flora behaves as antigens in an animal, they induce low levels of antibodies that cross-react with similar antigens on pathogens, preventing infection or invasion. In the large intestine, some synthesize vitamins that are absorbed as nutrients by the host (e.g., vitamin K & B12). Some bacteria produce substances that inhibit pathogenic species.

(40 Marks)

TIME: 2 Hours

SECTION 'A' (Multiple Choice Questions) (08)
This section consists of 16 part questions and all are to be answered. Each part question carries $\frac{1}{2}$ mark.
Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Choose the correct answer for each from the given options:

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Choose the correct answer for each from the given options:

Choose the correct answer for each from the given options:

Choose the correct answer for each from the given options:

Choose the correct answer for each from the given options:

Choose the correct answer for each from the given options:

- * Glycine ✓
- * Valine
- * Serine
- * Mitochondria
- * Cell Membrane
- * Placoid Scales ✓
- * Ventricular Systol

- * Atrial Systol
- * Ventricular Systol
- * Allosteric Inhibitor
- * Non-Competitive Inhibitor

- * Pharynx
- * Nasal Chambers ✓
- * Trachea
- * 39°C
- * 39°F

- * Spirometer ✓
- * Sphygmomanometer

- * Physalia
- * Sea Urchin ✓

- * Right Atrium
- * Right Ventricle

Choose the correct answer for each from the given options:

- * Macrophages
- * Neutrophils ✓
- xii) Deoxygenated blood to be carried to the lungs by:
 - * Dendritic Cells
 - * Monocytes
 - * Renal Artery
 - * Carotid Artery
- * Coronary Artery
- * Pulmonary Artery ✓
- xiii) The study of functioning and disorders of immune system is termed as:
 - * Ophthalmology
 - * Dermatology
- * Parasitology
- * Immunology ✓
- xiv) Number of milk teeth in human is:
 - * 10
 - * 15
 - * 20 ✓
 - * 25
- xv) Partial negative charge and partial positive charge of water molecules attract to each other due to:
 - * Hydrogen Bond ✓
 - * Covalent Bond
 - * Ionic Bond
 - * Peptide Bond
- xvi) Animals with dry scaly skin, egg with amniotic coat and three chambered heart are found in this class:
 - * Amphibia
 - * Aves
 - * Reptilia ✓
 - * Mammalia

SECTION "B" (Short Answer Question) Marks: 16

NOTE: Attempt any **EIGHT** part questions in all. Select **FOUR** Reasoning Questions and **FOUR** Non-Reasoning Questions. All part questions carry equal marks.

2.(a) REASONING QUESTIONS:

i) How does Bile help in digestion of fats?

ANSWER: Bile is yellowish-green in colour and it contains water, salts and bile pigments (brownish-yellow bilirubin and greenish biliverdin). These pigments break down large fat globules into smaller globules so that the pancreatic enzymes can easily act on them. The process is known as the emulsification of fats.

ii) Why are water molecules called amphoteric in nature?

ANSWER: The water molecules ionize into H^+ and OH^- . This reaction is reversible and also maintains equilibrium. Due to ionization property water behaves as an acid or base which is known as Amphoteric.

iii) Why are enzymes temperature sensitive?

ANSWER: Enzymes are temperature sensitive because of their proteinaceous composition. The temperature where enzymes work maximum or the rate of enzyme activity is maximum called optimum temperature, below and above this temperature the enzyme activity decreases.

Q: Artificial sweetener is not harmful to diabetic patient. Why?
 A: Because artificial sweetener usually are left handed sugars, whereas natural sugars are right handed and they are specific to break down the right handed sugars.

Q: Antipyretic therapy is used when fever is a non-specific response against microbial infections. The antipyretics can inhibit the secretion by the hypothalamus thereby reducing fever.

Q: SA. Node called pacemaker of the heart?
 A: Because the SA node generates electrical impulses to trigger a series of electrical events in the heart, which is controlling the sequence of contraction that pumps blood out of the heart. Since the SA node and maintains the rhythmicity of the heart, it is known as the pacemaker of the heart.

Q: Haemoglobin called as respiratory pigment?
 A: Haemoglobin is called respiratory pigment because it is the oxygen-carrying capacity of blood and contains iron-bound atoms which combine readily with oxygen.

Q: The stomach is protected from HCl?
 A: The stomach itself is protected against the strong HCl by the mucus membrane. The mucosa has mucous-secreting mucoid cells. The gastric mucous is a glycoprotein which is a means of protecting food as well as protecting the stomach against the action of HCl enzyme.

REASONING QUESTIONS:

Q: Explain the role of arteriole in vasodilation and vasoconstriction.

ANSWER: OF ARTERIOLE IN VASODILATION AND VASOCONSTRICTION

Arteriole walls are under the influence of nerves, hormones and chemicals produced by nearby tissues. They can therefore contract and relax in response to the changing needs of the tissues and the blood they supply. On the hot summer day, however, you become flushed because the arterioles in your skin expand (vasodilation) and bring more blood to the capillaries to dissipate excess heat to the outside and to maintain your internal temperature. In contrast in extremely cold weather, the arterioles and toes can become frostbitten because the arterioles supplying them constrict.

Q: List the four basic characteristics of Chordates.

BASIC CHARACTERISTICS OF CHORDATES

- (i) Notochord
- (ii) Nerve cord
- (iii) Gill slits
- (iv) Post anal tail
- xi) Differentiate between Polyp and Medusa.

ANSWER: Polyps are cylindrical with a mouth and tentacles while the Medusa umbrella is shaped with mouth and tentacles.

- xii) Write four characteristics of enzymes.

ANSWER:

CHARACTERISTICS OF ENZYMES

The characteristics of enzymes are as follows:

1. All enzymes are protein in nature; they can be destroyed by heat.
2. They act best within a narrow, temperature range.
3. They work efficiently in narrow range of acidity or alkalinity.
4. A particular enzyme forms the same end-product, because it acts on particular/specific substrate.

- xiii) Define Sinuses and write their types.

ANSWER:

SINUSES

Sinuses are hollow air-filled cavities. These hollow cavities found in the skull and linked to the nasal air passage. Human has four pairs of nasal cavities.

TYPES OF SINUSES

1. Frontal sinus, in the forehead region
2. Maxillary sinus, in the cheek
3. Ethmoid sinus, between the eyes
4. Sphenoid sinus, located in deep behind the ethmoids

- xiv) Write note on Subclass Ratitae and Carinatae.

ANSWER:

SUBCLASS RATITAE

Ratitae cannot fly because they have comparatively heavy weight and wings are either vestigial or rudimentary. They have a flat sternum without a keel. Their flight muscles are poorly developed. The distribution of ratitae is also restricted. None of them is found in Pakistan. Common flightless birds are Ostrich.

SUBCLASS CARINATAE

They are usually small, lightweight birds whose wings are highly developed and the feathers of their wings have an interlocking mechanism. The sternum is provided with a crest like keel to accommodate the highly developed pectoral flight muscles. The flying birds are distributed all around the world. Common flying birds are pigeons.

- xv) Define Diploblastic and Triploblastic animals with examples.

ANSWER:

PROBLASTIC ANIMALS
Problasts are animals in which the body wall is created from two embryonic germ layers- an inner layer (endoderm or mesendoderm) and an outer layer (ectoderm). They have a nonliving middle layer between the endoderm and ectoderm.

TRIOBLASTIC ANIMALS
Trioblasts are animals having three embryonic germ layers, an inner layer (endoderm), an outer layer (ectoderm), and a middle layer (mesoderm). The ectoderm forms the brain, spinal cord, lens of the eye, hairs, connective tissues, and circulatory systems and endoderm leads to formation of the stomach, liver, colon, urinary bladder, and lungs. Differentiate between single and double circuit circulation with examples.

ANSWER:

SINGLE CIRCUIT CIRCULATION	DOUBLE CIRCUIT CIRCULATION
A circulatory system in which blood flows once through the heart in one complete cycle of the body.	A circulatory system in which blood flows twice through the heart in one complete cycle of the body.
Single circulatory system is found in typical fish species and reptiles.	Occurs in all other vertebrates which include mammals and humans.
Organisms that possess a single circulatory system have a double chambered heart.	Animals with double circulatory system have a four-chambered heart.
This circulatory system is referred to as "single" circulatory because it is made up of only one circuit referred to as systemic circulatory system.	These circulatory system is referred to as "double" circulatory system because it is made up of two circuits referred to as the pulmonary and systemic circuits.
Deoxygenated blood passes through the heart.	Mixed or oxygenated or venous blood passes through the heart.
EXAMPLES: The heart of fish is an example of single circuit circulation	EXAMPLES: The heart of birds and mammals are example of double circuit circulation

NOTE: Attempt any TWO question from this section.

All questions carry equal marks.

3. Explain lower respiratory tract diseases.

LOWER RESPIRATORY INFECTIONS

(i) PNEUMONIA

Pneumonia is a lower respiratory infection due to the number of bacteria and fungi. In this infection air sacs filled with secretions and fluid.

(i) Lobar pneumonia occur in the lobes of lungs

(ii) Bronchial pneumonia (bronchopneumonia) patches appear on lungs.

The symptoms of bacterial pneumonia are headache, lips and fingers become bluish in color, fever with cough and yellowish green or black mucus may produce with cough, confused mental state, heavy sweating. Loss of appetite, rapid breathing and shortness of breath.

Viral and bacterial pneumonia have same symptoms in early stage which may cause shortness of breath, headache, muscle pain and weakness.

In Mycoplasma pneumonia symptoms are observed acute cough and mucus.

In most cases of bacterial pneumonia, oral antibiotics can be used. Antibiotic medications don't work on viruses. Patient should not drink refreshments. Humidification and steamy baths also very helpful to open airways blockage and easy respiring.

(ii) PULMONARY TUBERCULOSIS

Pulmonary tuberculosis (TB) is a bacterial chronic infection of the lung and caused by Mycobacterium tuberculosis. The treatment of Pulmonary TB is possible but late treatment or untreated patient may get in threatening situation.

Symptoms including coughing up mucus (phlegm), coughing up blood, low grade fever, chest pain and weight loss. Pulmonary TB is curable with early diagnosis and antibiotic treatment.

(iii) EMPHYSEMA

Emphysema is a condition of lung disease generally develops after many years of smoking. Emphysema is one of disease which belongs to group of chronic obstructive pulmonary disease (COPD). In this condition wall of the air sacs become damaged. So, the alveoli cannot support the bronchioles. Due to the blockage of tubes too much air traps inside the lungs. There are fewer alveoli, less oxygen transport into bloodstream. It is irreversible condition.

have no symptoms at the beginning but some specific symptoms are observed in later stages like shortness of breath especially during activity, tightness in chest, frequent coughing or wheezing and produces with cough. Some people with emphysema has respiratory distress such as cold and flu. In severe condition patient complains of lower body muscles and weight loss. Inhalators medicinal treatment recover the patient and relief from coughing and short breathing. Antibiotics may be used for the remedy steroid type drugs can also be inhaled like aerosol sprays which reduce and may help in easy breath.

LUNG CANCER

normal proliferation of affected tissues in the lungs. The main cause of lung cancer is Smoking. About 80% of lung cancer caused death. Symptoms are coughing with blood, chest infections that keep coming again when breathing or coughing, persistent breathlessness and chronic tiredness.

Lung cancer chemotherapy, radiation therapy, targeted therapy and surgery are acquires according to consultant.

Q-Define Lipids or Carbohydrates. Write their classifications and give examples.

ANSWER:

Lipids are an important, diverse group of biological molecules widely distributed among plants and animals. Like carbohydrates, they are composed of C, H, and O but contain a much lesser ratio of oxygen. Due to high quantity of carbon and hydrogen, they contain almost double the amount of energy as carbohydrates.

CLASSIFICATION AND FUNCTIONS OF LIPIDS

Lipids are classified into following four major types and each type is further divided into sub-groups:

TRIGLYCEROL (FATS AND OIL):

They are the condensation products of glycerol and three fatty acids. They are commonly called fats and oils. They can be defined as the esters of glycerol and fatty acids. Ester is the bond or linkage formed between glycerol and organic acid by removing water.

Glycerol is a trihydroxy alcohol, made of three carbon atoms each having one OH- group, while a fatty acid is a type of organic acid having one carboxylic acid group with a long hydrocarbon chain. When three fatty acids combine with glycerol each at one OH, they form ester bonds. A compound called Triglycerol (triglyceride) is formed. Triglycerol is neutral because all three OH groups of glycerol become

bonded with fatty acids and no charge-bearing OH is left. There are two types of acylglycerol.

(a) SATURATED ACYLGLYCEROL (FATS):

They contain saturated fatty acid i.e. do not contain any double bond between carbon atoms of the hydrocarbon chain. e.g. stearin.

(b) UNSATURATED ACYLGLYCEROL (OILS):

They contain unsaturated fatty acids. i.e. contain one or more than one double bond between carbon atoms of the hydrocarbon chain. They have a stable physical state, usually found in liquid state at ordinary temperature, found in plants, also called oil e.g. lanolin found in the seed.

2. PHOSPHOLIPIDS:

This type of lipid is the condensation product of Glycerol, two acids, choline and one phosphate. The Phospholipid is the most important group of lipids from the biological point of view. A phospholipid is similar to acylglycerol, except that one fatty acid is replaced by phosphate which is attached to a nitrogenous compound choline. It contains two ends, one which is made up of fatty acid, which form non-polar part therefore behave as water repellant so called hydrophobic. The phosphate and choline form another end, which form polar part, therefore attract water molecule so behave as water loving i.e. hydrophilic. Phospholipids are present in all living cells and form membrane. They are related to various functions such as regulation of cell permeability and transport processes. Properties of cell membrane depend on its phospholipid component.

3. WAXES:

They are esters of long-chain mono-alcohol and long-chain fatty acids. These are simple lipids and are found as a protective coating on stems, stalks, leaves, petals, fruit skin, animal skin, fur and feathers etc, these are water-repellant and non-reactive due to their non-polar nature. Hydrophobic compounds. These are chemically inert and resistant to atmospheric oxidation. There are two types of waxes i.e. Natural, like beeswax and cutin form cuticles of leaves and synthetic waxes, generally derived from petroleum or polyethylene.

4. TERPENOIDS:

Terpenoid is a large and important class of lipids which is made up of isoprenoid units (C_5H_8). Terpenes, steroids, carotenoids and prostaglandins are types of Terpenoids. These are found in cell membranes as cholesterol, as pigment like chlorophyll, fragrance as menthol etc.

(i) TERPENES:

Terpenes are the type of terpenoid that contain a few isoprenoid units like diterpenes and triterpenes. These small-size terpenes are volatile and produce special fragrance. Some of these are used in perfume e.g. Myrcene.

Geraniol from rose, Limonene from lemon oil, and some components of vitamin A, and A. molecules as well as some other molecules utilized in the rubber and latex.

TERPENES:
are a type of Terpenoid which forms steroid nuclei made up of units containing 17 Carbon atoms arranged in four attached rings of them are hexagonal and one is pentagonal. The radicals to them as side chains distinguish them from one another. is one type of steroid. Cholesterol is the precursor for the several steroids i.e. testosterone, progesterone and estrogens.

PROTENOIDS:
consists of a long chain of isoprenoid units which contain rings at both or one terminal. These compounds are pigments of red, orange, yellow and brown colour in plants. Some important are plant pigments, like chlorophyll, cytochromes, carotenes, latex, rubber etc.

PROSTAGLANDINS:
are a group of lipids made by mammalian tissues at the site of tissue damage or infection that are involved in dealing with injury. They control different physiological processes inflammation, sensation of pain, blood flow, and formation of blood clots, and the induction of labour. We use aspirin to reduce fever and pain depend on the inhibition of prostaglandin synthesis.

HYDRATES
The meaning of the word carbohydrate is hydrated carbon i.e. water containing carbon. Thus biomolecules contain C, H and O as elements. Hydrogen and oxygen are present in the simple ratio of 2:1 as in water. The general formula of carbohydrate molecules is $C_nH_{2n}O_n$ (Whereas 'n' is the whole number) According to I.U.P.A.C. Carbohydrates are defined as "The polyhydroxy carbonyl compounds", which are aldehydes or Ketones.

CLASSIFICATION AND FUNCTIONS OF CARBOHYDRATES
As discussed earlier that carbohydrate molecules are also called as these Saccharides are classified into three group.
(i) Monosaccharide
(ii) Oligosaccharide

MONOSACCHARIDE:
are of carbohydrate molecules which contain only one sugar unit. They cannot hydrolyse due to this reason. The empirical formula of their molecules is $C_nH_{2n}O_n$ e.g. Ribose ($C_5H_{10}O_5$). Fructose

$C_6H_{12}O_6$, etc; all are found in white crystalline solid with sweet taste, soluble in water. Monosaccharide can further classified on the basis of number of atoms present in them, the suffix 'Ose' use with no. of C atoms present in them.

Glycerose and Dihydroxy acetone are important triose, produced during respiration. Tetrose are rare in nature, it occurs in some bacteria, polysaccharide form basic skeleton of nucleic acid. Hexose are most important sugars from biological point of view. Glucose found in ripe fruit, corn and honey. It is also found in all known polysaccharide in condensed state. Fructose another hexose present in fruit so called fruit sugar, but they are found in ring structures but we can also draw their structure in open chain form:

OLIGOSACCHARIDES:

The type of carbohydrate which are made up to 2 to 10 monosaccharides. These are comparatively less sweet in taste and less soluble in water. They can hydrolyze. The most common type is disaccharide, on hydrolysis yield two monosaccharides. The covalent bond between these two monosaccharides is Glycosidic bond or linkage. A glycoside is simply a ring shaped sugar molecule that is attach to another molecule, the sugar may be either 5 membered ring or a six membered ring. For example sucrose is a disaccharide, composed of two sugar units a glucose and fructose.

The disaccharide may be reducing or non-reducing sugar. The reducing sugar is any carbohydrate which is capable of being oxidized and causing the reduction of other substances without hydrolysis. It is due to the presence of free aldehyde or free ketone group. Examples are maltose, lactose etc. The non-reducing sugars are carbohydrate which are unable to be oxidized and do not reduce other substance. It is due to absence of free aldehyde or ketone groups, e.g. sucrose or raffinose etc.

Living organisms especially plants transport their sugar from source (leaves) to sink (fruit) tissues in the form of non-reducing sugar where glycosidic bonds are formed between 'carbonyl' groups of both sugars. Sucrose is the sugar which is non-reducing. It contains more energy i.e, it is energy efficient in transport and storage. During transport it is not oxidized and react with other substance so no intermediate reaction with other molecules occur.

(III) POLYSACCHARIDE:

These are high molecular weight carbohydrates which on hydrolysis yield many monosaccharides. These are formed by the condensation of hundreds or thousands of Monosaccharide units, e.g. starch, glycogen, cellulose and chitin.

GENERAL CHARACTERISTICS OF PHYLUM ANNELIDA

They may be aquatic (marine and fresh water) or terrestrial; free-living, sometimes parasitic.
They exhibit organ-system level of body organisation and bilateral symmetry.

They are triploblastic, metamerically segmented and coelomate animals. They possess longitudinal and circular muscles which help in movement.

Aquatic annelids like *Nereis* possess lateral appendages, parapodia, which help in swimming.

Open circulatory system is present.

Nephridia help in osmoregulation and excretion.

Nervous system consists of paired ganglia (sing. ganglion) connected by nerves to a double ventral nerve cord.

In an aquatic form, is dioecious, but earthworms and leeches are hermaphrodites.

EVOLUTIONARY ADAPTATIONS

Annelids are supposed to have evolved from a primitive flat worm like *Planaria* in the sea.

DIET:

They have a complete digestive system with both mouth and anus, complete system.

RESPIRATION:

Respiratory system is not found, diffusion is sufficient for gases exchange.

CIRCULATION:

Open type circulatory system with blood confined to the heart and blood vessels.

EXCRETION: Excretory organs are nephridia which are found in most segments.

NERVOUS SYSTEM:

Nervous system consists of simple ganglionic brain in the head.

GENERAL CHARACTERISTICS OF PHYLUM ARTHROPODA

These animals are multicellular triploblastic, bilaterally symmetrical. Some of the anterior segments show cephalization forming a cephalic region.

Body is segmented. Segments are limited. Each segment bears a pair of jointed appendages. They are jointed.

Body is covered by exoskeleton. It is made by chitin. The exoskeleton is shed periodically. The projections are useful for muscle attachment.

Exoskeleton is cast off periodically and new one is secreted. It is called ecdysis.

5. Tubular dorsal heart is present.
6. True coelom is greatly reduced. It is represented only by the cavity of the reproductive and excretory organs.
7. The blood is colorless.
8. Circulatory system is open type or lacunar type.
9. The nervous system shows nerve ring and double ventral nerve cord.
10. Nephridia are absent. Excretion is by malpighian tubules, glands and coxal glands.

EVOLUTIONARY ADAPTATIONS

DIGESTION:

The Alimentary canal is well developed with a mouth and anus. The movement is assisted by jaws, the digestive system is complete.

RESPIRATION:

Arthropods' efficient gas exchange is accomplished by gills in aquatic forms such as crustaceans, and tracheae and by either book lungs or tracheae in terrestrial forms.

TRANSPORT:

Arthropods have well developed circulatory system, blood not only travels through vessels but also empties into the haemocoel, where it bathes the internal organs directly this arrangement is known as an open-type circulatory system.

5. Explain first line of defense with reference of Digestive Tracts and Respiratory Tracts.

FIRST LINE OF DEFENSE

The first line of defence is considered non-specific since it prevents the entry of all kinds of pathogens into our body. The first line of defence is manifested by physical as well as biochemical barriers. The physical barriers are skin, mucous membrane and ciliated epithelium. The biochemical barriers consist of saliva, tears, mucous, gastric juice, etc.

ROLE OF DIGESTIVE TRACTS AS THE FIRST LINE OF DEFENCE:

Our gut is lined internally with mucosa. In addition, our stomach has gastric glands which secrete HCl and a variety of enzymes to destroy intruding noxious microorganisms. The internal lining of the mouth, nose, digestive tract, respiratory passage and lungs, urinary tract, etc., contains mucous-secreting cells also known as mucous membranes or mucosa. These internal linings are made up of epithelial cells. The mucosa not only serves as a non-specific physical barrier due to its compact nature but also provides a biochemical barrier through its mucous which contains many antimicrobial peptides. Moreover, the mechanical action of the peristalsis of the gut also moves out the sloughed mucous along with the undigested food (faeces).

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CHAPTER XI (5)
The small intestine is also equipped with a substantial amount of lymphoid tissue, which is divided into three sectors, viz. Tonsils in the pharyngeal region, Patches in small intestines and the Appendix. The lymphoid tissue contains macrophages and lymphocytes for the protection against microorganisms which they encounter. Another sector of the lymphoid tissue is the W.B.Cs (e.g., Plasma Cells and Lymphocytes) which are present in the epithelial cells of mucosa offering a barrier to the pathogenic other micro-organisms.

RESPIRATORY TRACTS AS THE FIRST LINE OF DEFENCE:

In the respiratory system, where there is a great risk of entry of pathogenic microorganisms through the air, the mucosa traps the microorganisms and mucous flushes off the contaminated mucous outside to avoid the entry of dirt, pathogenic microorganisms inside thereby playing a major role in the first line of defence.

The cilia present in our nose trap the dust, dirt, etc., from the air, making sure that the air passing into the rest of the airway is free of such contaminations.

The mucous membrane of the respiratory system is composed of a layer of epithelial cells layer. It contains numerous mucous glands. Due to mucous, the dust and germs stick to it and are removed by the synchronized upward movement of cilia towards the pharynx. It is either pushed into the oesophagus or removed through the nose.

Explain the Thrombus, Atherosclerosis and Angina Pectoris. Explain Thrombolytics and Angioplasty.

EFFECTS, AND PREVENTION OF CORONARY DISEASES

A blood clot (thrombus) within an intact blood vessel is called a thrombus. The plaques, when destroying the inner lining of the blood vessels, and platelets gather at the damaged site to form a blood clot. Thrombus and embolus are potentially serious blood vessel problems that increase the risk of having a heart attack. Thrombus happens when a blood clot or thrombus forms in a blood vessel. This can reduce blood flow. An embolus is any blood clot that travels within the body. If it becomes stuck and blocks the flow of blood. The issue is called an embolism.

EFFECTS AND PREVENTION OF ATHEROSCLEROSIS

Atherosclerosis is a disease of the arterial wall (intima) which loses its elasticity and the inner layer thickens causing the narrowing of the artery. This impairs the blood flow. The narrowing is due to the

formation of fatty lesions called atheromatous plaques in the arteries. These plaques consist of low-density lipoproteins (cholesterol and protein) decaying muscle cells, fibrous tissue, blood platelets and sometimes calcium. The arteries become hard and the disease is called arteriosclerosis or simply the hardening of the arteries.

FACTOR CAUSING ATHEROSCLEROSIS

The possible causes of atherosclerosis are smoking, hypertension, gender, obesity, physical inactivity, a high serum cholesterol level, diabetes, family history of arterial disease and possibly an aggressive lifestyle. The risk of atherosclerosis increases with age.

CAUSE, EFFECTS, AND PREVENTION OF ANGINA PECTORIS

The strain on the heart by increasing resistance to blood flow. The heart may enlarge in response to this added demand, and the blood supply may not increase proportionately. The heart muscle is inadequately supplied with blood, especially during exercise. Insufficient oxygen to the heart can cause chest pain and a radiating pain to the left arm called angina pectoris.

CORONARY BYPASS

It is a medical procedure to improve blood flow to the heart. It is needed when the arteries supplying blood to heart tissue, called coronary arteries, are narrowed or blocked. The coronary bypass uses blood vessels from another part of the body. It connects them to blood vessels above and below the narrowed artery, bypassing the narrowed or blocked coronary arteries.

ANGIOPLASTY

It is a medical procedure used to improve the flow of blood through narrowed arteries. It involves placing a long, thin tube (catheter) into a blood vessel to place a small balloon at the site of the narrowing. When the balloon is inflated, the blood vessels will be open and allow a healthy flow of blood.

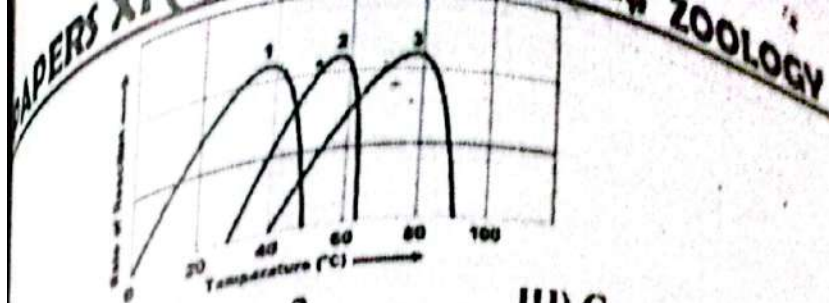
ZOOLOGY 2023

TIME: 2 Hours

(40 Marks)

SECTION 'A' (Multiple Choice Questions) (80 Marks)

1. Choose the correct answer for each from the given options:
- i. This curve represents the behavior of an enzyme taken from a bacterium that lives in hot springs:



II) Curve 2

III) Curve 3

- * II only ✓
- * Pharynx

- * III only
- * I and II
- * Larynx
- * Trachea

exchange takes place at the surface of:

animals show resemblance except:

- * Brittle Star
- * Sea Urchin

volume is measured by a special instrument

- * Spirometer ✓
- * Thermometer

transfer of antibodies from mother to the foetus is:

- * Natural Passive Immunity ✓
- * Artificial Passive Immunity

Active Immunity

the double circuit circulation of blood is found in

- * Reptilians
- * Amphibians and Reptilians both ✓

enzymes are highly active at this temperature

- * 37°C ✓
- * 50°C
- * 50°F

are produced by:

- * T - Cells
- * Macrophages

the correct matching pair:

- * Collar Cells ✓
- * Spicules - Ponfera
- * Gametes Formation
- * Parazoa - Porifera

red blood is carried by:

- * Pulmonary Artery ✓
- * Renal Artery

plant carbohydrate found in nature is:

- * Maltose
- * Cellulose ✓
- * Glycogen

water does not match the phylum Cnidaria:

- * Diploblastic
- * Bilateral Symmetry ✓

portal system of human is formed between:

- * Small Intestine and Liver ✓
- * Kidney and Stomach

xiv. It is an example of polynucleotide:

- * DNA ✓
- * ATP
- * ADP

xv. The protein part of enzyme is called.

- * Prosthetic Group
- * Holoenzyme
- * Cofactor
- * Apoenzyme ✓

xvi) All Amino acids have same formula except:

- * Alpha Carbon
- * Radical Group ✓
- * Hydroxyl Group
- * Amino Group

SECTION "B" (Short Answer Question)

NOTE: Attempt any EIGHT part questions in all. Select any FOUR Reasoning Questions and FOUR Non-Reasoning Questions. Each part question carries equal marks.

2.(a) REASONING QUESTIONS:

i) Why is SA node called pace maker of the heart?

ANSWER: (See Question No. 2 [a (vi)], 2024)

ii) How does stomach protect against HCl?

ANSWER: (See Question No. 2 [a (vii)], 2024)

iii) Why is artificial sweetener not harmful to diabetic patient?

ANSWER: (See Question No. 2 [a (iv)], 2024)

iv) Why are water molecules called amphoteric in nature?

ANSWER: (See Question No. 2 [a (ii)], 2024)

v) Why are Annelids, Molluscs and Arthropods considered as protostomes?

ANSWER: Annelids, Molluscs and Arthropods have unique characteristics but no matter how different they appear on the outside, they have same basic characteristics inside that make them protostomes. They are bilaterally symmetrical, triploblastic, coelomates and protostomes.

vi) Why do capillaries have single layer of Endothelium?

ANSWER: Because this very thin layer of endothelium helps in transferring oxygen in an efficient manner and has very little resistance to the diffusion of dissolved substances in or out.

2.(b) NON-REASONING QUESTIONS:

vii) Differentiate Cartilaginous and Bony fishes.

ANSWER:

CARTILAGINOUS FISH	BONY FISH
The exoskeleton of this group of fish is made up of bones.	The exoskeleton of this group of fish is made up of cartilage.
The gills are not covered with operculum in this group of fish.	The gills are covered with operculum in this group of fish.
They are marine fishes.	They are found in sea as well as in fresh water.

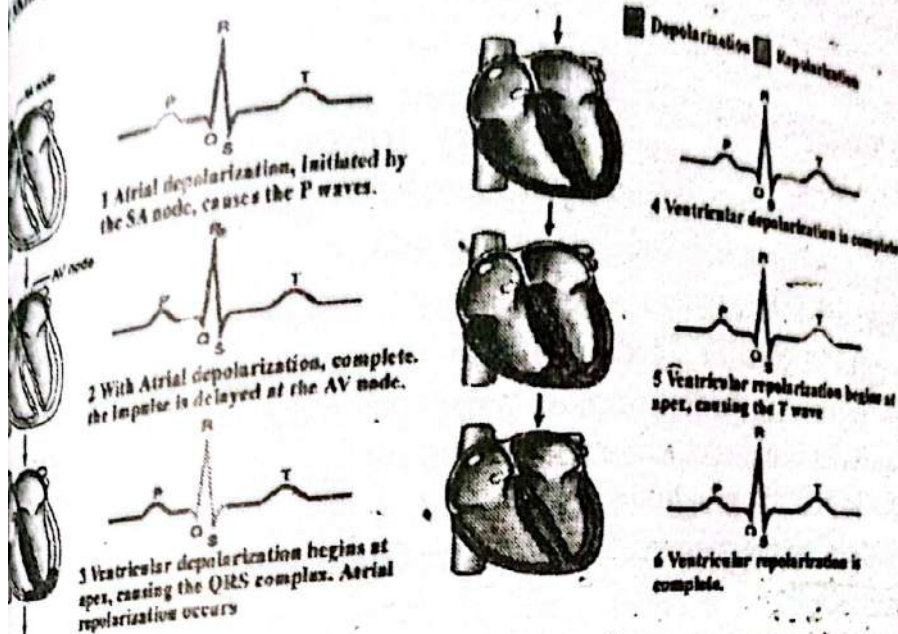
Write the characteristics of the heart. (See Question No. 2 [b (xii)], 2024)

ANSWER: (See Question No. 2 [b (xii)], 2024)

Write a note on ECG.

ELECTROCARDIOGRAM

Impulses that travel through cardiac muscle during the heart cycle produce electrical currents that are conducted through body fluids to the surface where the currents can be detected by electrodes placed on the skin and recorded as an electrocardiogram (ECG)



USES OF ECG:

With the help of an electrocardiogram, we can see the electrical changes in the heart. The heart cycle consists of two main phases: depolarization and repolarization. Depolarization of the atria produces the P wave. The ventricles of the heart are depolarized during the expression of the P wave on the ECG recording. The interval between the start of the P wave to the beginning of the QRS complex is the P-R interval. This interval indicates the amount of time required for the impulse to reach the ventricles. The QRS complex begins as a sharp upward spike (R) and a downward deflection (Q). The QRS complex indicates the depolarization of the ventricles during this interval, the ventricles are in contraction and blood is being ejected from the heart. The time duration known as the S-T segment represents the period between the completion of ventricular depolarization and the initiation of repolarization. The T wave is produced by ventricular repolarization. A normal ECG indicates that the heart is functioning properly. The second wave or the QRS complex occurs just prior to ventricular contraction the third, or T wave occurs just after the ventricles relax.

ANSWER: Write the types of structural proteins with examples.

TYPES OF STRUCTURAL PROTEIN

A: According to structure, proteins are of four types

1: PRIMARY STRUCTURE:

A polypeptide chain having a linear sequence of amino acid due to peptide bond is called primary structure

EXAMPLE: Insulin

2: SECONDARY STRUCTURE:

When a polypeptide chain of amino acid become coiled is called secondary structure of protein

EXAMPLE: keratin

3: TERTIARY STRUCTURE:

When the arrangements of secondary structure turns into three dimensional or super folded structure having peptide, hydrogen, ionic and disulphide bond is called Tertiary structure

EXAMPLE: lysozyme

4: QUATERNARY STRUCTURE:

The association of two or more polypeptide chains into large sized molecules is called Quaternary structure

EXAMPLE: hemoglobin

xi) Mention the causes of Sinusitis or Otitis Media.

ANSWER:

CAUSES OF SINUSITIS

Following reasons that can cause sinus blockage include:

1. Common cold
2. Allergic rhinitis, swelling in the lining of the nose.
3. Nasal polyps, nose lining show small growths.
4. Nasal septum physical deviation of nose, involving a dislocation of the nasal septum.

CAUSES OF OTITIS MEDIA

Otitis media is or infection in the middle ear. Cold, sore throat or upper respiratory infection can cause this disease. This infection usually a result of a failure of the eustachian tube.

xii) Write a note on Allergy or Autoimmune disease.

ANSWER:

ALLERGY

Sometimes, our immune system responds to certain normally harmless substances (eg. pollens, moulds, pet dander, certain drugs, some food, etc, called allergens. Such response is termed as allergy. It is considered to be an exaggerated response of the immune system against the normal substances taken by the WBCs as antigens. The common allergies of man

allergens enter our body, a type of allergin specific antibody, IgE, is produced by the plasma cells. These antibodies upon secretion into the blood bind to the allergin and then to the receptor on the mast cells or basophilic granulocytes, mast cells and basophils upon activation by the allergin complex, release histamine to produce inflammatory symptoms like itching, rashes, runny nose, sore throat, cough, etc. The treatment of allergies usually employs antihistamines and steroids.

IMMUNE DISORDER

An abnormality of the immune system in which the immune system recognizes one or more own tissues as "self". As a result, it initiates an immune response against its own cells. Diabetes mellitus type-1, rheumatoid arthritis, psoriasis, etc are some examples of autoimmune diseases in humans. In such cases, misdirected self-antibodies are produced which destroy their own tissue.

Autoimmune diseases can develop from a variety of factors with no known cause for the onset of the disease. Some suspected sources are bacterial or viral infections, effects from certain drugs, toxins and chemicals. In humans, some of the commonly affected tissues are muscle, blood vessels and connective tissues.

Section C (Detailed - Answer Question) (16)

Attempt any TWO question from this section.

Questions carry equal marks.

Q1. Cytotoxicity? Explain the mechanism of NK cells and TC cells.

Ans:

CYTOTOXICITY

Cytotoxicity refers to the ability of a substance or certain cell, either to kill or cause the death of a target cell. It is one of our protective mechanisms exhibited against virally infected cells, cancer to tumour cells as well as our cells which are being infected by some pathogenic organisms. The process of cytotoxicity could be a consequence of Natural Killer (NK) Cells or Cytotoxic T (Tc) Cells. Although both NK as well as Tc Cells are lymphoid in cell lineage and originate in the bone marrow, yet they differ in their mode of activation against their target cells.

STRUCTURE AND COMPARISON

NK cells are one of the types of lymphocytes involved in detecting and killing virally infected cells as well as cancer cells, while Cytotoxic T cells are another group of immune cells which belong to the Lymphocyte group. W.B.Cs.

2. Natural Killer Cells are like a policeman checking the identity card of a person for verification of his identity, the NK cell checks the cell surface, a conjugate molecule called Major Histocompatibility Complex-I (MHC-I), which serves as the identification card of "self". In the case of intact MHC-I, the NK cell remains inactive. In case of any modification or absence of MHC-I on the target cell, it is activated to destroy the target cell, while when the T-cells encounter infected cells, they recognize them immediately through the antigens expressed on the MHC-I of the infected cells. Their programmed response, is to release some protective proteins or cytokines such as Tumor Necrosis Factor-alpha (TNF- α) and Interferon-gamma (IFN- γ) which act on other immune cells such as macrophage and dendritic cells to initiate immune responses.

3. In Natural Killer Cells the lysis of the target cell is brought about by releasing cytotoxic granules having perforin protein and granzyme. The former forms a pore in the plasma membrane of the target cell and the latter enters the target cell and causes its lysis, while at this stage T-cells like NK Cells, they release perforin proteins and granzyme to cause the lysis of the infected target cell. Unlike NK Cells, the Tc cells then proceed to kill other infected cells to cause their lysis. In this way, they serve as "serial killers".

3 (OR) Describe the structure of Human Heart with the help of a diagram. Explain heart attack and heart failure.

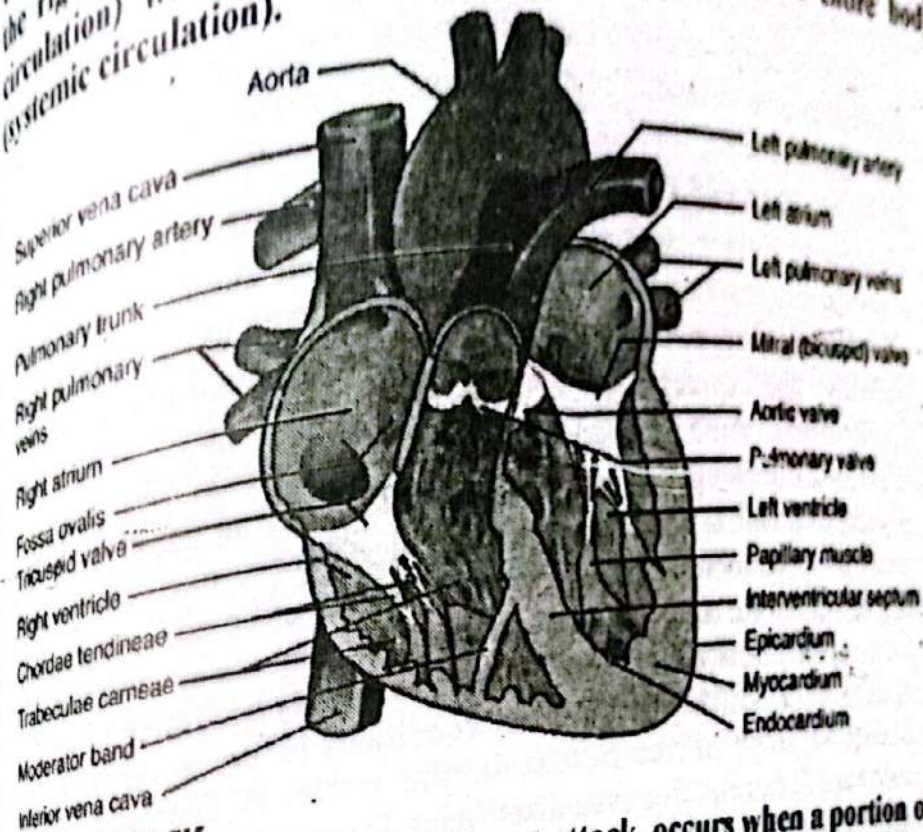
ANSWER:

STRUCTURE OF HEART

The human heart consists of four chambers, two upper thin-walled atria and two lower, thick-walled ventricles. Two large veins, the superior and inferior vena cava enter the right atrium and two pairs of pulmonary veins open into the left atrium. Similarly, two large arteries emerge, one from the right ventricle, the pulmonary aorta and the other from the left ventricle, the systemic aorta. The right and left atria are separated by a membranous interatrial septum, and the right atrium and left ventricle are also separated by a thick muscular interventricular septum. Valves in the heart each consisting of flaps of connective tissue prevent backward flow of blood. The right atrium opens into the right ventricle by an atrioventricular valve (Tricuspid valve). The left atrium opens into the left ventricle by the atrioventricular valve (Bicuspid valve or mitral valve). The AV valves are anchored by strong fibres that prevent them from turning inside out. The pressure generated by the powerful contraction of the ventricles closes the AV valves keeping blood from flowing back into the atria. Semilunar valves are located at the two exits of the heart, the aorta and the pulmonary artery. The aorta leaves the left ventricle and the pulmonary artery leaves the right ventricle.

MODEL PAPER 475 **ZOOLOGY**

the right ventricle. The blood is pumped out into the arteries through the semilunar valves, which are forced open by pressure created by ventricular contraction when the ventricles relax blood starts to flow back towards the heart, closing the semilunar valves, which prevents blood from flowing back into the ventricles. The cavity of the left ventricle is narrower than the right ventricle because of more muscular walls. It is due to the fact that the right ventricle has to pump blood into the lungs only (pulmonary circulation) while the left ventricle pumps blood to the entire body (systemic circulation).



HEART ATTACK

A myocardial infarction also called heart attack, occurs when a portion of the heart muscle dies due to lack of oxygen it is due to the blockage of any of the coronary artery either by thrombus or embolus, the blood supply to some cardiac muscles stops.

HEART FAILURE

Heart failure is characterized by the heart's inability to pump an adequate supply of blood to the body without sufficient blood flow all major body functions are disrupted.

4. Write the general characteristics, evolutionary adaptation and economic importance of Phylum Mollusca or Arthropoda.

ANSWER:

GENERAL CHARACTERS OF PHYLUM MOLLUSCA

Molluscs are soft-bodied animals, but most are protected by a hard shell made of calcium carbonate, slug, squids and octopuses have reduced shells, most of which are internal, or they have lost their shell completely during evolution. All the molluscs are triploblastic, are bilaterally symmetrical, coelomates and protostomes with organ system grade of body organization.

All molluscs have a similar body plan with three main parts, a muscular foot usually used for movement, a visceral mass containing most of the internal organs and a mantle a fold of tissue that drapes over the visceral mass and secretes a shell. Many molluscs feed by using a strap-like rasping organ called a radula to scrape. Reproduction is always sexual, some species have separate sexes others are hermaphrodites, and they all pass through a trochophore larvae stage. Common Molluscs are Unio, Octopus and Pearl oyster.

ECONOMIC IMPORTANCE OF MOLLUSCA:

A variety of molluscs called shellfish, together with crustaceans are still an essential source of food. Their shells are decorative, some members make the valuable pearl.

GENERAL CHARACTERS OF PHYLUM ARTHROPODA

Arthropoda is the largest phylum of the animal kingdom, they are found everywhere on earth wherever life is possible, even in oil wells. Arthropods are bilaterally symmetrical, triploblastic, coelomates and protostomes. The diversity and success of arthropods are largely related to their segmentation, hard exoskeleton, and joint appendages. The body of arthropods is completely covered with cuticles, an exoskeleton, made of protein and chitin. The exoskeleton protects the animals and provides the point of attachment for the muscles that move the appendages. Sexes are usually separate and metamorphosis is a common occurrence. Common Arthropods: Scorpion Prawn, Mosquito, Honey bee House fly etc.

Arthropods exist in the harshest conditions in the world from very cold places to some of the hottest in the world. A Scorpion, which is an arthropod, can survive even after being frozen solid.

ECONOMIC IMPORTANCE OF ARTHROPODS:

Arthropods are of great economic importance, the predominant of arthropods, insects not only help in pollination but are also predators of plant pests. Many cause diseases in plants and animals, by transmitting bacteria and viruses. Human beings are responsible for the transmission of Trypanosoma, plasmodium and germs of cholera. Arthropods are an important source of food for many animals and carnivorous plants. Seafood that is not fish or molluscs is generally arthropods.

5. Explain main components of the respiratory system of man and draw neat and labelled diagram.

ANSWER:

MAIN COMPONENTS OF THE RESPIRATORY SYSTEM

NOSE

The nose is the primary upper respiratory organ in which air enters into and exits from the body. Cilia and mucus line the nasal cavity and traps

germs and foreign particles that enter in through the nose. In addition, air that passes through the nasal cavity is humidified and moistened. The nasal septum divides the nose into two narrow nasal cavities; one area is responsible for smell and the other area is responsible for respiration. Within the walls of the nasal cavity there are frontal, nasal, ethmoid, maxillary, and sphenoid bones. Cartilage helps form the shape of the nose.

PHARYNX

Besides the nose, air can enter into the lungs through the mouth. The pharynx is a tubular structure, positioned behind the oral and nasal cavities, that allows air to pass from the mouth to the lungs. The pharynx contains three parts: The nasopharynx, which connects the upper part of the throat with the nasal cavity; the oropharynx, positioned between the epiglottis and the soft palate; and the laryngopharynx, located below the epiglottis.

LARYNX

From the pharynx, air enters into the larynx, commonly called the voice box. The larynx is part of the upper respiratory tract that has two main functions: a passageway for air to enter into the lungs, and a source of vocalization. The larynx is made up of the hyoid bone and cartilage, which helps regulate the flow of air. The epiglottis is a flap-like cartilage structure contained in the larynx that protects the trachea against food aspiration.

BRONCHI

The bronchi allow the passage of air to the lungs. The trachea is made of C-shaped ringed cartilage that divides into the right and left bronchus. The right main bronchus is shorter and wider than the left main bronchus. The right bronchus is subdivided into three lobar bronchi, while the left one is divided into two lobar bronchi.

LUNGS

The lungs are spongy, air-filled organs located on both sides of the chest cavity. The left lung is divided into a superior and inferior lobe, and the right lung is subdivided into a superior, middle, and inferior lobe. Pleura, a thin layer of tissue, line the lungs to allow the lungs to expand and contract with ease.

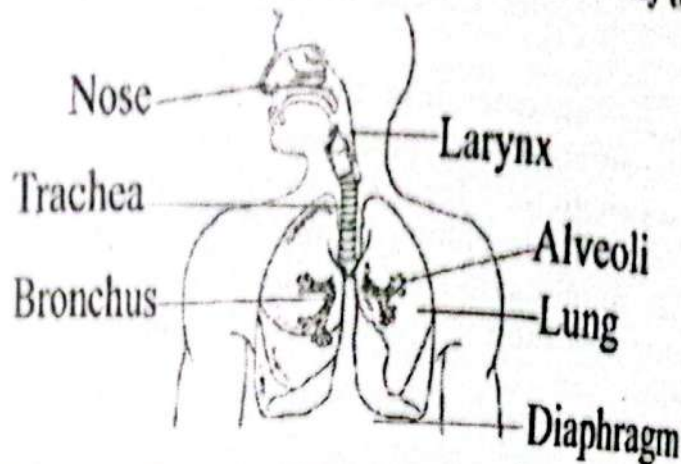
Respiration is the primary function of the lungs, which includes the transfer of oxygen found in the atmosphere into the blood stream and the release of carbon dioxide into the air.

ALVEOLI

The average adult has about 600 million alveoli, which are tiny grape-like sacs at the end of the respiratory tree. The exchange of oxygen and carbon dioxide gases occurs at the alveolar level. Although effort is required to inflate the alveoli (similar to blowing up a balloon), minimal effort is required to deflate the alveoli (similar to the deflating of a balloon).

DIAPHRAGM

The diaphragm is a muscular structure located between the thoracic and abdominal cavity. Contraction of the diaphragm causes the chest or thoracic cavity to expand, which occurs during inhalation. During exhalation, the release of the diaphragm causes the chest or thorax cavity to contract.



NEW MODEL QUESTION PAPER

Annual Examination 2023

ZOOLOGY (THEORY)

CLASS XI (SCIENCE PRE-MEDICAL GROUP)

Total Time: 2 Hours

Total Marks: 40

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

Time: 15 Minutes

(8 Marks)

NOTE: This Section consists of 16 MCQs and all are to be answered. Each MCQ carry 0.5 marks.

1. Choose the correct answer from the given options:

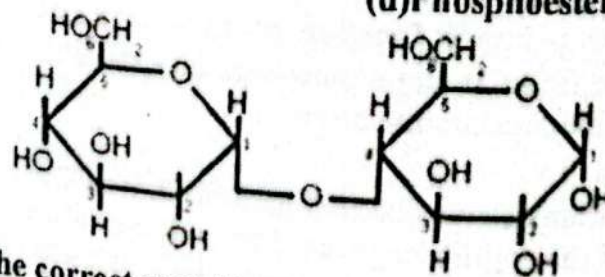
1. The Linkage between these two monosaccharides is formed by:

(a) Glycosidic bond ✓

(b) Peptide bond

(c) Ester bond

(d) Phosphoester bond



2. This is not the correct matching pair:

(a) Pinacocytes Collar cells ✓

(b) Euplectella Porifera

... Gametes formation

(d) Parazoa

... of the enzyme is due to:

- (b) Hydrogen bonding
- (d) High molecular weight
- (b) Carboxylase
- (d) Dehydrogenase

... of Carbon dioxide by RBCs requires an enzyme.

... of the characteristic of Deuterostome:

- (b) Blastopore becomes anus
- (d) Radial cleavage

... development

... is functionally similar to Keratin protein:

- (b) Amylopectin
- (d) Glycogen

... substance that inhibits the secretion of acid in the

(b) Lipase

- (c) Protease
- (d) Serotonin

... detect the blood pressure:

- (b) Chemoreceptors
- (d) Thigmoreceptors

... animals having pseudo coelom with sexual dimorphism

- (b) Annelids
- (d) Round Worms

... H^+ concentration in the blood causes:

... the breathing rate

... the breathing rate

... the breathing rate

... of more CO_2 to the tissues

... most closely resembling to the:

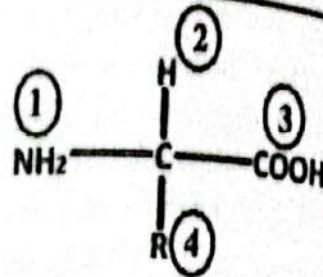
- (b) Urine
- (c) Plasma
- (d) Interstitial fluid

... of the hypothalamus that increases the temperature of

... fever during an infection is:

- (b) Vasopressin
- (d) Somatostatin

... in the given amino acid diagram combine to form a



- (a) 1 and 2 (b) 1 and 3 ✓ (c) 2 and 3 (d) 2 and 4
14. The effect of increasing the substrate concentration of an enzyme controlled reaction is:
- (a) Competitive inhibition decreased; Non-competitive inhibition increase
 (b) Competitive inhibition decreased; Non-competitive inhibition no change ✓
 (c) Competitive inhibition increased; Non-competitive inhibition decrease
 (d) Competitive inhibition shows no change; Non-competitive inhibition is an increase
15. Hepatic portal system is formed between:
- (a) Oral cavity and Spleen (b) Liver and Small intestine
 (c) Kidney and stomach (d) Heart and kidney
16. Vaccine given after dog biting develops this type of immunity:
- (a) Active Natural Immunity (b) Active Artificial Immunity ✓
 (c) Passive natural Immunity (d) Passive Artificial Immunity

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 10x2=20)

NOTE: Attempt any FIVE questions from PART A (Reasoning questions) and five from PART B (Non- Reasoning questions).

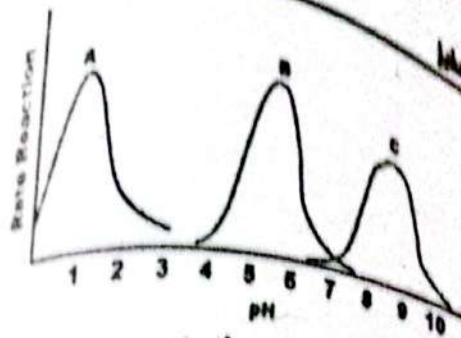
(REASONING QUESTIONS)

2. (i) Why do a double circuit heart better than a single circuit heart?

ANSWER: In a Double Circulatory system, blood is pumped twice. Once from the heart to the lungs and then from heart to rest of the body. It helps in maintaining pressure of the blood unlike Single Circulatory system where blood is pumped once and pressure of the blood is not maintained.

2. (ii) Why are water molecules amphoteric and buffer?

ANSWER: (See Question No. 2(iv), 2023)



The above graph represents the effect of pH on the activity of three enzymes A, B and C. write the name of these enzymes; why do these enzymes show their optimum activity at specific pH?

ENZYMES	NAME OF ENZYMES
A	Pepsin
B	Salivary amylase
C	Alkaline phosphatase

Enzyme has a very narrow range of it works at optimum pH. The pH has marked effect on the charges of amino acid side chains which form primary and quaternary structures of the protein (enzyme). Due to the change in shape rate of reaction falls down. For example, pepsin works at acidic condition i.e. its optimum pH is 1.4 but works in the range of 1-3. Similarly, amylase is the enzyme released from the pancreas that works at highly alkaline i.e. 8.5 pH.

Why does excess use of spicy food; use of aspirin cause peptic ulcers?

Ans: Because spicy food and aspirin affect the stomach in that HCl of the gastric juice etches away the protective layer of mucous lining cause painful conditions and peptic ulcers.

What does the pre-capillary sphincter; regulates the flow of blood in it?

Ans: The flow of blood in capillaries is regulated by tiny rings of smooth muscles pre-capillary sphincters. That surrounds the junctions of arterioles and capillaries. These sphincters open and close in response to local changes that signal the needs of nearby tissues for the accumulation of carbon dioxide, lactic acid or other cellular products the need for increased blood flow to the tissues. These signals regulate the pre-capillary sphincters as well as the muscles in nearby arterioles to increase blood flow through the capillaries.

What disease has inflammation in the sinus and failure in the Eustachian tube are these diseases? Write symptoms of any one disease.

Ans: If a person has inflammation in the sinus it means that he is suffering from Sinusitis. The symptoms of this disease are following

Symptoms of SINUSITIS

2. Fever

3. Cough

4. Tooth pain

6 fatigue.

If a person suffers from a disease in which the Eustachian tube is a failure so this disease is known as Otitis media. The symptoms of this disease are following

SYMPTOMS OF OTITIS MEDIA

1. Fever

4. Ear pain

2. (vii) Why is artificial sweetener not harmful to diabetic patients?

ANSWER: (See Question No. 2(iii), 2023)

2. (viii) Reptiles and birds are vertebrates; why do birds can fly while reptiles can't fly?

ANSWER: Because in birds have a different respiratory system which is supplemented by air sacs. These air sacs supply oxygenated air to the lungs. Other reptiles don't have air sacs in their lungs that's why they can't fly

(NON-REASONING QUESTIONS)

2. (ix) Mention the economic importance of Cnidaria OR Mollusca.

ANSWER:**ECONOMIC IMPORTANCE OF CNIDARIA**

Coral reefs which are usually restricted to warm shallow water provide a heaven to many marine species. Jewelry and other decorative items are carved from red corals. Haakeem uses red corals named MARJAN in preparing eastern medicine.

ECONOMIC IMPORTANCE OF MOLLUSCA

A variety of molluscs called shell fish, together with crustaceans are still an essential source of food. Their shells are decorative, some members make valuable pearl.

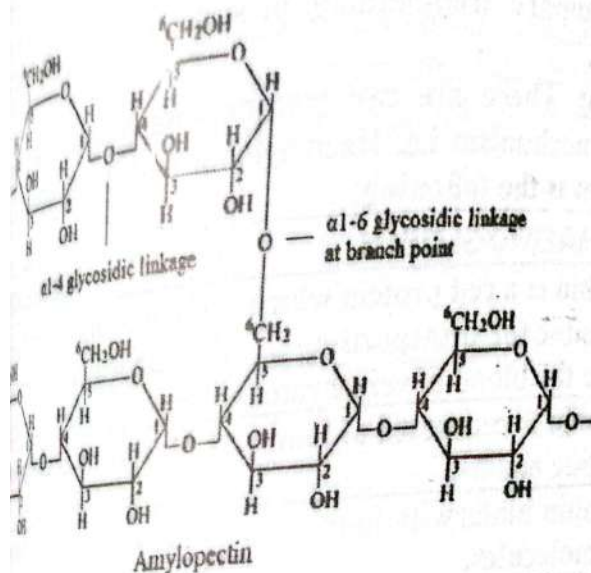
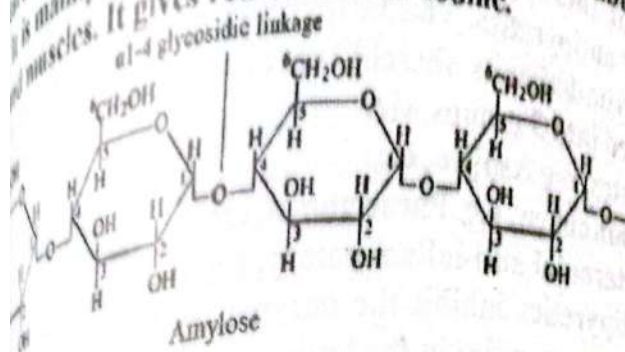
2. (x) Compare the inhibitor and activator with examples.

ANSWER:

INHIBITOR	ACTIVATOR
Enzyme inhibitor refers to a molecule that binds to an enzyme, decreasing the activity.	Enzyme activator refers to a molecule that binds to an enzyme, increasing the activity
enzyme activators can be either proteins, peptides, lipids, small organic molecules or ions	the two main types of enzyme inhibitors are reversible and irreversible inhibitors
Some examples of enzyme activators are calcium and magnesium ions, calmodulin, EDTA and EGTA	enzyme inhibitors are N-ethylmaleimide, DFMO, DFP, and most of the pharmaceutical drugs.

on animal starch and its functions.

(glycogen) is a polymer of glucose. Its molecular structure is similar to starch but found in animal therefore it is commonly called animal starch. It is mainly found in bacteria, fungi, in animals abundantly in muscles. It gives red color with Iodine.



stored reservoir of energy in animals.

acts as a glucose storage facility in the body.

branched and is stored in the liver of animals and is used for metabolic need.

exists between cartilaginous fishes and bony fishes.

Question No. 2(vii), 2023)

the LUB-DUB sounds initiated in the heart; Write their functions in monitoring the heartbeat.

ANSWER

The heart rate, systole lasts about 0.3 seconds and diastole lasts about 0.4 seconds. When the heart beats the familiar LUB-DUB sound is produced. The LUB sound is caused by vibrations of the heart close, the LUB is caused by vibrations of

the heart when the atrioventricular valves close, and this vibration occurs due to the closing of the semilunar valves. This heart sound is significant for clinical diagnoses purpose.

2. (xiv) Write a note on antipyretic therapy and its uses.

ANSWER:

ANTIPYRETIC THERAPY AND ITS USES

Antipyretic therapy consists of those drugs which reduce or are called antipyretics. The antipyretics inhibit the prostaglandin synthesis by the hypothalamus thereby reducing fever. The antipyretics are categorized into 3 groups, viz.

- i) Salicylates, e-g Aspirin, etc.
- ii) Acetaminophen, e.g. Paracetamol, etc.
- iii) Non-steroidal anti-inflammatory, e.g. Ibuprofen, etc.

Most antipyretics inhibit the enzyme, cyclooxygenase which reduces the level of prostaglandins in the brain.

2. (xv) Compare iron-binding proteins that involve in the transport mechanism.

ANSWER: There are two iron-binding proteins involved in the transport mechanism i.e. Haemoglobin and Myoglobin. The comparison among them is the following.

HAEMOGLOBIN	MYOGLOBIN
Hemoglobin is a red protein which is responsible for transporting oxygen in the blood of vertebrates.	Myoglobin is a red protein which contains haem which carries and stores oxygen in the muscle cells.
Hemoglobin is composed of four polypeptide chains.	Myoglobin is composed of a single polypeptide chain.
Hemoglobin binds with four oxygen molecules.	Myoglobin only binds with a single oxygen molecule.
Hemoglobin takes oxygen from lungs and transports to the rest of the body.	Myoglobin stores oxygen in the muscle cells and releases when needed.
Hemoglobin is found in the blood stream.	Myoglobin is found inside muscle cells.

2. (xvi) Relate hepatitis and jaundice with the functions of the liver.

ANSWER: Hepatitis and jaundice are two major medical conditions affecting the liver, having different causing factors. The main difference between Jaundice and Hepatitis is that jaundice occurs as a result of an increase in bilirubin levels in the blood whereas hepatitis is a result of an invasion of the liver tissue by hepatitis viral entities.

FUNCTIONS OF LIVER

The function of the liver is to purify blood coming from the system, secretion of bile, detoxification of chemicals, excretion of drugs, and in protein production.

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: $6 \times 2 = 12$)

Attempt any three (2) questions from this section. Questions carry equal marks:

1. Describe the characters and uses of acylglycerols, lipids, and terpenoids.

Lipids are organic compounds that contain hydrocarbons which are the basis for the structure and function of living cells. The Lipids are a diverse group of compounds, including fats, oils, steroids, waxes, and other compounds, which are related more by their physical than by chemical properties.

CHARACTERISTICS OF LIPIDS

Lipids may be either liquids or non-crystalline solids at room temperature.

Fats and oils are colorless, odorless, and tasteless.

Lipids are energy-rich organic molecules.

Lipids are insoluble in water.

Lipids are soluble in organic solvents like alcohol, chloroform, acetone, benzene, etc.

Lipids are electrically neutral.

Glycerols (Fats) have high proportions of saturated fatty acids.

Triglycerols (Oils) have high proportions of unsaturated fatty acids.

ACYLGLYCEROLS

Lipids provide energy for different metabolic activities and is very important chemical energy, twice in the amount of energy content than carbohydrates. It is estimated that a person of average size contains about 144 x 10³ KCal of energy.

PHOSPHOLIPIDS

Phospholipids are present in all living cells and form membrane. They are important for various functions such as regulation of cell permeability and other processes. Properties of cell membrane depend on its lipid component.

USES OF TERPENOIDS

Terpenoid is a large and important class of lipids, made up of units (C_5H_8). Terpenes, steroids, carotenoids and prostaglandins are of Terpenoids. These are found in cell membrane as cholesterol like chlorophyll, fragrance as menthol etc.

Q 1 (OR) Describe the main structural features and components of the human respiratory system.

ANSWER:**STRUCTURAL FEATURES OF HUMAN RESPIRATORY SYSTEM**

The human respiratory system consists of a complex set of organs and tissues. This system carries air from the environment to the lungs and vice versa. The human lungs are the paired, soft, spongy and highly vascularized organ in structure, located inside the thorax (chest) and are the primary organ of the respiratory system. However, the external nostril (nose), pharynx, larynx, and trachea are the other respiratory accessory organs.

The human respiratory system is anatomically differentiated into upper and lower respiratory air passages. The upper airways consist of the nasal cavity, and Pharynx while the lower air passage is the trachea, bronchial branches and the pair of lungs. The respiratory system starts from the nose. During respiration, air enters the nose through the opening called the external nostrils or nares. The nose is the only part of our respiratory system. Interiorly nose consists of the nasal cavity, a bony partition that separates the nasal cavity into nasal chambers, the midline cartilaginous nasal septum. The hairs of the nasal cavity and ciliated cells trap the dust particle and decontaminate the gas. The nasal chambers run forward through internal nares into a tube called the pharynx behind the mouth. This cavity is called the pharynx. The muscular pharynx is long about 13 cm, which serves as a common pathway for the respiratory and digestive systems. The external air enters the nasal cavity to the pharynx. Then air moves from the pharynx to the larynx, nasopharynx and larynx from the laryngopharynx. Nasal air enters the pharynx through the nose, but it may also enter through the mouth if the nasal sections are blocked. Larynx or voice box is the component of the lower respiratory tract located at the entrance of the trachea. The flap-like pairs of tissues called vocal cords, present in the larynx, produce sounds when air is expelled from it. From the larynx, air enters the trachea or windpipe. Trachea is the main airway to the lungs. It is a fibro-cartilaginous tube about 10-11 cm long. Trachea has 16 to 20 "C" shaped cartilaginous rings, providing rigidity and preventing the trachea from collapsing. Before entering the lungs, the trachea divides into two bronchi. A singular right bronchus is broader, shorter,

lighter than the left bronchus. Each bronchi branches into smaller bronchioles. These bronchioles form the bronchial tree which terminates at the alveoli, the microscopic air sacs where gases are exchanged between air and blood.

FUNCTIONS OF THE RESPIRATORY SYSTEM

The respiratory system performs the following major functions:

OXYGEN SUPPLIER:

It provides a continuous supply of oxygen to all tissues.

WITHDRAWAL:

Removal of by-product, carbon dioxide (CO_2).

CONSERVATION OF GAS:

The mechanism of gas exchange between the internal and external environment of the body is regulated through respiratory surfaces.

HUMIDIFIER:

The respiratory system performs as a humidifier. It has the capability to humidify and keep the air warm which is inhaled from the external environment.

Q 2. Describe the general characters and evolutionary adaptations of Phylum Arthropoda OR Phylum Annelida.

ANSWER: (See Question No. 4, 2024)

Q 3. Write the cause, effects, and prevention of Atherosclerosis, Myocardial infarction, Angina pectoris, Hypertension, and coronary thrombosis.

ANSWER:

CAUSE, EFFECTS, AND PREVENTION OF ATHEROSCLEROSIS

It is a disease of the arterial wall (intima) which loses its elasticity gradually as its inner layer thickens causing the narrowing of the artery and consequently impairing the blood flow. The narrowing is due to the formation of fatty lesions called atheromatous plaques in the inner lining of the arteries. These plaques consist of low-density lipoproteins (LDL or cholesterol and protein) decaying muscle cells, fibrous tissue, clumps of blood platelets and sometimes calcium. The arteries become extremely hardened and the disease is called arteriosclerosis or simply the hardening of the arteries.

FACTOR CAUSING ATHEROSCLEROSIS

The possible causes of atherosclerosis are smoking, hypertension, male gender, obesity, physical inactivity, a high serum cholesterol level, severe diabetes, family history of arterial disease and possibly an anxious or depressive state. The risk of atherosclerosis increases with age.

CAUSE, EFFECTS, AND PREVENTION OF MYOCARDIAL

INFARCTION

A myocardial infarction also called a heart attack, occurs when a portion of the heart muscle dies due to lack of oxygen it is due to the blockage of any of the coronary arteries either by a thrombus or embolus, the blood supply to some cardiac muscles stops.

CAUSE, EFFECTS, AND PREVENTION OF ANGINA PECTORIS

The strain on the heart by increasing resistance to blood flow. Although the heart may enlarge in response to this added demand, and its own blood supply may not increase proportionately. The heart muscle is then inadequately supplied with blood, especially during exercise. Lack of sufficient oxygen to the heart can cause chest pain and a radiating pain in the left arm called angina pectoris.

CAUSE, EFFECTS, AND PREVENTION OF HYPERTENSION

When the mean arterial pressure is greater than the upper range, women of any age are considered to have hypertension if their blood pressure reading is 160/95 or above. For a man under age 45, a reading above 140/95 is considered hypertensive. Hypertension is called a "Silent Killer", because the affected individuals may show no outward symptoms until a stroke or heart attack occurs. Several factors such as heredity, higher intake of salts in diet, smoking, obesity and disorders of the kidney or adrenal gland are responsible for hypertension.

CAUSE, EFFECTS, AND PREVENTION OF CORONARY THROMBOSIS

The formation of a blood clot (thrombus) within an intact blood vessel is initiated by atherosclerotic plaques. The plaques when destroying the endothelium of the blood vessels, and platelets gather at the damaged site to initiate the process of the clot. Thrombus and embolus are potentially life-threatening blood vessel problems that increase the risk of having a heart attack or stroke. Thrombus happens when a blood clot or thrombus grows in blood vessels. This can reduce blood flow. An embolus is any foreign material that travels within the body. If it becomes stuck and severely blocks the flow of blood. The issue is called an embolism.

Q.3. (OR) What is an autoimmune disorder? Explain the role of T-cells and B-cells in transplant rejection.

ANSWER:**AUTOIMMUNE DISORDER**

It is the abnormality of the immune system in which the immune system fails to recognize one or more own tissues as "self". As a result, it initiates an immune response against its own cells. Diabetes mellitus type-1, rheumatoid arthritis, psoriasis, etc are some examples of autoimmune

In such cases, misdirected self-antibodies are produced by humans. In such cases, misdirected self-antibodies are produced by their own tissue. Some diseases can develop from a variety of factors with no known cause for the onset of the disease. Some suspected sources are bacterial or viral infections, effects from certain drugs, toxins and in humans, some of the commonly affected tissues are muscle, blood vessels and connective tissues. Symptoms are malaise (sick feeling), low fever, fatigue, and rashes on different areas of the body. Physicians prescribe antihistamine therapy to patients with skin rashes.

It is already learnt that during the allergic reaction, mast cells and histamine to induce inflammation in order to reduce the response, antihistamine drugs are used to block the receptors of mast cells, muscles, endothelial cells, etc. and the allergy is controlled. Itching, sneezing, runny nose, and skin rashes are reduced in this way.

T-CELLS IN TRANSPLANT REJECTION:

In case of transplant of solid organs from individuals of the same allograft, the major threats are the T cells of the recipient. Dendritic cells of the donor tissue migrate into the recipient tissue where they are recognized by the recipient's TH cells. Once activated, TH cells proliferate and activate NK cells to attack the grafted tissue.

B-CELLS IN TRANSPLANT REJECTION:

Activated T cells signal the B-cells to transform into specific antibody-producing Plasma Cells. The antibodies thus produced play a very role in destroying the grafted tissue.

USUIFS MODEL QUESTION PAPER 1

Annual Examination 2023

ZOOLOGY (THEORY)

CLASS XI (SCIENCE PRE-MEDICAL GROUP)

Time: 2 Hours

Total Marks: 40

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

(8 Marks)

10 Minutes

This Section consists of 16 MCQs and all are to be answered. Each MCQ carry 0.5 marks.

1. Choose the correct answer from the given options.
(i) Most important organic compound of the cell which carry out virtually all of the cell's activities.

- (a) Protein
(b) Carbohydrates
(c) Nucleic acid ✓
(d) Lipids

(ii) Competitive inhibitors block the entry of substrate into the site of an enzyme. On which of the following properties of an enzyme does this primarily depend?

- (a) The ability of an enzyme to form a template for holding and moving molecules. ✓
(b) The enzyme's ability to stretch reactants and move them to transition state.
(c) The enzyme providing an appropriate microenvironment conducive to a reaction's occurrence
(d) The enzyme forming covalent bonds with the reactants.

(iii) Water movement through sponge?

- (a) Ostia → Spongocoel → Osculum. ✓
(b) Spongocoel → Osculum → Ostia.
(c) Ostia → Osculum → Spongocoel.
(d) Osculum → Spongocoel → Ostia.

(iv) The thecodont condition refers to:

- (a) Presence of two sets of teeth during life time
(b) All alike teeth
(c) Sharp, pointed teeth
(d) Embedded teeth

(v) Where are similar valves to be found in the mammalian heart?

- (a) Where blood goes from Atria to ventricles
(b) On the right side of the heart only.
(c) Where the Pulmonary vein attach to the heart.
(d) At the places where blood leaves via the aorta and pulmonary arteries. ✓

(vi) It is not involved in antigen presentation:

- (a) Macrophage
(b) Dendritic cell
(c) Neutrophil
(d) Both a and b ✓

(vii) The tiny air sacs present in human lungs is called:

- (a) Alveoli ✓
(b) Bronchus
(c) Bronchioles
(d) All of the above

(viii) The chemical substance that inhibits the secretion of acid in stomach is:

- (a) Gastrin ✓
(b) Lipase
(c) Protease
(d) Serotonin

(ix) Increase in H^+ concentration in the blood causes:

1. Increase in the breathing rate ✓
 2. No change in the breathing rate
 3. Transport of more CO_2 to the tissues
 4. The largest invertebrate of the world is:
 (a) Whale (b) Elephant (c) Giraffe (d) Squid ✓
 5. Amoebiasis is caused by:
 (a) Virus (b) Entamoeba gingivalis
 (c) Fungi (d) Entamoeba histolytica ✓
 6. Water vascular system is found in:
 (a) Cockroach (b) Starfish ✓ (c) Earthworm (d) Sponge
 7. The air always present in the human lungs is called:
 (a) Tidal Volume (b) Vital Volume
 (c) Residual Volume (d) Total Lung Capacity
 8. Human sex hormones are synthesized by this:
 (a) Protein (b) Triglyceride (c) Steroid ✓ (d) Carbohydrate
 9. Antibodies are produced by:
 (a) B Cells ✓ (b) T Cells (c) Macrophages (d) Neutrophils
 10. RBCs are produced in:
 (a) Kidney (b) Heart (c) Brain (d) Bone Marrow ✓
 11. SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 10x2=20)

NOTE: Attempt any FIVE questions from PART A (Reasoning questions) and five from PART B (Non-Reasoning questions).

(REASONING QUESTIONS)

Q.2 (i) Why circulatory system of arthropods is called open type?
 ANSWER: Because in Arthropods blood not only travels through vessels but also empties into the haemocoel, where it bathes the internal organs directly this arrangement.

Q.2 (ii) How are saturated acylglycerols different from unsaturated acylglycerols?

ANSWER: Saturated Acylglycerol contains no double bond while unsaturated acylglycerols contains unsaturated fatty acids i.e. they contain one or more than one double bond between carbon atom ($C=C$). Saturated acylglycerols melt at higher temperature than unsaturated acylglycerols.

Q.2 (iii) Why enzymes are called temperature sensitive?

ANSWER: (See Question No. 2 [a (iii)], 2024)

Q.2 (iv) List out the stages in holozoic nutrition

ANSWER:

STAGES OF THE HOLOZOIC NUTRITION PROCESS

The Holozoic nutrition consists of the following five stages:

1. Ingestion
2. Digestion
3. Absorption
4. Assimilation
5. Excretion.

Q.2 (v) A person has inflammation in the sinus and failure in the Eustachian tube; what are these diseases? Write symptoms of any one disease.

ANSWER: (See Question No. 2(vi) New Model Paper)

Q.2 (vi) Why rate of breathing increases in human beings?

ANSWER: The high concentrations of carbon dioxide (CO_2) and H^+ ions in the blood are stimuli to increase the rate of breathing. The lower part of the brain medulla oblongata is responsible to detect any change in carbon dioxide and H^+ concentration in cerebrospinal fluid. So this impulse (messages) is sent to inter-costal muscles and diaphragm to increase the breathing rate, accordingly.

Q.2 (vii) Why monosaccharides are sweet whereas polysaccharides are not?

ANSWER: Monosaccharides are sweet in nature but polysaccharides are not because the polysaccharides do not bind to the taste receptors of the tongue as the monosaccharides.

Q.2 (viii) Why sponges place in animal kingdom, when it lacks tissue organization?

ANSWER: Because sponges do not have chlorophyll and cannot prepare their own food rather they capture different organisms for nutrition. Most sponges are hermaphrodites whereas in a few members sexes are separate. Sponges can reproduce asexually by fragmentation or by budding or gemmules. During sexual reproduction eggs and sperm are formed by amoebocytes.

(NON-REASONING QUESTIONS)

Q.2 (ix) Differentiate between invertebrate chordates and vertebrates.

ANSWER:

INVERTEBRATE CHORDATES	VERTEBRATES
Chordates refer to an animal phylum that contains a notochord and a dorsally situated central nervous system.	Vertebrates refer to a large group of animals, which consist of a backbone.
Chordates consist of both primitive and advanced chordates.	Vertebrates consist of advanced chordates.
Chordates possess a notochord at some point of their life.	Vertebrates possess a notochord as well as a brain case.
Invertebrate chordates do not have a vertebral column.	Vertebrates have a vertebral column surrounding the nerve

vertebrate chordates lack a cartilaginous or bony skeleton.
vertebrate chordates are

cord.

Vertebrates contain a cartilaginous or bony skeleton.
Vertebrate possess limbs or fins.

ZOOLOGY

(i) Different between the activator and inhibitors.

ANSWER: (See Question No. 2(x) New Model Paper)

(ii) Write note on condensation.

ANSWER:

CONDENSATION

Condensation reaction is the formation of a larger molecule from two smaller ones, which also form another, smaller molecule by losing functional groups in order to join together. Formation of water is an example of condensation process.

EXAMPLE: Water (H_2O) is formed from one molecule losing hydrogen (H) and another losing a hydroxyl group ($-OH$)

Condensation reactions are used to make crucial large molecules called macromolecules in the body, such as carbohydrates, lipids, and proteins. Carbohydrates are simple sugars such as glucose, which is used for energy in the body.

(iii) Differentiate between cartilaginous fishes and Bony fishes.

ANSWER: (See Question No. 2(vii), 2023)

(iii) What are LUB and DUB?

ANSWER:

LUB AND DUB

LUB is the sound of the heart heard as the valves of the heart close. The first sound is LUB produced when the atrioventricular valves i.e. tricuspid and bicuspid valves close at the start of ventricular systole, while DUB is the second sound produced at the beginning of ventricular diastole as the pulmonary and aortic semilunar valves close.

(iv) What is phagocytosis? Name some WBCs acting as phagocytes.

ANSWER:

PHAGOCYTOSIS:

Phagocytosis is the process through which some specific types of WBCs engulf and digest the foreign particle or microorganism.

WBCs THAT ACT AS PHAGOCYTES

Following WBCs act as phagocytosis

1. Neutrophils
2. Macrophages (monocytes)

and dendritic cells

(v) Differentiate between Artery and Vein.

ANSWER:

DIFFERENCE BETWEEN ARTERY AND VEIN

ARTERY	VEIN
Arteries carry blood away from the heart to the tissues of the body	Veins carry blood from the tissues of the body back to the heart
Arteries are usually positioned deeper within the body	Veins are usually positioned closer beneath the surface of the skin
Arteries are more muscular than veins, which helps in transporting blood that is full of oxygen efficiently to the tissues	Veins are less muscular than arteries, but contain valves to help keep blood flowing in the right direction, usually toward the heart
Arteries would generally remain open if blood flow stopped, due to their thick muscular layer.	Veins would collapse if blood flow stops.

Q.2 (xvi) Enlist the role of large intestine.

ANSWER:

ROLE OF LARGE INTESTINE

1. Reabsorption of water and nutrients
2. Synthesis of some vitamins,
3. Reabsorption of some electrolytes such as sodium and chloride
4. Formation of feces and then its egestion outside the body.
5. Maintaining a resident population of over 500 species of bacteria that are most beneficial to digestion.

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: 6x2=12)

Note: Attempt any three (2) questions from this section.

All questions carry equal marks:

Q.1 What is amino acid? Explain peptide linkage formation.

ANSWER:

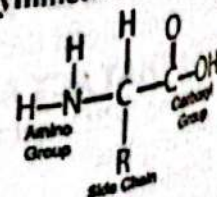
AMINO ACID

Amino acids are organic compounds which contain at least one amino group ($-NH_2$) which works as a base and one carboxylic acid, which works as an acid, both are chemically bonded to an asymmetric carbon.

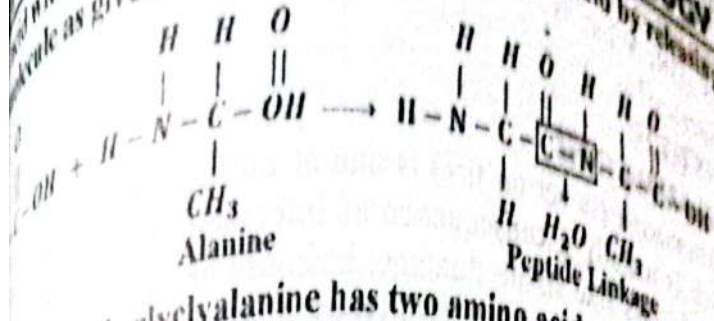
These biomolecules are involved in several biological and chemical functions in the human body and are the necessary ingredients for the growth and development of human beings. There are about 300 amino acids that occur in nature.

PEPTIDE LINKAGE FORMATION

The protein or polypeptide chain is formed by linking amino acids by a peptide bond. The peptide bonds are formed by linking the amino group of

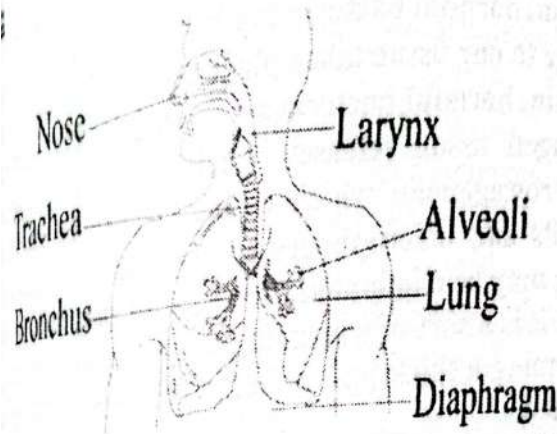


cond with the carboxylic acid of another amino acid by releasing water molecule as given below



A dipeptide is glycylalanine has two amino acid subunits called. A dipeptide has an amino group at one end and a carboxylic group at the other end of the molecule. In this way, both reactive parts are available for further peptide linkage to produce tri, tetra, Penta and so on polypeptide chains.

Draw a labelled diagram of human respiratory system?



Draw evolutionary adaptations of phylum annelid.

EVOLUTIONARY ADAPTATIONS OF PHYLUM ANNELID

Annelids are supposed to have evolved from a primitive flat worm like planaria.

FOOD: They have a complete digestive system with both mouth and anus, complete system.

RESPIRATION: A respiratory system is not found, diffusion is sufficient for gases exchange.

CIRCULATION: A circulatory system with blood confined to the heart and blood vessels.

EXCRETION: Excretory organs are nephridia which are found in most annelids.

NERVOUS SYSTEM: The nervous system consists of simple ganglionic brain in the head. The structure of human heart.

ANSWER: (See Question No. 3(OR), 2023)**Q.3 (OR) What is inflammation? Why it is developed and controlled?****ANSWER:****INFLAMMATION**

Inflammation (to set on fire) is one of the non-specific immune response which is usually a consequence of infection. However, it may be caused by other factors like tissue damage, irritants or autoimmune disorders. It is classically categorized into two types, acute and chronic. The acute inflammation is a short-term inflammation characterized by redness, heat, swelling and loss of the function of the affected organ. The chronic inflammation is of long-term duration and is usually a consequence of diseases like Cardiovascular Disorder (CVD), allergy, etc.

DEVELOPMENT AND CONTROL OF INFLAMMATION

Whenever damage to our tissue takes place either through an accident like a pinprick to the skin, harmful bacteria are introduced into our tissues. Whenever damage to our tissue takes place either through an accident like a pinprick to the skin, harmful bacteria are introduced into our tissues. As a result, the damaged tissue releases specific complex proteins such as Histamine and Prostaglandin which dilate the nearby blood vessels to attract neutrophils and macrophages to the site of infection so that the intruder bacteria may be eliminated through phagocytosis. It may result in an abscess. Since it is a sort of war going on in the infected tissue, our cells may die forming a thick paste-like, white, yellow, orange or green fluid, Pus is formed.

Normally the immune cells respond to foreign cells, or particles such as bacteria, viruses, etc. only. However, in a few cases like autoimmune disorders such as Rheumatoid Arthritis, the immune cells respond against their self (own) cells/antigens. The exact reasons for this misdirection are not certain in many specific cases.

YOUSUIFS MODEL QUESTION PAPER 2**Annual Examination 2023****ZOOLOGY (THEORY)****CLASS XI (SCIENCE PRE-MEDICAL GROUP)****Total Time: 2 Hours****Total Marks: 40****SECTION 'A' (COMPULSORY)****MULTIPLE CHOICE QUESTIONS (M.C.Qs)****Time: 15 Minutes****(8 Marks)**

This Section consists of 16 MCQs and all are to be answered.
Each MCQ carry 0.5 marks.

Choose the correct answer from the given options:

Amino acids have same formula except:
(b) Hydroxyl group
(d) Amino group

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(b) Hydroxyl group
(d) Amino group

According to the induced fit hypothesis of enzyme catalysis, which
following is correct?
Binding of the substrate depends on the shape of active site.
Enzyme change their structure when activators bind to the
Competitive inhibitor can outcompete the substrate for the active
Binding of the substrate changes the shape of the enzyme's

Which of the following is the odd one out?

- (b) Brittle star ✓
- (d) Sea urchin

Which of the following are meant for cutting are:

- (b) Canines
- (d) Molars

Which of the following is the S.A node in humans:

Major contributor to heart attacks ✓

What block conductance between the bundle branches and the
fibers.

What have a negative effect on peripheral resistance.

What disrupt the rate and timing of cardiac muscle contractions.
Responsible for specific immune response:

Immune system

Two hands of defense

Which of the following is the function of the trachea?

Gas exchange ✓

Removes the air from the body

Which fossil bird of the world is:

Archaeopteryx ✓

Platypus

Which pigment in nucleus is:

Chromatin

- (b) Platypus
- (d) Echidna

- (b) Haemoglobin ✓
- (d) Myoglobin

- (x) The secretion of the hypothalamus that increases the temperature of the body to cause fever during an infection is:
 (a) Oxytocin (b) Vasopressin
 (c) Prostaglandin ✓ (d) Somatostatin
- (xi) Hepatic portal system is formed between:
 (a) Oral cavity and Spleen (b) Liver and Small intestine
 (c) Kidney and stomach (d) Heart and kidney
- (xii) The non-protein part of enzyme is:
 (a) Apoenzyme (b) Holoenzyme
 (c) Prosthetic group ✓ (d) Ribozyme
- (xiii) Nematocytes are present in:
 (a) Ascon (b) Sycon (c) Leucon (d) Hydra ✓
- (xiv) Digestion takes place by:
 (a) Acids (b) Alkaline solutions (c) Enzymes ✓ (d) Minerals
- (xv) Coenzyme consists of:
 (a) Protein (b) Carbohydrate (c) Lipid (d) Vitamin ✓
- (xvi) Enzymes of human body are more active at:
 (a) 40°C (b) 37°F (c) 37°C ✓ (d) 55°C

SECTION 'B' (SHORT-ANSWER QUESTIONS)
 (Marks: 10x2=20)

NOTE: Attempt any FIVE questions from PART A (Reasoning questions) and five from PART B (Non-Reasoning questions).

(REASONING QUESTIONS)

Q.2 (i) Why blood flow faster in arteries?

ANSWER: Blood flows faster in large arteries because individual capillary is much narrower the capillary beds have an enormous number of such capillaries so the total diameter of these vessels is much greater than the arteries. As blood leaves the capillary bed and passes to the veins and veins it speeds up due to the reduction of the total cross-sectional area.

Q.2 (ii) Why water molecules are called amphoteric in nature?

ANSWER: (See Question No. 2(iv), 2023)

Q.2 (iii) How does substrate concentration affect the activity of enzymes?

ANSWER: The rate of reaction increases with an increase in concentration of substrate. As the concentration of substrate increases, the enzyme becomes saturated with substrate. This increase continues until the available enzyme becomes saturated with substrate. At very high concentrations the substrate exerts retarding effects on enzyme action. It means the substrate and enzyme concentration are directly proportional up to a certain maximum velocity after which further increase in substrate concentration has no effect on the rate of reaction.

Q.2 (iv) How stomach itself is protected against the strong HCl.

Q. (See Question No. 2(ii), 2023)
Why the capillaries have a single layer of endothelium?

Q. (See Question No. 2(vi), 2023)
How CO₂ from cell transport to lungs?

Q. **TO TRANSPORT CARBON DIOXIDE**
How three ways to transport carbon dioxide into the bloodstream;

1. Dissolved gas.
2. As carbamino-hemoglobin (bound to haemoglobin or other proteins),
3. As bicarbonate.

Q. How monosaccharide's are classified?

Ans: Monosaccharides can be classified on the basis of Carbon present in them, the suffix 'Ose' is used with the number of Carbon present in them. For example Monosaccharides containing three carbons are referred to as trioses, while those with four carbons are pentoses and those with five are called pentoses etc.

Q. Why protostomes are named so?

Ans: Because it used to be thought that in their embryos the dentum is formed at the mouth while the anus was formed later, at the opening made by the end of the gut.

(NON-REASONING QUESTIONS)

Q. What adaptation made the Birds capable of flight?

Ans:

ADAPTATION OF BIRDS

1. Having adaptation made the birds capable of flight.
2. Have short body with compact bones. This reduces the weight of body and makes it light.

3. There are different kinds of feathers present in birds: outline feather decrease the drag of the body by making it streamlines; primary feathers help to provide shape to the wings and make the flight possible; secondary feathers are used as rudder for balance of the body.

4. Bones in birds are either hollow or fused to reduce the weight of the body.
5. The fusion of caudal bone to form pygostyle and hollow bones of the wing are good example of it.

6. Absence of teeth and present of beak is also one of the important adaptations for flight in birds.

7. Forelimbs are modified into wings and hence aerodynamics in shape. 8. Center of mass is close to the centre of gravity.

9. Powerful muscles present in birds helps in the movement of legs, wings, and tail. Mainly wings are made of strong muscle so as to provide lift while flying which are called pectoral and breast muscles.

8. Birds generally eat foods that are small and light but which high of energy. They have gizzard that helps to grind the eaten food and absorb it. Beside this, some birds also have crop that aids in absorbing of eaten food.

Q.2 (x) What do you mean by prosthetic group?

ANSWER: The prosthetic group is a non-amino acid component closely connected to the apoprotein and is part of the structure of heteroproteins or conjugated proteins. It is bonded covalently as a permanent part of the enzyme i.e. do not detached

Q.2 (xi) Distinguished between saturated acylglycerol and unsaturated acylglycerols.

ANSWER:

SATURATED ACYLGLYCEROL	UNSATURATED ACYLGLYCEROLS
They contain saturated fatty acid i.e. do not contain any double bond between carbon atoms of hydrocarbon chain	They contain unsaturated fatty acids i.e. contain one or more than one double bond between carbon atoms of hydrocarbon chain
They are Solid at room temperature.	They are Liquid at room temperature.
It has a straight chain.	It has bend chains at the double bond.
Their sources are animal fats, palm oil, and coconut oil.	Their sources are plant and vegetable oil, avocado, sunflower oil, walnuts, flax, canola oil and fish oil.

Q.2 (xii) Mention the fundamental characters of Chordates.

ANSWER:

CHARACTERS OF CHORDATES

1. PRESENCE OF NOTOCHORD:

It is an elastic, longitudinal stiff rod present between the nerve cord and alimentary canal. It is covered by an outer chodal sheath and inner elastic sheath or elastic-intern. Below it vacuolated, non-nucleated cells are present.

Notochord is present in the embryos of the vertebrates, but is replaced by vertebral column. In less-developed vertebrates the notochord is present throughout the life.

2. PRESENCE OF PHARYNGEAL GILL SLITS:

Gill slits are present on either side of the pharynx. Each gill slit develops in the embryonic stage by evaginations of endoderm in pharynx with corresponding invagination of ectoderm on the outside of the body. They are useful for respiration. In reptiles, birds and mammals, there are

presence of gill slits in embryonic life, but they are not functional hence followed.
SENSE OF DORSAL TUBULAR NERVE CORD:
 The nerve cord in chordates is a hollow tube, situated dorsal to the notochord and the notochord. It develops from the ectoderm. The nerve cord is called neurocoel.

OBBLASTIC NATURE:
 Chordates possess three germ layers,
 1. Ectoderm,
 2. Mesoderm,
 3. Endoderm.

DEVELOPMENT OF TRUE COELOME:
 Chordates a true coelome is present. It develops from mesoderm. The coelome is developed by enterocoelic method.

SENSE OF BILATERAL SYMMETRY:
 The chordates shows bilateral symmetry. Their body can be divided into two equal halves, through a sagittal plane which passes vertically.

FUNCTION OF METAMERIC SEGMENTATION:
 In the chordates, metamerism is seen in the internal structures. The metamerism is seen in lower chordates and segmentation is seen in the condition of higher vertebrates.

Distinguish between single circuit circulation and double circuit circulation.

SINGLE CIRCUIT CIRCULATION	DOUBLE CIRCUIT CIRCULATION
A circulatory system in which blood flows once through the heart in one complete cycle of the body.	A circulatory system in which blood flows twice through the heart in one complete cycle of the body.
This circulatory system is found in fish species and reptiles.	Occurs in all other vertebrates which include mammals and humans.
Animals that possess a single circulatory system have a double heart.	Animals with double circulatory system have a four-chambered heart.
This circulatory system is referred to as "single" circulatory system because it is made up of only one circuit referred to as systemic circuit.	These circulatory system is referred to as "double" circulatory system because it is made up of two circuits referred to as the pulmonary and systemic circuits.

Only venous blood passes through the heart.

Mixed or oxygenated or venous blood passes through the heart.

Q.2 (xiv) List out the ways of second line of defense.

ANSWER: Following are the ways of second line of defense

1. Phagocytes
2. Natural Killer Cells,
3. Inflammation
4. Antimicrobial Proteins
5. Fever

Q.2 (xv) Why are valves absent in the lumen of artery?

ANSWER: The main function of the valves in veins is to prevent flow of blood.

Blood in the arteries, after being pumped by the heart, are under higher pressure than blood in veins so there is no need for valves in arteries to prevent back flow.

Blood in veins is under much lower pressure as it is returning to the heart from the various organs and tissues and so has a risk of back flow. Valves are not present to prevent it.

Q.2 (xvi) What are the health risk involved in obesity?

ANSWER: Obesity has many health risks. It increases the risk of diabetes, cardiovascular diseases, hypertension and some cancers.

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTION)

(Marks: 6x2=12)

Note: Attempt any three (2) questions from this section.

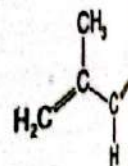
All questions carry equal marks:

Q.1 Explain terpenoids and its types.

ANSWER:

TERPENOIDS:

Terpenoid is a large and important class of lipids, made up of isoprenoid units (C_5H_8). These are found in the cell membrane as cholesterol, as pigment like chlorophyll, fragrance as menthol etc.



TYPES OF TERPENOIDS

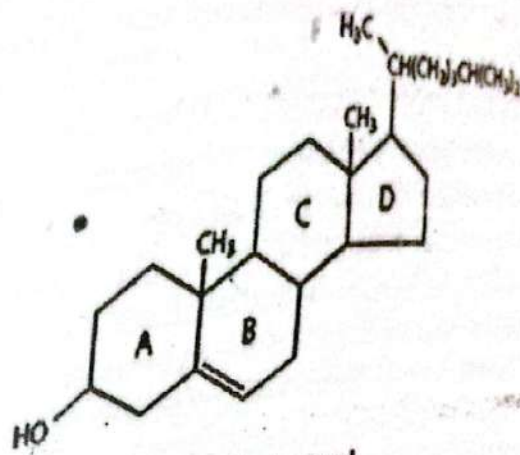
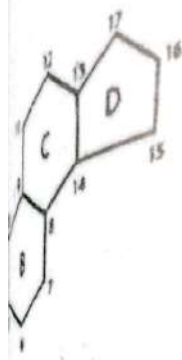
Terpenoids have following four types

1. Terpenes
2. Steroids
3. Carotenoids
4. Prostaglandins

1. TERPENES:

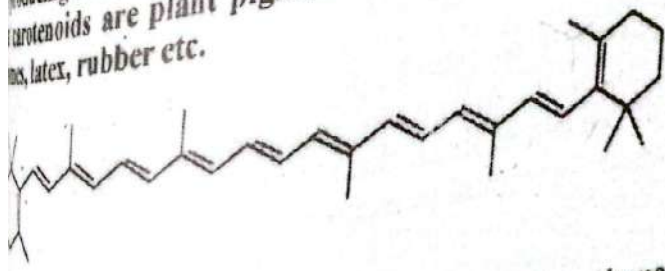
Terpenes are the type of terpenoids which contain a few isoprenoid units like diterpenes, and Triterpenes. These small-size terpenes are volatile in nature and produce special fragrances. Some of these are used in perfumes e.g. Myrcene from the oil of bay. Geraniol from rose. Limonene from lemon oil, Menthol from peppermint oil. Some components of vitamins

are a type of Terpenoid which form a steroid nucleus made up of four fused rings containing 17 Carbon atoms arranged in four attached units. The rings of them are hexagonal and one is pentagonal in shape. The side chains distinguish them from one another. Cholesterol is the precursor for the synthesis of a number of steroids i.e. testosterone, progesterone and



(b) Cholesterol

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(INHALATION)

Basically, inspiration is an active process (an energy-consuming mechanism). During this process, the thoracic cavity is increased (volume) due to the contraction of external intercostal muscles, and relaxation of internal intercostal muscles. Contraction of external intercostal muscles moves the ribs outward and as well as sternum progressively moves in the upward direction while shrinkage of diaphragm makes it levelled. Subsequently, the capacity is increased in thoracic cavity, so negative pressure is developed inside the thoracic cavity, eventually in the lungs. The expansion of the lungs decreases the pressure inside the lungs. So ultimately air rushes into the lungs up to alveolar sac through the respiratory passage where gaseous exchange takes place.

EXPIRATION (EXHALATION)

It is the passive phase of respiration. In this process, the gaseous exchange is completed in the lungs and the air is thrown out which is just reverse of inspiration. Expiration takes place due to the decreased volume of thoracic cavity. It is due to the relaxation of external inter-costal and contraction of internal intercostal muscles, which move ribs and sternum inward and downward respectively. Comparably, the diaphragm relaxes which makes it dome-shaped, thus the volume of the thoracic cavity is reduced. As an outcome, the lungs are pressed so the air along with water vapours expire outside through the respiratory pathway.

Q.2 Describe the general characters of phylum mollusca.

ANSWER: (See Question No. 4, 2023)

Q.3 What is circulation? Explain needs of circulation.

ANSWER:

CIRCULATION

The word circulation refers to the continuous movement of blood throughout the body, driven by the pumping action of the heart. The circulation of the blood links the sites of oxygen utilization and uptake in the body; thus, it is critical to ensure that oxygen-rich blood reaches tissues and that oxygen-poor blood is transported away from tissues and back to the lungs. The rapidity of circulation is determined by the output of the heart, which in turn is responsive to overall body requirements.

NEEDS OF CIRCULATION

Diffusion alone is not adequate for transporting chemicals over macroscopic distances in animals for example to transport glucose from the digestive tract and oxygen from the lungs to the brain of animals. The time it takes for a substance to diffuse from one place to another is proportional to the square of the distance the chemical will travel. The circulatory system solves this problem by ensuring that no substance has to diffuse very far to enter or leave a cell. By transporting fluid throughout

body, it functionally connects the aqueous environment of the body to the organs that exchange gases, and nutrients and dispose of wastes. In the lungs of a mammal, for example, oxygen from inhaled air diffuses through a thin epithelium and into the blood, while carbon dioxide diffuses in the opposite direction. The circulatory system then carries the oxygenated blood to all parts of the body. As the blood streams through the tissues via microscopic vessels called capillaries, chemicals are transported between the blood and the interstitial fluid that directly bathes the cells. (OR) Discuss the structure features of antibody? How it is helpful in getting rid of pathogenic organisms.

ANSWER:

STRUCTURE FEATURES OF ANTIBODY

An antibody molecule is essentially identical to the antigen receptor of a B cell that produced it. The basic structure of these proteins consists of two pairs of polypeptide chains (lengths of amino acids linked by peptide bonds) that form a flexible Y shape. The stem of the Y consists of one end of each of two identical heavy chains, while each arm is composed of the remaining portion of a heavy chain plus a smaller protein called the light chain. The two light chains also are identical. Within particular classes of antibodies the stem and the bottom of the arms are fairly similar and thus called the constant region. The tips of the arms, however, are highly variable in sequence. It is these tips that bind antigen. Thus each antibody has two identical antigen-binding sites, one at the end of each arm, and the antigen-binding sites vary greatly among antibodies.

Antibodies are grouped into five classes according to their constant region. Each class is designated by a letter attached to an abbreviation of the word immunoglobulin: IgG, IgM, IgA, IgD, and IgE. The classes of antibodies differ not only in their constant region but also in activity. For example, IgG, the most common antibody, is present mostly in the blood and tissue fluids, while IgA is found in the mucous membranes lining the respiratory and gastrointestinal tracts.

HELPFUL IN GETTING RID OF PATHOGENIC ORGANISMS

An antigen could be any foreign macromolecule, especially proteins which elicit an immune response. In response to the particular antigens, B-lymphocytes produce a small, soluble and specific protein called an antibody which combines with specific antigens and helps to destroy the antigen-displaying pathogenic organisms. The immune system of a vertebrate can produce billions of different antibodies. Not only the immune system destroys in this way the pathogenic organism but it also remembers it and can produce antibodies against the same pathogen in case of reinfection. The distinction between the self and non-self is the hallmark of the immune system.